





# Introduction to Order Theory

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# Notation

|                                 |                                       |
|---------------------------------|---------------------------------------|
| $\mathcal{M}_+(X, \mathcal{A})$ | finite measures on $(X, \mathcal{A})$ |
| $v$                             | velocity                              |



# Preface

cc-by-nc-sa



# Part I

## Sets



# Chapter 1

## Relations

### 1.1 Introduction

**Definition 1.1.0.1.** Let  $n \in \mathbb{N}$  and  $(A_j)_{j \in [n]}$  a collection of sets.

- Let  $R \subset \prod_{j \in [n]} A_j$ . Then  $R$  is said to be a  **$n$ -ary relation over  $(A_j)_{j \in [n]}$** .
- We define  $\mathcal{R}(A_j : j \in [n]) := \mathcal{P}(\prod_{j \in [n]} A_j)$

## 1.2 Homogeneous Relations

**Definition 1.2.0.1.** Let  $n \in \mathbb{N}$  and  $A$  a set.

- Let  $R \subset A^n$ . Then  $R$  is said to be a **homogeneous  $n$ -ary relation on  $A$**
- We define  $\mathcal{R}^n(A) := \mathcal{P}(A^n)$ .

### 1.2.1 Subrelations

**Definition 1.2.1.1.** Let  $n \in \mathbb{N}$ ,  $A, B$  sets,  $R \in \mathcal{R}^n(A)$  and  $S \in \mathcal{R}^n(B)$ . Then

- $S$  is said to be a **restriction of  $R$**  if  $B \subset A$ ,
- $R$  is said to be an **extension of  $S$**  if  $S$  is a restriction of  $R$ .

**Definition 1.2.1.2.** Let  $n \in \mathbb{N}$ ,  $A$  a set,  $B \subset A$  and  $R \in \mathcal{R}^n(A)$ . We define the **restriction of  $R$  to  $B$** , denoted  $R|_B$ , by  $R|_B := R \cap B^n$ .

**Exercise 1.2.1.3.** Let  $n \in \mathbb{N}$ ,  $A$  a set,  $B \subset A$  and  $R \in \mathcal{R}^n(A)$ . Then  $R|_B$  is a restriction of  $R$ .

*Proof.* Clear. **fill out later**

□

### 1.2.2 Product Relations



## 1.3 Binary Relations

**Definition 1.3.0.1.** Let  $A, B$  be sets.

- Let  $R \subset A \times B$ . Then  $R$  is said to be a **relation over  $A$  and  $B$** .
- We define  $\mathcal{R}(A, B) := \mathcal{P}(A \times B)$

**Definition 1.3.0.2.** Let  $A, B, C$  be sets,  $R \in \mathcal{R}(A, B)$ ,  $S \in \mathcal{R}(B, C)$ . We define the **composition relation of  $S$  with  $R$** , denoted  $S \circ_{\text{Rel}} R \in \mathcal{R}(A, C)$ , by

$$S \circ_{\text{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

**Note 1.3.0.3.** When the context is clear, we write  $S \circ R$  in place of  $S \circ_{\text{Rel}} R$ .

**Exercise 1.3.0.4.** Let  $A, B, C, D$  be sets,  $R$  a relation between  $A$  and  $B$ ,  $S$  relation between  $B$  and  $C$  and  $T$  relation between  $C$  and  $D$ . Then  $(T \circ S) \circ R = T \circ (S \circ R)$ .

*Proof.*

- Let  $(a, d) \in (T \circ S) \circ R$ . Then there exists  $b \in B$  such that  $(a, b) \in R$  and  $(b, d) \in T \circ S$ . Since  $(b, d) \in T \circ S$ , there exists  $c \in C$  such that  $(b, c) \in S$  and  $(c, d) \in T$ . Since  $(a, b) \in R$  and  $(b, c) \in S$ ,  $(a, c) \in S \circ R$ . Since  $(a, c) \in S \circ R$  and  $(c, d) \in T$ , we have that  $(a, d) \in T \circ (S \circ R)$ . Since  $(a, d) \in (T \circ S) \circ R$  is arbitrary, we have that for each  $(a, d) \in (T \circ S) \circ R$ ,  $(a, d) \in T \circ (S \circ R)$ . Hence  $(T \circ S) \circ R \subset T \circ (S \circ R)$ .
- Similarly,  $T \circ (S \circ R) \subset (T \circ S) \circ R$ .

Since  $(T \circ S) \circ R \subset T \circ (S \circ R)$  and  $T \circ (S \circ R) \subset (T \circ S) \circ R$ , we have that  $(T \circ S) \circ R = T \circ (S \circ R)$ . □

**Definition 1.3.0.5.** Let  $A$  be a set. We define the **diagonal relation on  $A$** , denoted  $\Delta_A \in \mathcal{R}(A, A)$ , by

$$\Delta_A := \{(a, a) : a \in A\}.$$

**Exercise 1.3.0.6.** Let  $A, B$  be sets and  $R \in \mathcal{R}(A, B)$ . Then

1.  $R \circ \Delta_A = R$
2.  $\Delta_B \circ R = R$

*Proof.*

1.
  - Let  $(a, b) \in R \circ \Delta_A$ . Then there exists  $a' \in A$  such that  $(a, a') \in \Delta_A$  and  $(a', b) \in R$ . By definition of  $\Delta_A$ ,  $a = a'$ . Therefore  $(a, b) \in R$ . Since  $(a, b) \in R \circ \Delta_A$  is arbitrary, we have that for each  $(a, b) \in R \circ \Delta_A$ ,  $(a, b) \in R$ . Hence  $R \circ \Delta_A \subset R$ .
  - Let  $(a, b) \in R$ . Then  $a \in A$ . Therefore  $(a, a) \in \Delta_A$ . Since  $(a, a) \in \Delta_A$  and  $(a, b) \in R$ , we have that  $(a, b) \in R \circ \Delta_A$ . Since  $(a, b) \in R$  is arbitrary, we have that for each  $(a, b) \in R$ ,  $(a, b) \in R \circ \Delta_A$ . Hence  $R \circ \Delta_A \supset R$ .

Since  $R \circ \Delta_A \subset R$  and  $R \circ \Delta_A \supset R$ , we have that  $R \circ \Delta_A = R$ .

2. Similar to (1). □

**Exercise 1.3.0.7.** Let  $A, B, C$  be sets.

1. Let  $R \in \mathcal{R}(A, B)$  and  $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(B, C)$ . Then

$$(a) \quad \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R),$$

$$(b) \quad \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$$

2. Let  $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$  and  $S \in \mathcal{R}(B, C)$ . Then

$$(a) \quad S \circ \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) = \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma).$$

$$(b) \quad S \circ \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma).$$

*Proof.*

1. (a) We note that

$$\begin{aligned} (a, c) \in \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\ &\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, c) \in S_{\gamma_0} \circ R \\ &\iff (a, c) \in \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R). \end{aligned}$$

$$\text{Hence } \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R).$$

(b) We note that

$$\begin{aligned} (a, c) \in \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\ &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and for each } \gamma \in \Gamma \text{ and } (b, c) \in S_\gamma \\ &\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_\gamma \\ &\iff \text{for each } \gamma \in \Gamma, (a, c) \in S_\gamma \circ R \\ &\iff (a, c) \in \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R). \end{aligned}$$

$$\text{Hence } \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R).$$

2. (a) We note that

$$\begin{aligned} (a, c) \in S \circ \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \text{ and } (b, c) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, c) \in S \circ R_{\gamma_0} \\ &\iff (a, c) \in \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma). \end{aligned}$$

$$\text{Hence } S \circ \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) = \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma).$$

(b) We note that

$$\begin{aligned} (a, c) \in S \circ \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma, (a, b) \in R_\gamma \text{ and } (b, c) \in S \\ &\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a, b) \in R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{for each } \gamma \in \Gamma, (a, c) \in S \circ R_\gamma \\ &\iff (a, c) \in \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma). \end{aligned}$$

Hence  $S \circ \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$ .

□

**Exercise 1.3.0.8.** Let  $A, B, C$  be sets.

1. Let  $R_1, R_2 \in \mathcal{R}(A, B)$  and  $S \in \mathcal{R}(B, C)$ . If  $R_1 \subset R_2$ , then  $S \circ R_1 \subset S \circ R_2$ .
2. Let  $R \in \mathcal{R}(A, B)$  and  $S_1, S_2 \in \mathcal{R}(B, C)$ . If  $S_1 \subset S_2$ , then  $S_1 \circ R \subset S_2 \circ R$ .
3. Let  $R_1, R_2 \in \mathcal{R}(A, B)$  and  $S_1, S_2 \in \mathcal{R}(B, C)$ . If  $R_1 \subset R_2$  and  $S_1 \subset S_2$ , then  $S_1 \circ R_1 \subset S_2 \circ R_2$ .

*Proof.*

1. Suppose that  $R_1 \subset R_2$ . Then

$$\begin{aligned} (a, c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R_1 \text{ and } (b, c) \in S \\ &\implies \text{there exists } b \in B \text{ such that } (a, b) \in R_2 \text{ and } (b, c) \in S \\ &\iff (a, c) \in S \circ R_2. \end{aligned}$$

Hence  $S \circ R_1 \subset S \circ R_2$ .

2. Suppose that  $S_1 \subset S_2$ . Then

$$\begin{aligned} (a, c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_1 \\ &\implies \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_2 \\ &\iff (a, c) \in S_2 \circ R. \end{aligned}$$

Hence  $S_2 \circ R \subset S_1 \circ R$ .

3. Suppose that  $R_1 \subset R_2$  and  $S_1 \subset S_2$ . Then **the previous parts** imply that

$$\begin{aligned} S_1 \circ R_1 &\subset S_2 \circ R_1 \\ &\subset S_2 \circ R_2. \end{aligned}$$

□

**Definition 1.3.0.9.** Let  $A, B$  be sets and  $R \in \mathcal{R}(A, B)$ . We define the **converse of  $R$** , denoted  $R^* \in \mathcal{R}(B, A)$ , by

$$R^* := \{(b, a) \in B \times A : (a, b) \in R\}.$$

**Exercise 1.3.0.10.** Let  $A, B$  be sets and  $R$  a relation between  $A$  and  $B$ . Then  $(R^*)^* = R$ .

*Proof.*

$$\begin{aligned} (a, b) \in (R^*)^* &\iff (b, a) \in R^* \\ &\iff (a, b) \in R. \end{aligned}$$

Hence  $(R^*)^* = R$ .

□

**Exercise 1.3.0.11.** Let  $A, B, C$  be sets,  $R \in \mathcal{R}(A, B)$  and  $S \in \mathcal{R}(B, C)$ . Then  $(S \circ R)^* = R^* \circ S^*$ .

*Proof.* We note that

$$\begin{aligned} (c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\ &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\ &\iff (c, a) \in R^* \circ S^*. \end{aligned}$$

Hence  $(S \circ R)^* = R^* \circ S^*$ .

□

**Exercise 1.3.0.12.** Let  $A$  and  $B$  be sets and  $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$ . Then

1.  $\left( \bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2.  $\left( \bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

*Proof.*

1. We note that

$$\begin{aligned}
 (b, a) \in \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\
 &\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*.
 \end{aligned}$$

$$\text{Hence } \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*.$$

2. We note that

$$\begin{aligned}
 (b, a) \in \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (b, a) \in R_\gamma^* \\
 &\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*.
 \end{aligned}$$

$$\text{Hence } \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*.$$

□

### 1.3.1 Subrelations

**Definition 1.3.1.1.** Let  $n \in \mathbb{N}$ ,  $A, B, E, F$  sets,  $R \in \mathcal{R}(A, B)$  and  $S \in \mathcal{R}(E, F)$ . Then

- $S$  is said to be a **restriction of  $R$**  if  $E \subset A$  and  $F \subset B$ ,
- $R$  is said to be an **extension of  $S$**  if  $S$  is a restriction of  $R$ .

**Definition 1.3.1.2.** Let  $A, B$  be sets,  $E \subset A$ ,  $F \subset B$  and  $R \in \mathcal{R}(A, B)$ . We define the **restriction of  $R$  over  $E$  and  $F$** , denoted  $R|_{E \times F} \in \mathcal{R}(E, F)$ , by  $R|_{E \times F} := R \cap (E \times F)$ .

**Exercise 1.3.1.3.** Let  $A, B$  be sets,  $E \subset A$ ,  $F \subset B$  and  $R \in \mathcal{R}(A, B)$ . Then  $R|_B$  is a restriction of  $R$ .

*Proof.* Clear. fill out later

□

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain  $f|_E$  or  $f|_F$ , just get rid of this subsection

**Exercise 1.3.1.4.** Let  $A, B$  be sets,  $E \subset A$ ,  $F \subset B$  and  $R \in \mathcal{R}(A, B)$ . Suppose that  $E \subset A$  and  $F \subset B$ . Then  $(R^*)|_{F \times E} = (R|_{E \times F})^*$ .

*Proof.* We note that for each  $b, a \in B \times A$ ,

$$\begin{aligned} (b, a) \in (R^*)|_{F \times E} &\iff (b, a) \in R^* \cap (F \times E) \\ &\iff (a, b) \in R \cap (E \times F) \\ &\iff (a, b) \in R|_{E \times F}. \end{aligned}$$

Hence  $(R^*)|_{F \times E} = R|_{E \times F}$ . □

### 1.3.2 Products of Relations

**Definition 1.3.2.1.** Let  $(A_j)_{j \in [n]}, (B_j)_{j \in [n]}$  be sets and  $(R_j)_{j \in [n]} \in \prod_{j \in [n]} \mathcal{R}(A_j, B_j)$ . We define the **product relation of  $(R_j)_{j \in [n]}$** ,

denoted  $\prod_{j \in [n]}^{\text{Rel}} R_j \in \mathcal{R}\left(\prod_{j \in [n]} A_j, \prod_{j \in [n]} B_j\right)$ , by

$$\prod_{j \in [n]}^{\text{Rel}} R_j := \left\{ ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \left( \prod_{j \in [n]} A_j \right) \times \left( \prod_{j \in [n]} B_j \right) : \text{for each } j \in [n], (a_j, b_j) \in R_j \right\}.$$

**Note 1.3.2.2.**

- When  $n = 2$ , we may write  $R_1 \times^{\text{Rel}} R_2$  in place of  $\prod_{j \in [2]}^{\text{Rel}} R_j$ ,
- If for each  $j \in [n]$ ,  $R_j = R$ , we may write  $R^{\times^{\text{Rel}} n}$  in place of  $\prod_{j \in [n]}^{\text{Rel}} R$ ,
- When the context is clear we may write  $\prod_{j \in [n]} R_j$  and  $R_1 \times R_2$  in place of  $\prod_{j \in [n]}^{\text{Rel}} R_j$  and  $R_1 \times^{\text{Rel}} R_2$ .

**Exercise 1.3.2.3.** Let  $A, B, C, D$  be sets.

1. Let  $R \in \mathcal{R}(A, B)$  and  $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(C, D)$ . Then

$$(a) \quad R \times \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$$

$$(b) \quad R \times \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$$

2. Let  $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$  and  $S \in \mathcal{R}(C, D)$ . Then

$$(a) \quad \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)$$

$$(b) \quad \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)$$

*Proof.*

1. (a) We note that

$$\begin{aligned} ((a, c), (b, d)) \in R \times \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\ &\iff (a, b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c, d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (c, d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R \times S_{\gamma_0} \\ &\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R \times S_\gamma). \end{aligned}$$

$$\text{Hence } R \times \left( \bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma).$$

(b) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in R \times \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\
 &\iff (a, b) \in R \text{ for each } \gamma \in \Gamma, (c, d) \in S_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R \text{ and } (c, d) \in S_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, ((a, c), (b, d)) \in R \times S_\gamma \\
 &\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).
 \end{aligned}$$

$$\text{Hence } R \times \left( \bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).$$

2. (a) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S &\iff (a, b) \in \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) \text{ and } (c, d) \in S \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \text{ and } (c, d) \in S \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R_{\gamma_0} \times S \\
 &\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)
 \end{aligned}$$

$$\text{Hence } \left( \bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S).$$

(b) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S &\iff (a, b) \in \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \text{ and } (c, d) \in S \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R_\gamma \text{ and } (c, d) \in S \\
 &\iff \text{for each } \gamma \in \Gamma, ((a, c), (b, d)) \in R_\gamma \times S \\
 &\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)
 \end{aligned}$$

$$\text{Hence } \left( \bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S).$$

□

### 1.3.3 Subrelations

**Exercise 1.3.3.1.** Let  $A$  be a set,  $B \subset A$  and  $R \in \mathcal{R}^2(A)$ . Then  $(R^*)|_B = (R|_B)^*$

*Proof.* Clear by [prev ex in section on binary hetero relations](#).

□

## Chapter 2

# Operations





# Chapter 3

## Equivalence Relations

### 3.1 Introduction

**Note 3.1.0.1.** Let  $X$  be a set and  $\nu \in \mathcal{R}^2(X)$ . When the context is clear, we may write  $a\nu b$  in place of  $(a, b) \in \nu$ .

**Definition 3.1.0.2.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ .

- Let  $\theta \in \mathcal{R}^2(X)$ . Then  $\theta$  is said to be
  - **reflexive** if for each  $a \in X$ ,  $(a, a) \in \theta$
  - **symmetric** if for each  $a, b \in X$ ,  $(a, b) \in \theta$  implies that  $(b, a) \in \theta$
  - **transitive** if for each  $a, b, c \in X$ ,  $(a, b) \in \theta$  and  $(b, c) \in \theta$  implies that  $(a, c) \in \theta$
  - an **equivalence relation on  $X$**  if
    - 1.  $\theta$  is reflexive,
    - 2.  $\theta$  is symmetric,
    - 3.  $\theta$  is transitive.
- We define  $\text{Equiv}(X) := \{\theta \in \mathcal{R}^2(X) : \theta \text{ is an equivalence relation on } X\}$

**Exercise 3.1.0.3.** Let  $X$  be a set and  $\theta \in \mathcal{R}^2(X)$ . Then

1.  $\theta$  is reflexive iff  $\Delta_X \subset \theta$
2.  $\theta$  is symmetric iff  $\theta^* \subset \theta$
3.  $\theta$  is transitive iff  $\theta \circ \theta \subset \theta$

*Proof.*

1.
  - ( $\implies$ ):  
Suppose that  $\theta$  is reflexive. Let  $(a, a) \in \Delta_X$ . Then  $a \in X$ . Since  $\theta$  is reflexive,  $(a, a) \in \theta$ . Since  $(a, a) \in \Delta_X$  is arbitrary, we have that for each  $(a, a) \in \Delta_X$ ,  $(a, a) \in \theta$ . Hence  $\Delta_X \subset \theta$ .
  - ( $\impliedby$ ):  
Suppose that  $\Delta_X \subset \theta$ . Let  $a \in X$ . By definition and assumption,

$$\begin{aligned} (a, a) &\in \Delta_X \\ &\subset \theta. \end{aligned}$$

Since  $a \in X$  is arbitrary, we have that for each  $a \in X$ ,  $(a, a) \in \theta$ . Hence  $\theta$  is reflexive.

2.
  - ( $\implies$ ):  
Suppose that  $\theta$  is symmetric. Let  $(b, a) \in \theta^*$ . By definition,  $(a, b) \in \theta$ . Since  $\theta$  is symmetric,  $(b, a) \in \theta$ . Since  $(b, a) \in \theta^*$  is arbitrary, we have that for each  $(b, a) \in \theta^*$ ,  $(b, a) \in \theta$ . Hence  $\theta^* \subset \theta$ .

- ( $\Leftarrow$ ):

Suppose that  $\theta^* \subset \theta$ . Let  $a, b \in X$ . Suppose that  $(a, b) \in \theta$ . By definition and assumption,

$$\begin{aligned} (b, a) &\in \theta^* \\ &\subset \theta. \end{aligned}$$

Since  $a, b \in X$  with  $(a, b) \in \theta$  are arbitrary, we have that for each  $a, b \in X$ ,  $(a, b) \in \theta$  implies that  $(b, a) \in \theta$ . Hence  $\theta$  is symmetric.

3. • ( $\Rightarrow$ ):

Suppose that  $\theta$  is transitive. Let  $(a, c) \in \theta \circ \theta$ . By definition, there exists  $b \in X$  such that  $(a, b) \in \theta$  and  $(b, c) \in \theta$ . Since  $\theta$  is transitive,  $(a, c) \in \theta$ . Since  $(a, c) \in \theta \circ \theta$  is arbitrary, we have that for each  $(a, c) \in \theta \circ \theta$ ,  $(a, c) \in \theta$ . Hence  $\theta \circ \theta \subset \theta$ .

- ( $\Leftarrow$ ):

Suppose that  $\theta \circ \theta \subset \theta$ . Let  $a, b, c \in X$ . Suppose that  $(a, b) \in \theta$  and  $(b, c) \in \theta$ . By definition and assumption,

$$\begin{aligned} (a, c) &\in \theta \circ \theta \\ &\subset \theta. \end{aligned}$$

Since  $a, b, c \in X$  with  $(a, b) \in \theta$  and  $(b, c) \in \theta$  are arbitrary, we have that for each  $a, b \in X$ ,  $(a, b) \in \theta$  and  $(b, c) \in \theta$  implies that  $(a, c) \in \theta$ . Hence  $\theta$  is transitive.

□

**Definition 3.1.0.4.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . We define the **universal relation on  $X$** , denoted  $\nabla_X \in \mathcal{R}^2(X)$ , by  $\nabla_X := X^2$ .

**Exercise 3.1.0.5.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Then  $\nabla_X \in \text{Equiv}(X)$ .

*Proof.* Let  $a, b, c \in X$ .

1. Since  $a \in X$ ,  $(a, a) \in \nabla_X$ .
2. Suppose that  $(a, b) \in \nabla_X$ . Since  $a, b \in X$ ,  $(b, a) \in X^2$ . Therefore  $(a, b) \in \nabla_X$  implies that  $(b, a) \in \nabla_X$ .
3. Suppose that  $(a, b), (b, c) \in \nabla_X$ . Since  $a, c \in X$ ,  $(a, c) \in X^2$ . Therefore  $(a, b), (b, c) \in X^2$  implies that  $(a, c) \in \nabla_X$ .

□

**Exercise 3.1.0.6.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Then  $\Delta_X \in \text{Equiv}(X)$ .

*Proof.* Let  $a, b, c \in X$ .

1. Since  $a \in X$ ,  $(a, a) \in \Delta_X$ .
2. Suppose that  $(a, b) \in \Delta_X$ . Then  $a = b$  and

$$\begin{aligned} (b, a) &= (a, a) \\ &\in \Delta_X. \end{aligned}$$

Therefore  $(a, b) \in \Delta_X$  implies that  $(b, a) \in \Delta_X$ .

3. Suppose that  $(a, b), (b, c) \in \Delta_X$ . Then  $a = b$  and  $b = c$ . Hence  $a = c$  and

$$\begin{aligned} (a, c) &= (a, a) \\ &\in \Delta_X. \end{aligned}$$

Therefore  $(a, b), (b, c) \in \Delta_X$  implies that  $(a, c) \in \Delta_X$ .

□

**Exercise 3.1.0.7.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Then for each  $(\theta_\alpha)_{\alpha \in A} \subset \text{Equiv}(X)$ ,  $\bigcap_{\alpha \in A} \theta_\alpha \in \text{Equiv}(X)$ .

*Proof.* Let  $(\theta_\alpha)_{\alpha \in A} \subset \text{Equiv}(X)$ . Set  $\theta := \bigcap_{\alpha \in A} \theta_\alpha$ . Let  $a, b, c \in X$ .

1. Let  $\alpha \in A$ . Since  $\theta_\alpha \in \text{Equiv}(X)$ ,  $(a, a) \in \theta_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $(a, a) \in \theta_\alpha$ . Hence  $(a, a) \in \theta$ .
2. Suppose that  $(a, b) \in \theta$ . Let  $\alpha \in A$ . Then  $(a, b) \in \theta_\alpha$ . Since  $\theta_\alpha \in \text{Equiv}(X)$ , we have that  $(b, a) \in \theta_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $(b, a) \in \theta_\alpha$ . Hence  $(b, a) \in \theta$ . Therefore  $(a, b) \in \theta$  implies that  $(b, a) \in \theta$ .
3. Suppose that  $(a, b) \in \theta$  and  $(b, c) \in \theta$ . Let  $\alpha \in A$ . Then  $(a, b), (b, c) \in \theta_\alpha$ . Since  $\theta_\alpha \in \text{Equiv}(X)$ ,  $(a, c) \in \theta_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $(a, c) \in \theta_\alpha$ . Hence  $(a, c) \in \theta$ . Therefore  $(a, b) \in \theta$  and  $(b, c) \in \theta$  implies that  $(a, c) \in \theta$ .

Hence  $\theta \in \text{Equiv}(X)$ . □

**Definition 3.1.0.8.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Let  $\nu \in \mathcal{R}^2(X)$ .

- We define the **equivalence relations containing**  $\nu$ , denoted  $\mathcal{E}(\nu)$ , by  $\mathcal{E}(\nu) := \{\theta \in \text{Equiv}(X) : \nu \subset \theta\}$ .
- We define the **equivalence relation on  $X$  generated by**  $\nu$ , denoted  $\epsilon(\nu) \in \mathcal{R}^2(X)$  by  $\epsilon(\nu) := \bigcap_{\theta \in \mathcal{E}(\nu)} \theta$ .

**Note 3.1.0.9.**

- A previous ex implies that  $\nabla_X \in \mathcal{E}(\nu)$ .
- A previous ex (about intersections of equiv relats) implies that  $\epsilon(\nu) \in \mathcal{E}(\nu)$ .

**Exercise 3.1.0.10.** Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Let  $\nu \in \mathcal{R}^2(X)$ . Define  $(\nu_n)_{n \in \mathbb{N}_0} \subset \mathcal{R}^2(X)$  by

$$\nu_n := \begin{cases} \Delta_X \cup \nu \cup \nu^*, & n = 0 \\ \nu_{n-1} \circ \nu_{n-1}, & n \geq 1. \end{cases}$$

Then

1. (a) for each  $n \in \mathbb{N}$ ,  $\nu_n$  is reflexive  
 (b) for each  $n \in \mathbb{N}$ ,  $\nu_n$  is symmetric  
 (c) for each  $n \in \mathbb{N}_0$ ,  $\nu_n \subset \nu_{n+1}$
2.  $\epsilon(\nu) = \bigcup_{n \in \mathbb{N}_0} \nu_n$ .

*Proof.*

1. (a) • **Base Case:**  
 Since

$$\begin{aligned} \Delta_X &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0, \end{aligned}$$

a previous ex implies that  $\nu_0$  is reflexive.

• **Induction Step:**

Let  $n \in \mathbb{N}_0$ . Suppose that  $\nu_n$  is reflexive. Then that same prev ex implies that  $\Delta_X \subset \nu_n$ . two prev ex in section on relations about composition with identity and subsets then imply that

$$\begin{aligned} \Delta_X &= \Delta_X \circ \Delta_X \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}. \end{aligned}$$

Another application of that prev ex implies that  $\nu_{n+1}$  is reflexive.

By induction, for each  $n \in \mathbb{N}_0$ ,  $\nu_n$  is reflexive.

(b) • **Base Case:**

a previous ex about converse relations and unions, double converse and since  $\Delta_X \in \text{Equiv}(X)$  and an ex about symmetric relations implies that

$$\begin{aligned}
 \nu_0^* &= (\Delta_X \cup \nu \cup \nu^*)^* \\
 &= \Delta_X^* \cup \nu^* \cup (\nu^*)^* \\
 &= \Delta_X^* \cup \nu^* \cup \nu \\
 &\subset \Delta_X \cup \nu^* \cup \nu \\
 &= \Delta_X \cup \nu \cup \nu^* \\
 &= \nu_0.
 \end{aligned}$$

that same previous ex about symmetric relations implies that  $\nu_0$  is symmetric.

• **Induction Step:**

Let  $n \in \mathbb{N}_0$ . Suppose that  $\nu_n$  is symmetric. Then that same prev ex about symmetric and converse relations implies that  $\nu_n^* \subset \nu_n$ . Therefore prev ex imply

$$\begin{aligned}
 \nu_{n+1}^* &= (\nu_n \circ \nu_n)^* \\
 &= \nu_n^* \circ \nu_n^* \\
 &\subset \nu_n \circ \nu_n. &= \nu_{n+1}.
 \end{aligned}$$

Another application of that prev ex implies that  $\nu_{n+1}$  is symmetric.

By induction, for each  $n \in \mathbb{N}_0$ ,  $\nu_n$  is symmetric.

(c) Let  $n \in \mathbb{N}_0$ . Since  $\nu_n$  is reflexive, prev ex implies that  $\Delta_X \subset \nu_n$ . Therefore prev ex implies

$$\begin{aligned}
 \nu_n &= \Delta_X \circ \nu_n \\
 &\subset \nu_n \circ \nu_n \\
 &= \nu_{n+1}.
 \end{aligned}$$

Since  $n \in \mathbb{N}_0$  is arbitrary, we have that for each  $n \in \mathbb{N}_0$ ,  $\nu_n \subset \nu_{n+1}$ .

2. Set  $\theta_0 := \bigcup_{n \in \mathbb{N}_0} \nu_n$ .

• – **Base Case:**

Since  $\epsilon(\nu) \in \mathcal{E}(\nu)$ , we have that  $\epsilon(\nu) \in \text{Equiv}(X)$  and  $\nu \subset \epsilon(\nu)$ . Since  $\epsilon(\nu) \in \text{Equiv}(X)$ , we have that  $\epsilon(\nu)$  is reflexive and symmetric. A ref ex then implies that  $\Delta_X \subset \epsilon(\nu)$  and  $\epsilon(\nu)^* \subset \epsilon(\nu)$ . Therefore a prev ex about converse relations and subsets

$$\begin{aligned}
 \nu_0 &= \Delta_X \cup \nu \cup \nu^* \\
 &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu)^* \\
 &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu) \\
 &= \epsilon(\nu).
 \end{aligned}$$

– **Induction Step:**

Let  $n \in \mathbb{N}_0$ . Suppose that  $\nu_n \subset \epsilon(\nu)$ . Then ref prev ex

$$\begin{aligned}
 \nu_{n+1} &= \nu_n \circ \nu_n \\
 &\subset \epsilon(\nu) \circ \epsilon(\nu) \\
 &= \epsilon(\nu).
 \end{aligned}$$

By induction, for each  $n \in \mathbb{N}_0$ ,  $\nu_n \subset \epsilon(\nu)$ . Therefore

$$\begin{aligned}
 \theta_0 &= \bigcup_{n \in \mathbb{N}_0} \nu_n \\
 &\subset \epsilon(\nu).
 \end{aligned}$$

- – Since  $\nu_0$  is reflexive, we have that

$$\begin{aligned}\Delta_X &\subset \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Hence  $\theta_0$  is reflexive.

- Since for each  $n \in \mathbb{N}_0$ ,  $\nu_n$  is symmetric, we have that that **ref exs** for each  $n \in \mathbb{N}_0$ ,  $\nu_n^* \subset \nu_n$ . Therefore

$$\begin{aligned}\theta_0^* &= \left( \bigcup_{n \in \mathbb{N}_0} \nu_n \right)^* \\ &= \bigcup_{n \in \mathbb{N}_0} \nu_n^* \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Hence  $\theta_0$  is symmetric.

- Let  $a, b, c \in X$ . Suppose that  $(a, b) \in \theta_0$  and  $(b, c) \in \theta_0$ . Then there exist  $n_1, n_2 \in \mathbb{N}_0$  such that  $(a, b) \in \nu_{n_1}$  and  $(b, c) \in \nu_{n_2}$ . Set  $n_0 := \max\{n_1, n_2\}$ . Since for each  $n \in \mathbb{N}$ ,  $\nu_n \subset \nu_{n+1}$ , we have that

$$\begin{aligned}(a, b) &\in \nu_{n_1} \\ &\subset \nu_{n_0}.\end{aligned}$$

and

$$\begin{aligned}(b, c) &\in \nu_{n_2} \\ &\subset \nu_{n_0}.\end{aligned}$$

Since  $(a, b), (b, c) \in \nu_{n_0}$ , we have that

$$\begin{aligned}(a, c) &\in \nu_{n_0} \circ \nu_{n_0} \\ &= \nu_{n_0+1} \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Since  $a, b, c \in X$  with  $(a, b) \in \theta_0$  and  $(b, c) \in \theta_0$  are arbitrary, we have that for each  $a, b, c \in X$ ,  $(a, b) \in \theta_0$  and  $(b, c) \in \theta_0$  implies that  $(a, c) \in \theta_0$ . Hence  $\theta_0$  is transitive.

Since  $\theta_0$  is reflexive, symmetric and transitive,  $\theta_0 \in \text{Equiv}(X)$ . Since

$$\begin{aligned}\nu &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0,\end{aligned}$$

we have that  $\theta_0 \in \mathcal{E}(\nu)$ . Therefore

$$\begin{aligned}\epsilon(\nu) &= \bigcap_{\theta \in \mathcal{E}(\nu)} \theta \\ &\subset \theta_0.\end{aligned}$$

Since  $\theta_0 \subset \epsilon(\nu)$  and  $\epsilon(\nu) \subset \theta_0$ , we have that  $\theta_0 = \epsilon(\nu)$ .

□

### 3.2 Products

**Exercise 3.2.0.1.** Let  $(X_j)_{j \in [n]}$  be sets. Then for each  $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ ,  $\Pi_{j \in [n]}^{\text{Rel}} \theta_j \in \text{Equiv}(\prod_{j \in [n]} X_j)$ .

*Proof.* Set  $X := \prod_{j \in [n]} X_j$  and define  $\theta \in \mathcal{R}^2(X)$  by  $\theta := \Pi_{j \in [n]}^{\text{Rel}} \theta_j$ .

1. For each  $(a_j)_{j \in [n]} \in X$ ,

$$\begin{aligned} \text{for each } j \in [n], a_j \in X_j &\implies \text{for each } j \in [n], (a_j, a_j) \in \theta_j \\ &\implies ((a_j)_{j \in [n]}, (a_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus  $\theta$  is reflexive.

2. For each  $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$ ,

$$\begin{aligned} ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \theta &\iff \text{for each } j \in [n], (a_j, b_j) \in \theta_j \\ &\implies \text{for each } j \in [n], (b_j, a_j) \in \theta_j \\ &\iff ((b_j)_{j \in [n]}, (a_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus  $\theta$  is symmetric.

3. For each  $(a_j)_{j \in [n]}, (b_j)_{j \in [n]}, (c_j)_{j \in [n]} \in X$ ,

$$\begin{aligned} ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \theta \text{ and } ((b_j)_{j \in [n]}, (c_j)_{j \in [n]}) \in \theta &\iff \text{for each } j \in [n], (a_j, b_j) \in \theta_j \text{ and } (b_j, c_j) \in \theta_j \\ &\implies \text{for each } j \in [n], (a_j, c_j) \in \theta_j \\ &\iff ((a_j)_{j \in [n]}, (c_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus  $\theta$  is transitive.

Hence  $\theta \in \text{Equiv}(X)$ . □

**Exercise 3.2.0.2.**

*Proof.* □

### 3.3 Quotients

**Exercise 3.3.0.1.** Let  $X, Y \in \text{Set}$ ,  $\theta_X \in \text{Equiv}(X)$ ,  $\theta_Y \in \text{Equiv}(Y)$  and  $f : X \rightarrow Y$ . If  $f$  is  $(\theta_X, \theta_Y)$ -invariant, then there exists a unique  $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$  such that  $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$ , i.e. the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi_{\theta_X} \downarrow & & \downarrow \pi_{\theta_Y} \\ X/\theta_X & \xrightarrow{\bar{f}} & Y/\theta_Y \end{array}$$

*Proof.* Suppose that  $f$  is  $(\theta_X, \theta_Y)$ -invariant.

• **Existence:**

Define  $\bar{f} \subset (X/\theta_X) \times (Y/\theta_Y)$  by  $\bar{f} := \{(\pi_{\theta_X}(a), \pi_{\theta_Y}(f(a))) : a \in X\}$ .

- Let  $c \in X/\theta_X$ . Then there exists  $a \in X$  such that  $c := \pi_{\theta_X}(a)$ . Define  $d \in Y/\theta_Y$  by  $d := \pi_{\theta_Y}(f(a))$ . Then

$$\begin{aligned} (c, d) &= (\pi_{\theta_X}(a), \pi_{\theta_Y}(f(a))) \\ &\in \bar{f}. \end{aligned}$$

Since  $c \in X/\theta_X$  is arbitrary, we have that for each  $c \in X/\theta_X$ , there exists  $d \in Y/\theta_Y$  such that  $(c, d) \in \bar{f}$ .

- Let  $(c_1, d_1), (c_2, d_2) \in \bar{f}$ . Then there exists  $a_1, a_2 \in X$  such that  $(c_1, d_1) = (\pi_{\theta_X}(a_1), \pi_{\theta_Y}(f(a_1)))$ ,  $(c_2, d_2) = (\pi_{\theta_X}(a_2), \pi_{\theta_Y}(f(a_2)))$ . Suppose that  $c_1 = c_2$ . Then

$$\begin{aligned}\pi_{\theta_X}(a_1) &= c_1 \\ &= c_2 \\ &= \pi_{\theta_X}(a_2).\end{aligned}$$

Hence  $a_1 \theta_X a_2$ . Since  $f$  is  $(\theta_X, \theta_Y)$ -invariant,  $f(a_1) \theta_Y f(a_2)$ . Thus

$$\begin{aligned}d_1 &= \pi_{\theta_Y}(f(a_1)) \\ &= \pi_{\theta_Y}(f(a_2)) \\ &= d_2.\end{aligned}$$

Since  $(c_1, d_1), (c_2, d_2) \in \bar{f}$  with  $c_1 = c_2$  are arbitrary, we have that for each  $(c_1, d_1), (c_2, d_2) \in \bar{f}$ ,  $c_1 = c_2$  implies that  $d_1 = d_2$ .

Thus  $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$ . By construction, for each  $a \in X$ ,

$$\begin{aligned}\pi_{\theta_Y} \circ f(a) &= \pi_{\theta_Y}(f(a)) \\ &= \bar{f}(\pi_{\theta_X}(a)) \\ &= \bar{f} \circ \pi_{\theta_X}(a).\end{aligned}$$

Thus  $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$ .

• **Uniqueness:**

Let  $g : X/\theta_X \rightarrow Y/\theta_Y$ . Suppose that  $\pi_{\theta_Y} \circ f = g \circ \pi_{\theta_X}$ . Let  $c \in X/\theta_X$ . Then there exists  $a \in X$  such that  $c = \pi_{\theta_X}(a)$ . Then

$$\begin{aligned}g(c) &= g(\pi_{\theta_X}(a)) \\ &= g \circ \pi_{\theta_X}(a) \\ &= \pi_{\theta_Y} \circ f(a) \\ &= \bar{f} \circ \pi_{\theta_X}(a) \\ &= \bar{f}(\pi_{\theta_X}(a)) \\ &= \bar{f}(c).\end{aligned}$$

Since  $c \in X/\theta_X$  is arbitrary, we have that for each  $c \in X/\theta_X$ ,  $g(c) = \bar{f}(c)$ . Hence  $g = \bar{f}$ .

□

**Definition 3.3.0.2.** Let  $X, Y \in \text{Set}$ ,  $\theta_X \in \text{Equiv}(X)$ ,  $\theta_Y \in \text{Equiv}(Y)$  and  $f : X \rightarrow Y$ . Suppose that  $f$  is  $(\theta_X, \theta_Y)$ -invariant. We define the **quotient of  $f$  by  $(\theta_X, \theta_Y)$** , denoted  $f/(\theta_X, \theta_Y) : X/\theta_X \rightarrow Y/\theta_Y$ , to be the unique  $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$  such that  $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$ .

**Exercise 3.3.0.3.** Let  $X, Y \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $f : X \rightarrow Y$ . If  $f$  is  $\theta$ -invariant, then there exists a unique  $\bar{f} : X/\theta \rightarrow Y$  such that  $f = \bar{f} \circ \pi_\theta$ , i.e. the following diagram commutes:

$$\begin{array}{ccc} X & & \\ \pi_\theta \downarrow & \searrow f & \\ X/\theta & \xrightarrow{\bar{f}} & Y \end{array}$$

*Proof.* Suppose that  $f$  is  $\theta$ -invariant.

- **Existence:**

Since  $f$  is  $\theta$ -invariant, we have that  $f$  is  $(\theta, \Delta_Y)$ -invariant. **make ref ex** implies that  $\pi_{\Delta_Y} \in \text{Iso}_{\text{Set}}(Y, Y/\Delta_Y)$ . Define  $\bar{f} : X/\theta \rightarrow Y$  by  $\bar{f} := \pi_{\Delta_Y}^{-1} \circ f/(\theta, \Delta_Y)$ . Then

$$\begin{aligned} \bar{f} \circ \pi_\theta &= (\pi_{\Delta_Y}^{-1} \circ f/(\theta, \Delta_Y)) \circ \pi_\theta \\ &= \pi_{\Delta_Y}^{-1} \circ (f/(\theta, \Delta_Y) \circ \pi_\theta) \\ &= \pi_{\Delta_Y}^{-1} \circ (\pi_{\Delta_Y} \circ f) \\ &= (\pi_{\Delta_Y}^{-1} \circ \pi_{\Delta_Y}) \circ f \\ &= \text{id}_Y \circ f \\ &= f. \end{aligned}$$

- **Uniqueness:**

Let  $g : X/\theta \rightarrow Y$ . Suppose that  $g \circ \pi_\theta = f$ . Let  $c \in X/\theta$ . Then there exists  $a \in X$  such that  $c = \pi_\theta(a)$ . Then

$$\begin{aligned} g(c) &= g(\pi_\theta(a)) \\ &= g \circ \pi_\theta(a) \\ &= f(a) \\ &= \bar{f} \circ \pi_\theta(a) \\ &= \bar{f}(\pi_\theta(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since  $c \in X/\theta_X$  is arbitrary, we have that for each  $c \in X/\theta_X$ ,  $g(c) = \bar{f}(c)$ . Hence  $g = \bar{f}$ . □

**Definition 3.3.0.4.** Let  $X, Y \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $f : X \rightarrow Y$ . Suppose that  $f$  is  $(\theta)$ -invariant. We define the **quotient of  $f$  by  $\theta$** , denoted  $f/\theta : X/\theta \rightarrow Y$ , to be the unique  $\bar{f} : X/\theta \rightarrow Y$  such that  $f = \bar{f} \circ \pi_\theta$ .

**Exercise 3.3.0.5.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$  and  $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ . Define  $X, X' \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $\pi' : X \rightarrow X'$  by  $X := \prod_{j \in [n]} X_j$ ,  $X' := \prod_{j \in [n]} (X_j/\theta_j)$ ,  $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$  and  $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$ . Then  $\pi'$  is  $\theta$ -invariant.

*Proof.* For each  $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$ ,

$$\begin{aligned} (a_j)_{j \in [n]} \sim_\theta (b_j)_{j \in [n]} &\iff \text{for each } j \in [n], a_j \sim_{\theta_j} b_j \\ &\iff \text{for each } j \in [n], \pi_{\theta_j}(a_j) = \pi_{\theta_j}(b_j) \\ &\iff (\pi_{\theta_j}(a_j))_{j \in [n]} = (\pi_{\theta_j}(b_j))_{j \in [n]} \\ &\iff \pi'((a_j)_{j \in [n]}) = \pi'((b_j)_{j \in [n]}). \end{aligned}$$

Hence  $\pi'$  is  $\theta^{\times n}$ -invariant. □

**Exercise 3.3.0.6.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$  and  $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ . Define  $X, X' \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $\pi' : X \rightarrow X'$  by  $X := \prod_{j \in [n]} X_j$ ,  $X' := \prod_{j \in [n]} (X_j/\theta_j)$ ,  $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$ ,  $\pi := \pi_\theta$  and  $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$ . Then  $(\pi')/\theta \in \text{Iso}_{\text{Set}}(X/\theta, X')$ .

*Proof.* Set  $\alpha := (\pi')/\theta$ . **A prev ex** implies that  $\pi' = \alpha \circ \pi_\theta$

- **Injectivity:**

Let  $c, d \in X/\theta$ . Then there exist  $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$  such that  $c = \pi_\theta((a_j)_{j \in [n]})$  and  $d = \pi_\theta((b_j)_{j \in [n]})$ . Suppose that  $\alpha(c) = \alpha(d)$ . Then

$$\begin{aligned} \pi'((a_j)_{j \in [n]}) &= \alpha(\pi_\theta((a_j)_{j \in [n]})) \\ &= \alpha(c) \\ &= \alpha(d) \\ &= \alpha(\pi_\theta((b_j)_{j \in [n]})) \\ &= \pi'((b_j)_{j \in [n]}). \end{aligned}$$



Thus for each  $j \in [n]$ ,  $\pi_{\theta_j}(a_j) = \pi_{\theta_j}(b_j)$ . Hence for each  $j \in [n]$ ,  $a_j \sim_{\theta_j} b_j$ . Therefore  $(a_j)_{j \in [n]} \sim_{\theta} (b_j)_{j \in [n]}$ . Thus

$$\begin{aligned} c &= \pi_{\theta}((a_j)_{j \in [n]}) \\ &= \pi_{\theta}((b_j)_{j \in [n]}) \\ &= d. \end{aligned}$$

Since  $c, d \in X/\theta$  with  $\alpha(c) = \alpha(d)$  are arbitrary, we have that for each  $c, d \in X/\theta$ ,  $\alpha(c) = \alpha(d)$  implies that  $c = d$ . Hence  $\alpha$  is injective.

• **Surjectivity:**

Since for each  $j \in [n]$ ,  $\pi_{\theta_j}$  is surjective, **ref ex** implies that  $\pi'$  is surjective. Since  $\pi' = \alpha \circ \pi_{\theta}$ , **ref ex** implies that  $\alpha$  is surjective.

**FINISH!!!** (ref exercises) □

**Exercise 3.3.0.7.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$ ,  $Y \in \text{Set}$ ,  $(\theta_{X_j})_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ ,  $\theta_Y \in \text{Equiv}(Y)$  and  $f : \prod_{j \in [n]} X_j \rightarrow Y$ . Define  $X, X' \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $\pi' : X \rightarrow X'$  by  $X := \prod_{j \in [n]} X_j$ ,  $X' := \prod_{j \in [n]} (X_j/\theta_{X_j})$ ,  $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_{X_j}$  and  $\pi' := \prod_{j \in [n]} \pi_{\theta_{X_j}}$ . If  $f$  is  $(\theta, \theta_Y)$ -invariant, then there exists a unique  $\bar{f} : X' \rightarrow Y/\theta_Y$  such that  $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi'$ , i.e. the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi' \downarrow & & \downarrow \pi_{\theta_Y} \\ X' & \xrightarrow{\bar{f}} & Y/\theta_Y \end{array}$$

*Proof.* Suppose that  $f$  is  $(\theta, \theta_Y)$ -invariant.

• **Existence:**

**A prev ex** implies that  $\pi'$  is  $\theta$ -invariant. **The prev ex** implies that  $(\pi')/\theta \in \text{Iso}_{\text{Set}}(X/\theta, X')$ . Define  $\bar{f} : X' \rightarrow Y/\theta_Y$  by  $\bar{f} := f/(\theta, \theta_Y) \circ [(\pi')/\theta]^{-1}$ . Then

$$\begin{aligned} \bar{f} \circ \pi' &= (f/(\theta, \theta_Y) \circ [(\pi')/\theta]^{-1}) \circ ((\pi')/\theta \circ \pi_{\theta}) \\ &= f/(\theta, \theta_Y) \circ ([(\pi')/\theta]^{-1} \circ (\pi')/\theta) \circ \pi_{\theta} \\ &= f/(\theta, \theta_Y) \circ \text{id}_{X/\theta} \circ \pi_{\theta} \\ &= f/(\theta, \theta_Y) \circ \pi_{\theta} \\ &= \pi_{\theta_Y} \circ f. \end{aligned}$$

• **Uniqueness:**

Let  $g : X' \rightarrow Y/\theta_Y$ . Suppose that  $\pi_{\theta_Y} \circ f = g \circ \pi'$ . Let  $c \in X'$ . Then there exists  $a \in X$  such that  $c = \pi'(a)$ . Then

$$\begin{aligned} g(c) &= g(\pi'(a)) \\ &= g \circ \pi'(a) \\ &= \pi_{\theta_Y} \circ f(a) \\ &= \bar{f} \circ \pi'(a) \\ &= \bar{f}(\pi'(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since  $c \in X'$  is arbitrary, we have that for each  $c \in X'$ ,  $g(c) = \bar{f}(c)$ . Hence  $g = \bar{f}$ . □

**Definition 3.3.0.8.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$ ,  $Y \in \text{Set}$ ,  $(\theta_{X_j})_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ ,  $\theta_Y \in \text{Equiv}(Y)$  and  $f : \prod_{j \in [n]} X_j \rightarrow Y$ . Suppose that  $f$  is  $\left(\prod_{j \in [n]}^{\text{Rel}} \theta_{X_j}, \theta_Y\right)$ -invariant. We define **multiquotient of  $f$  by  $((\theta_{X_j})_{j \in [n]}, \theta_Y)$** , denoted  $f / ((\theta_{X_j})_{j \in [n]}, \theta_Y) : \prod_{j \in [n]} X_j / \theta_{X_j} \rightarrow Y / \theta_Y$ , to be the unique  $\bar{f} : \prod_{j \in [n]} X_j / \theta_{X_j} \rightarrow Y / \theta_Y$  such that

$$\pi_{\theta_Y} \circ f = \bar{f} \circ \prod_{j \in [n]} \pi_{\theta_{X_j}}.$$

**Note 3.3.0.9.** If for each  $j \in [n]$ ,  $X_j = X$  and  $\theta_{X_j} = \theta_X$ , we may write  $f / ((n : \theta_X), \theta_Y)$  in place of  $f / ((\theta_{X_j})_{j \in [n]}, \theta_Y)$

**Exercise 3.3.0.10.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$ ,  $Y \in \text{Set}$ ,  $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$  and  $f : \prod_{j \in [n]} X_j \rightarrow Y$ . Define  $X, X' \in \text{Set}$ ,  $\theta \in \text{Equiv}(X)$  and  $\pi' : X \rightarrow X'$  by  $X := \prod_{j \in [n]} X_j$ ,  $X' := \prod_{j \in [n]} (X_j / \theta_j)$ ,  $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$  and  $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$ . If  $f$  is  $\theta$ -invariant, then there exists a unique  $\bar{f} : X' \rightarrow Y$  such that  $f = \bar{f} \circ \pi'$ , i.e. the following diagram commutes:

$$\begin{array}{ccc} X & & \\ \pi' \downarrow & \searrow f & \\ X' & \xrightarrow{\bar{f}} & Y \end{array}$$

*Proof.* Suppose that  $f$  is  $\theta$ -invariant.

• **Existence:**

Since  $f$  is  $\theta$ -invariant, we have that  $f$  is  $(\theta, \Delta_Y)$ -invariant. Define  $\bar{f}' : X' \rightarrow Y / \Delta_Y$  by  $\bar{f}' := f / ((\theta_j)_{j \in [n]}, \Delta_Y)$ . Then  $\pi_{\Delta_Y} \circ f = \bar{f}' \circ \pi'$ . **make ref ex** implies that  $\pi_{\Delta_Y} \in \text{Iso}_{\text{Set}}(Y, Y / \Delta_Y)$ . Define  $\bar{f} : X' \rightarrow Y$  by  $\bar{f} := \pi_{\Delta_Y}^{-1} \circ \bar{f}'$ . Then

$$\begin{aligned} \bar{f} \circ \pi' &= (\pi_{\Delta_Y}^{-1} \circ \bar{f}') \circ \pi' \\ &= \pi_{\Delta_Y}^{-1} \circ (\bar{f}' \circ \pi') \\ &= \pi_{\Delta_Y}^{-1} \circ (\pi_{\Delta_Y} \circ f) \\ &= (\pi_{\Delta_Y}^{-1} \circ \pi_{\Delta_Y}) \circ f \\ &= \text{id}_Y \circ f \\ &= f. \end{aligned}$$

• **Uniqueness:**

Let  $g : X' \rightarrow Y$ . Suppose that  $g \circ \pi' = f$ . Let  $c \in X'$ . Then there exists  $a \in X$  such that  $c = \pi'(a)$ . Then

$$\begin{aligned} g(c) &= g(\pi'(a)) \\ &= g \circ \pi'(a) \\ &= f(a) \\ &= \bar{f} \circ \pi'(a) \\ &= \bar{f}(\pi'(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since  $c \in X'$  is arbitrary, we have that for each  $c \in X'$ ,  $g(c) = \bar{f}(c)$ . Hence  $g = \bar{f}$ . □

**Definition 3.3.0.11.** Let  $n \in \mathbb{N}$ ,  $(X_j)_{j \in [n]} \subset \text{Set}$ ,  $Y \in \text{Set}$ ,  $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$  and  $f : \prod_{j \in [n]} X_j \rightarrow Y$ . Suppose that  $f$  is  $\prod_{j \in [n]}^{\text{Rel}} \theta_j$ -invariant. We define **multiquotient of  $f$  by  $(\theta_j)_{j \in [n]}$** , denoted  $f / (\theta_j)_{j \in [n]} : \prod_{j \in [n]} X_j / \theta_j \rightarrow Y$ , to be the unique

$\bar{f} : \prod_{j \in [n]} X_j / \theta_j \rightarrow Y$  such that

$$f = \bar{f} \circ \prod_{j \in [n]} \pi_{\theta_j}.$$

**Note 3.3.0.12.** If for each  $j \in [n]$ ,  $X_j = X$  and  $\theta_j = \theta$ , we may write  $f/(n : \theta)$  in place of  $f/(\theta_j)_{j \in [n]}$



# Part II

## Ordered Sets



## 3.4 To Do

- Relabel exercises
- finish generated element subsection in galois connection section

at this point, there should be a simple structure capturing posets and  $\downarrow S$ , measurable spaces and  $\sigma(\mathcal{A})$ , topological spaces and  $\tau(\mathcal{E})$ , groups/rings/ $\dots$ /algebras and  $\langle S \rangle / (E) / \dots / (F)$ . It seems like we need a category  $\mathcal{C}$  with some structure, another category  $\mathcal{C}_F$  which is like  $\mathcal{C}$ , but less structured and some forgetful-like functor  $F : \mathcal{C} \rightarrow \mathcal{C}_F$ . eg,  $F : \mathbf{Pro} \rightarrow \mathbf{Set}$  or  $F : \mathbf{Top} \rightarrow \mathbf{Set}$  and we need an associated poset  $\mathcal{P}$  containing the structure forgotten by  $F$ , e.g. the lower sets of  $X$  or the set of topologies on  $X$  and we need a “generating” or “contained” object  $\mathcal{E}$  in  $\mathcal{C}_F$  in the sense that there is at least one object  $A$  in  $\mathcal{C}$  and monomorphism  $\iota_A : \mathcal{E} \rightarrow F(A)$  in  $\mathcal{C}_F$ . We then need a minimal element in the structure poset  $\mathcal{P}$  in some universal sense relating to these monomorphisms. Ask people who know category theory





# Chapter 4

## Prosets

### 4.1 Introduction

**Definition 4.1.0.1. Proset:**

Let  $X$  be a set and  $\leq \subset X \times X$  a binary relation on  $X$ . Then

- $\leq$  is said to be a **preorder on  $X$**  if
  1. for each  $a \in X$ ,  $a \leq a$
  2. for each  $a, b, c \in X$ ,  $a \leq b$  and  $b \leq c$  implies that  $a \leq c$
- $(X, \leq)$  is said to be a **preordered set** or **proset** if  $\leq$  is a preorder on  $X$ .

**Definition 4.1.0.2.** maybe just ref def about converse relation in set theory appendices section on heterogeneous binary relations

Let  $(X, \leq)$  be a proset. We define the **dual order of  $\leq$  on  $X$** , denoted  $\leq^*$ , by  $a \leq^* b$  iff  $b \leq a$ .

**Exercise 4.1.0.3.** Let  $(X, \leq)$  be a proset. Then  $\leq^*$  is a preorder on  $X$ .

*Proof.*

1. Let  $a \in X$ . Since  $a \leq a$ , we have that  $a \leq^* a$ .
2. Let  $a, b, c \in X$ . Suppose that  $a \leq^* b$  and  $b \leq^* c$ . Then  $b \leq a$  and  $c \leq b$ . Hence  $c \leq a$ . Thus  $a \leq^* c$ .

Therefore  $\leq^*$  is a preorder on  $X$ . □

**Definition 4.1.0.4.** Let  $(X, \leq)$  be a proset. We define  $\sim_\leq \subset X \times X$  by  $a \sim_\leq b$  iff  $a \leq b$  and  $b \leq a$ .

**Exercise 4.1.0.5.** Let  $(X, \leq)$  be a proset. Then  $\sim_\leq$  is an equivalence relation on  $X$ .

*Proof.* Let  $x, y, z \in X$ .

1. Since  $x \leq x$ ,  $x \sim_\leq x$ .
2. Suppose that  $x \sim_\leq y$ . Then  $x \leq y$  and  $y \leq x$ . Thus  $y \sim_\leq x$ .
3. Suppose that  $x \sim_\leq y$  and  $y \sim_\leq z$ . Then  $x \leq y$ ,  $y \leq x$ ,  $y \leq z$  and  $z \leq y$ . Therefore  $x \leq z$  and  $z \leq x$ . Hence  $x \sim_\leq z$ . □

**Exercise 4.1.0.6.** Let  $(X, \leq)$  be a proset. Then for each  $x, y \in X$ ,  $x \sim_\leq y$  iff  $x \sim_{\leq^*} y$ .

*Proof.* We have that for each  $x, y \in X$ ,

$$\begin{aligned} x \sim_\leq y &\iff x \leq y \text{ and } y \leq x \\ &\iff y \leq^* x \text{ and } x \leq^* y \\ &\iff x \sim_{\leq^*} y. \end{aligned}$$
□

**Exercise 4.1.0.7.** Let  $X$  be a set. Then  $(\mathcal{P}(X), \subset)$  is a proset.

*Proof.*

1. For each  $A \in \mathcal{P}(X)$ ,  $A \subset A$ .
2. For each  $A, B, C \in \mathcal{P}(X)$ , if  $A \subset B$  and  $B \subset C$ , then  $A \subset C$ .

Thus  $\subset$  is a preorder on  $\mathcal{P}(X)$ . Hence  $(\mathcal{P}(X), \subset)$  is a proset. □

**Definition 4.1.0.8.** We define the

- **one point proset**, denoted  $(\mathbf{1}, \leq_1)$ , by
  - $\mathbf{1} := \{0\}$
  - $\leq_1 := \{(0, 0)\}$ .
- **two point proset**, denoted  $(\mathbf{2}, \leq_2)$ , by
  - $\mathbf{2} := \{0, 1\}$
  - $\leq_2 := \{(0, 0), (0, 1), (1, 1)\}$ .

**Note 4.1.0.9.** When the context is clear, we may write  $\leq$  in place of  $\leq_1$  and  $\leq_2$ .

**Exercise 4.1.0.10.** We have that

1.  $(\mathbf{1}, \leq_1)$  is a proset
2.  $(\mathbf{2}, \leq_2)$  is a proset

*Proof.* Clear, FINISH!!!

- 1.
- 2.

□

Talk about presets/posests where the order is induced by a map and another preset/poset, i.e. Let  $(Y, \leq)$  be a preset/poset and  $f : X \rightarrow Y$ . Define  $f^* \leq$  on  $X$  by  $x_1 f^* \leq x_2$  iff  $f(x_1) \leq f(x_2)$ . Similarly, If  $(X, \leq)$  is a preset/poset and  $f : X \rightarrow Y$ . Define  $f_* \leq$  on  $Y$  by  $y_1 f_* \leq y_2$  iff there exists  $x_1 \in f^{-1}(y_1)$  and  $x_2 \in f^{-1}(y_2)$  such that  $x_1 \leq x_2$ . When are these well defined? I think the posetal reflection is an example of the second case. An example of the first would be from a probability, define order on measurable sets  $A \leq B$  iff  $P(A) \leq P(B)$  so that one is larger than another if it is more likely.

## 4.2 Subprosets

**Definition 4.2.0.1.** ref defn in set theory appendices on restrictions of  $n$ -ary homogeneous relations

**Note 4.2.0.2.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . When the context is clear, we write  $(A, \leq)$  in place of  $(A, \leq|_A)$

## 4.3 Products

**Definition 4.3.0.1.** Let  $(A, \leq_A)$  and  $(B, \leq_B)$  be prosets. We define the

- **product preorder of  $\leq_A$  and  $\leq_B$  on  $A \times B$** , denoted  $\leq_A \otimes \leq_B$  by  $(a_1, b_1) \leq_A \otimes \leq_B (a_2, b_2)$  iff  $a_1 \leq_A a_2$  and  $b_1 \leq_B b_2$ .
- **product proset of  $(A, \leq_A)$  and  $(B, \leq_B)$** , denoted  $(A, \leq_A) \otimes (B, \leq_B)$  by  $(A, \leq_A) \otimes (B, \leq_B) := (A \times B, \leq_A \otimes \leq_B)$

**Exercise 4.3.0.2.** probably need to change notation since  $\otimes$  might be reserved for something else. Let  $(A, \leq_A)$  and  $(B, \leq_B)$  be prosets. Then

1.  $\leq_A \otimes \leq_B$  is a preorder on  $A \times B$ ,
2.  $(A, \leq_A) \otimes (B, \leq_B)$  is a proset.

*Proof.*

1. (a) Let  $(a, b) \in A \times B$ . Then  $a \leq_A a$  and  $b \leq_B b$ . Therefore  $(a, b) \leq_A \otimes \leq_B (a, b)$ .
- (b) Let  $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$ . Suppose that  $(a_1, b_1) \leq_A \otimes \leq_B (a_2, b_2)$  and  $(a_2, b_2) \leq_A \otimes \leq_B (a_3, b_3)$ . Then  $a_1 \leq_A a_2$ ,  $a_2 \leq_A a_3$ ,  $b_1 \leq_B b_2$  and  $b_2 \leq_B b_3$ . Therefore  $a_1 \leq_A a_3$  and  $b_1 \leq_B b_3$ . Hence  $(a_1, b_1) \leq_A \otimes \leq_B (a_3, b_3)$ .

Hence  $\leq_A \otimes \leq_B$  is a preorder on  $A \times B$ .

2. Since  $\leq_A \otimes \leq_B$  is a preorder on  $A \times B$ ,  $(X, \leq)$  is a proset.

□

## 4.4 Quotients



# Chapter 5

## Posets

### 5.1 Introduction

#### Definition 5.1.0.1. Poset:

Let  $(X, \leq)$  be a poset. Then

- $\leq$  is said to be a **partial order on  $X$**  if for each  $a, b \in X$ ,  $a \leq b$  and  $b \leq a$  implies that  $a = b$
- $(X, \leq)$  is said to be a **partially ordered set** or **poset** if  $\leq$  is a partial ordering on  $X$ .

**Exercise 5.1.0.2.** Let  $(X, \leq)$  be a poset and  $a, b \in X$ . Then  $a \sim_{\leq} b$  iff  $a = b$ .

*Proof.*

- $(\implies)$ :  
Suppose that  $a \sim_{\leq} b$ . Then  $a \leq b$  and  $b \leq a$ . Since  $\leq$  is a partial order,  $a = b$ .
- $(\impliedby)$ :  
Since  $\sim_{\leq}$  is reflexive,  $a = b$  implies that  $a \sim_{\leq} b$ .

□

**Definition 5.1.0.3.** Let  $(X, \leq)$  be a poset. We define the

- **posetal reflection partial order on  $X / \sim_{\leq}$** , denoted  $\leq_{X / \sim_{\leq}} \subset X / \sim_{\leq} \times X / \sim_{\leq}$ , by  $y_1 \leq_{X / \sim_{\leq}} y_2$  iff there exists  $x_1 \in y_1$  and  $x_2 \in y_2$  such that  $x_1 \leq x_2$ .
- **the posetal reflection of  $(X, \leq)$** , denoted  $\text{PosRefl}(X, \leq)$ , by  $\text{PosRefl}(X, \leq) := (X / \sim_{\leq}, \leq_{X / \sim_{\leq}})$

**Exercise 5.1.0.4.** Let  $(X, \leq)$  be a poset. Then  $\text{PosRefl}(X, \leq)$  is a poset.

*Proof.* Set  $Y := X / \sim_{\leq}$  and  $\leq_Y := \leq_{X / \sim_{\leq}}$ .

- (a) Let  $y \in Y$ . Then there exists  $x \in y$ . Since  $x \leq x$ , we have that  $y \leq_Y y$ .  
(b) Let  $y_1, y_2, y_3 \in Y$ . Suppose that  $y_1 \leq_Y y_2$  and  $y_2 \leq_Y y_3$ . Then there exist  $x_1 \in y_1$ ,  $x_2 \in y_2$  and  $x_3 \in y_3$  such that  $x_1 \leq x_2$  and  $x_2 \leq x_3$ . Therefore  $x_1 \leq x_3$ . Hence  $y_1 \leq_Y y_3$ .

Thus  $\leq_Y$  is a preorder on  $Y$ .

- Let  $y_1, y_2 \in Y$ . Suppose that  $y_1 \leq_Y y_2$  and  $y_2 \leq_Y y_1$ . Then there exist  $a_1, b_1 \in y_1$ ,  $a_2, b_2 \in y_2$  such that  $a_1 \leq a_2$  and  $b_2 \leq b_1$ . Since  $a_1, b_1 \in y_1$  and  $a_2, b_2 \in y_2$ ,  $a_1 \sim_{\leq} b_1$  and  $a_2 \sim_{\leq} b_2$ . Thus

$$\begin{aligned} a_1 &\leq a_2 \\ &\leq b_2. \end{aligned}$$

and

$$\begin{aligned} b_2 &\leq b_1 \\ &\leq a_1. \end{aligned}$$

Hence  $a_1 \sim_{\leq} b_2$  and

$$\begin{aligned} y_1 &= \pi(a_1) \\ &= \pi(b_2) \\ &= y_2. \end{aligned}$$

Therefore  $\leq_Y$  is a partial order on  $Y$ . □

**Exercise 5.1.0.5.** Let  $X$  be a set. Then  $(\mathcal{P}(X), \subset)$  is a poset.

*Proof.* Exercise 4.1.0.7 implies that  $(\mathcal{P}(X), \subset)$  is a poset. We note that for each  $A, B \in \mathcal{P}(X)$ ,  $A \subset B$  and  $B \subset A$  implies that  $A = B$  ref def of equality in set theory notes?. □

**Exercise 5.1.0.6.** We have that

1.  $(\mathbf{1}, \leq_1)$  is a poset
2.  $(\mathbf{2}, \leq_2)$  is a poset

*Proof.* Clear, FINISH!!! reference Exercise 4.1.0.10.

- 1.
- 2.

□

# Chapter 6

## Directed Sets

### 6.1 Introduction

**Definition 6.1.0.1. Directed Set:**

Let  $(X, \leq)$  be a proset. Then

- $\leq$  is said to be
  - an **upward direction** on  $X$  if for each  $a, b \in X$ , there exists  $c \in X$  such that  $a \leq c$  and  $b \leq c$ .
  - a **downward direction** on  $X$  if for each  $a, b \in X$ , there exists  $c \in X$  such that  $c \leq a$  and  $c \leq b$ .
- $(X, \leq)$  is said to be
  - an **upward directed set** if  $X \neq \emptyset$  and  $\leq$  is an upward direction on  $X$ .
  - a **downward directed set** if  $X \neq \emptyset$  and  $\leq$  is a downward direction on  $X$ .

**Note 6.1.0.2.** When the context is clear, we may say “direction” and “directed” instead of “upward direction” and “upward directed”.

**Exercise 6.1.0.3.** Let  $(X, \leq)$  be a proset. Then

1.  $\leq$  is an upward direction on  $X$  iff  $\leq^*$  is a downward direction on  $X$ ,
2.  $\leq^*$  is an upward direction on  $X$  iff  $\leq$  is a downward direction on  $X$ ,

*Proof.*

1.

$$\begin{aligned} \leq \text{ is an upward direction on } X &\iff \text{for each } a, b \in X, \text{ there exists } c \in X \text{ such that } a \leq c \text{ and } b \leq c \\ &\iff \text{for each } a, b \in X, \text{ there exists } c \in X \text{ such that } c \leq^* a \text{ and } c \leq^* b \\ &\iff \leq^* \text{ is an downward direction on } X \end{aligned}$$

2. We have that

$$\begin{aligned} \leq^* \text{ is an upward direction on } X &\iff (\leq^*)^* \text{ is a downward direction on } X \quad (\text{by prev part}) \\ &\iff \leq \text{ is a downward direction on } X \quad (\text{by ref ex in section on binary relations}). \end{aligned}$$

□

## 6.2 Subdirected Sets

**Definition 6.2.0.1.** Let  $(A, \leq)$  be a proset and  $B \subset A$ . Then  $B$  is said to be

- **$\leq$ -upward directed** if  $(B, \leq|_B)$  is an upward directed set,
- **$\leq$ -downward directed** if  $(B, \leq^*|_B)$  is a downward directed set.

**Note 6.2.0.2.** When the context is clear, we may say “upward directed” and “downward directed” instead of “ $\leq$ -upward directed” and “ $\leq$ -downward directed” respectively.

**Exercise 6.2.0.3.** Let  $(A, \leq)$  be a proset and  $B \subset A$ . Then

1.  $B$  is  $\leq$ -downward directed iff  $B$  is  $\leq^*$ -upward directed.
2.  $B$  is  $\leq$ -upward directed iff  $B$  is  $\leq^*$ -downward directed.

*Proof.*

1. We note that

$$\begin{aligned}
 B \text{ is } \leq\text{-downward directed} &\iff (B, \leq^*|_B) \text{ is an upward directed set} \quad (\text{by ref prev ex}) \\
 &\iff (B, (\leq|_B)^*) \text{ is an upward directed set} \quad (\text{by ref prev ex about restriction of binary/converse relation}) \\
 &\iff B \text{ is } \leq^*\text{-upward directed} \quad (\text{by Exercise 6.1.0.3})
 \end{aligned}$$

2. We note that

$$\begin{aligned}
 B \text{ is } \leq\text{-upward directed} &\iff B \text{ is } (\leq^*)^*\text{-upward directed} \quad \text{ref ex in set theory section on binary binary/converse relation} \\
 &\iff B \text{ is } \leq^*\text{-downward directed} \quad \text{ref previous part.}
 \end{aligned}$$

□

**Exercise 6.2.0.4.** Let  $(X, \leq)$  be a proset and  $x \in X$ . Then

1.  $\{x\}$  is upward directed
2.  $\{x\}$  is downward directed

*Proof.*

1.
  - Clearly  $\{x\} \neq \emptyset$ .
  - For each  $a, b \in \{x\}$ , we have that

$$\begin{aligned}
 a, b &\leq x \\
 &\in \{x\}.
 \end{aligned}$$

Thus for each  $a, b \in \{x\}$ , there exists  $c \in \{x\}$  such that  $a \leq c$  and  $b \leq c$ . Hence  $\{x\}$  is upward directed.

- 2.

□



## 6.3 Products of Directed Sets

**Exercise 6.3.0.1.** Let  $(A, \leq_A)$  and  $(B, \leq_B)$  be posets. If  $(A, \leq_A)$  and  $(B, \leq_B)$  are upward directed sets

1.  $\leq_A \otimes \leq_B$  is an upward direction on  $A \times B$ ,
2.  $(A \times B, \leq_A \otimes \leq_B)$  is an upward directed set.

*Proof.*

1. (a) Exercise ?? implies that  $\leq_A \otimes \leq_B$  is a preorder of  $A \times B$ .  
 (b) Let  $(a_1, b_1), (a_2, b_2) \in A \times B$ . Then there exist  $a \in A$  and  $b \in B$  such that  $a_1, a_2 \leq_A a$  and  $b_1, b_2 \leq_B b$ . Hence  $(a_1, b_1), (a_2, b_2) \leq_A \otimes \leq_B (a, b)$ .

Hence  $\leq_A \otimes \leq_B$  is an upward direction on  $A \times B$ .

2. (a) Since  $A \neq \emptyset$  and  $B \neq \emptyset$ , we have that  $A \times B \neq \emptyset$ .  
 (b) From above,  $\leq_A \otimes \leq_B$  is an upward direction on  $A \times B$ .

Hence  $(A \times B, \leq_A \otimes \leq_B)$  is a directed set.

□



# Chapter 7

## Monotone Maps

### 7.1 Introduction

**Definition 7.1.0.1.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be posets.

- $f : X \rightarrow Y$ . Then  $f$  is said to be
  - $(\leq_X, \leq_Y)$ -**order preserving** or  $(\leq_X, \leq_Y)$ -**monotone** if for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(a) \leq_Y f(b)$ .
  - $(\leq_X, \leq_Y)$ -**order reversing** or  $(\leq_X, \leq_Y)$ -**antitone** if for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(b) \leq_Y f(a)$ .
- We define
  - $\text{Mtn}((X, \leq_X), (Y, \leq_Y)) := \{f \in Y^X : f \text{ is } (\leq_X, \leq_Y)\text{-monotone}\},$
  - $\text{Antn}((X, \leq_X), (Y, \leq_Y)) := \{f \in Y^X : f \text{ is } (\leq_X, \leq_Y)\text{-antitone}\}.$

**Note 7.1.0.2.** When the context is clear

- we may say that  $f$  is “monotone” and “antitone” instead of “ $(\leq_X, \leq_Y)$ -monotone” and “ $(\leq_X, \leq_Y)$ -antitone” respectively.
- we may write  $\text{Mtn}(X, Y)$  and  $\text{Antn}(X, Y)$  in place of  $\text{Mtn}((X, \leq_X), (Y, \leq_Y))$  and  $\text{Antn}((X, \leq_X), (Y, \leq_Y))$  respectively.

need to go rewrite using notation  $\text{Mtn}$  and  $\text{Antn}$

**Exercise 7.1.0.3.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ . Then

1. the following are equivalent:
  - (a)  $f$  is  $(\leq_X, \leq_Y)$ -antitone
  - (b)  $f$  is  $(\leq_X, \leq_Y^*)$ -monotone
  - (c)  $f$  is  $(\leq_X^*, \leq_Y)$ -monotone
  - (d)  $f$  is  $(\leq_X^*, \leq_Y^*)$ -antitone
2. the following are equivalent:
  - (a)  $f$  is  $(\leq_X, \leq_Y)$ -monotone
  - (b)  $f$  is  $(\leq_X, \leq_Y^*)$ -antitone
  - (c)  $f$  is  $(\leq_X^*, \leq_Y)$ -antitone
  - (d)  $f$  is  $(\leq_X^*, \leq_Y^*)$ -monotone

*Proof.*

1. • (a)  $\implies$  (b):  
Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -antitone. Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq_X b &\implies f(b) \leq_Y f(a) \\ &\iff f(a) \leq_Y^* f(b). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(a) \leq_Y^* f(b)$ . Hence  $f$  is  $(\leq_X, \leq_Y^*)$ -monotone.

- (b)  $\implies$  (c):  
Suppose that  $f$  is  $(\leq_X, \leq_Y^*)$ -monotone. Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq_X^* b &\iff b \leq_X^* a \\ &\implies f(b) \leq_Y^* f(a) \\ &\iff f(a) \leq_Y f(b). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X^* b$  implies that  $f(a) \leq_Y f(b)$ . Hence  $f$  is  $(\leq_X^*, \leq_Y)$ -monotone.

- (c)  $\implies$  (d):  
Suppose that  $f$  is  $(\leq_X^*, \leq_Y)$ -monotone. Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq_X^* b &\implies f(a) \leq_Y f(b) \\ &\iff f(b) \leq_Y^* f(a). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X^* b$  implies that  $f(b) \leq_Y^* f(a)$ . Hence  $f$  is  $(\leq_X^*, \leq_Y^*)$ -antitone.

- (d)  $\implies$  (a):  
Suppose that  $f$  is  $(\leq_X^*, \leq_Y^*)$ -antitone. Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq_X b &\iff b \leq_X^* a \\ &\implies f(a) \leq_Y^* f(b) \\ &\iff f(b) \leq_Y f(a). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(b) \leq_Y f(a)$ . Hence  $f$  is  $(\leq_X, \leq_Y)$ -antitone.

2. Part (1) implies that

$$\begin{aligned} f \text{ is } (\leq_X, (\leq_Y^*)^*)\text{-monotone} &\iff f \text{ is } (\leq_X, \leq_Y^*)\text{-antitone} \quad (\text{by } 1(b) \iff 1(a)) \\ &\iff f \text{ is } (\leq_X^*, (\leq_Y^*)^*)\text{-antitone} \quad (\text{by } 1(a) \iff 1(d)) \\ &\iff f \text{ is } (\leq_X^*, \leq_Y^*)\text{-monotone} \quad (\text{by } 1(d) \iff 1(c)). \end{aligned}$$

Since  $(\leq_Y^*)^* = \leq_Y$ , we have that

$$\begin{aligned} f \text{ is } (\leq_X, \leq_Y)\text{-monotone} &\iff f \text{ is } (\leq_X, \leq_Y^*)\text{-antitone} \\ &\iff f \text{ is } (\leq_X^*, \leq_Y)\text{-antitone} \\ &\iff f \text{ is } (\leq_X^*, \leq_Y^*)\text{-monotone}. \end{aligned}$$

□

**Exercise 7.1.0.4.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be prosets and  $f : X \rightarrow Y$ . Then  $f$  is  $(\leq_X, \leq_Y)$ -monotone iff  $f$  is  $(\leq_X^*, \leq_Y^*)$ -monotone.

*Proof.*

- ( $\Rightarrow$ ):

Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -monotone. Let  $a, b \in X$ . Suppose that  $a \leq_X^* b$ . Then  $b \leq_X a$ . Hence  $f(b) \leq_Y f(a)$ . Thus  $f(a) \leq_Y^* f(b)$ . Since  $a, b \in X$  with  $a \leq_X^* b$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X^* b$  implies that  $f(a) \leq_Y^* f(b)$ . Therefore  $f$  is  $(\leq_X^*, \leq_Y^*)$ -monotone.

- ( $\Leftarrow$ ):

Suppose that  $f$  is  $(\leq_X^*, \leq_Y^*)$ -monotone. Since  $(\leq_X^*)^* = \leq_X$ , the previous part implies that  $f$  is  $(\leq_X, \leq_Y)$ -monotone.

□

**Definition 7.1.0.5.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ . Then  $f$  is said to be a  $(\leq_X, \leq_Y)$ -isomorphism if

1.  $f$  is a bijection,
2.  $f$  is  $(\leq_X, \leq_Y)$ -monotone and  $f^{-1}$  is  $(\leq_Y, \leq_X)$ -monotone.

**Definition 7.1.0.6.** Let  $X$  be a set and  $(Y, \leq)$  a poset. We define the **preorder on  $Y^X$  induced by  $\leq$** , denoted  $\leq_{(Y, \leq)^X}$ , by

$$f \leq_{(Y, \leq)^X} g \text{ iff for each } x \in X, f(x) \leq g(x)$$

**Note 7.1.0.7.** When the context is clear, we may write  $\leq$  in place of  $\leq_{(Y, \leq)^X}$ .

**Exercise 7.1.0.8.** Let  $X$  be a set and  $(Y, \leq)$  a poset. Then

1.  $(Y^X, \leq_{(Y, \leq)^X})$  is a poset.
2.  $(Y, \leq)$  a poset, implies that  $(Y^X, \leq_{(Y, \leq)^X})$  is a poset.

*Proof.* Set  $\leq' := \leq_{(Y, \leq)^X}$ . Since  $(Y, \leq)$  is a poset,  $\leq$  is reflexive and transitive.

1.
  - Let  $f \in Y^X$  and  $x \in X$ . Since  $\leq$  is reflexive, we have that  $f(x) \leq f(x)$ . Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $f(x) \leq f(x)$ . Hence  $f \leq' f$ . Since  $f \in Y^X$  is arbitrary, we have that for each  $f \in Y^X$ ,  $f \leq' f$ . Thus  $\leq'$  is reflexive.
  - Let  $f, g, h \in Y^X$  and  $x \in X$ . Suppose that  $f \leq' g$  and  $g \leq' h$ . Then  $f(x) \leq g(x)$  and  $g(x) \leq h(x)$ . Since  $\leq$  is transitive,  $f(x) \leq h(x)$ . Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $f(x) \leq h(x)$ . Thus  $f \leq' h$ . Since  $f, g, h \in Y^X$  with  $f \leq' g$  and  $g \leq' h$  are arbitrary, we have that for each  $f, g, h \in Y^X$ ,  $f \leq' g$  and  $g \leq' h$  implies that  $f \leq' h$ . Thus  $\leq'$  is transitive.

Hence  $\leq'$  is a preorder on  $Y^X$  and  $(Y^X, \leq')$  is a poset.

2. Suppose that  $(Y, \leq)$  is a poset. Then  $\leq$  is antisymmetric. Let  $f, g \in Y^X$  and  $x \in X$ . Suppose that  $f \leq' g$  and  $g \leq' f$ . Then  $f(x) \leq g(x)$  and  $g(x) \leq f(x)$ . Since  $\leq$  is antisymmetric,  $f(x) = g(x)$ . Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $f(x) = g(x)$ . Hence  $f = g$ . Since  $f, g \in Y^X$  with  $f \leq' g$  and  $g \leq' f$  are arbitrary, we have that for each  $f, g \in Y^X$ ,  $f \leq' g$  and  $g \leq' f$  implies that  $f = g$ . Thus  $\leq'$  is antisymmetric. Thus  $\leq'$  is a partial order on  $Y^X$  and  $(Y^X, \leq')$  is a poset.

□

## 7.2 Galois Connections

ToDo: If this comes after groups rings, etc, give examples of defining the 'algebraic object generated by less structured objects' in terms of a galois connection. e.g. the subgroup/subring/upperset/sublattice/ideal/filter/sigma-alg/topology generated, subsets of set/powerset

**Definition 7.2.0.1.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Then

- $(f, g)$  is said to be a  $(\leq_X, \leq_Y)$ -**monotone Galois connection** if for each  $x \in X$  and  $y \in Y$ ,

$$f(x) \leq_Y y \text{ iff } x \leq_X g(y),$$

- $(f, g)$  is said to be a  $(\leq_X, \leq_Y)$ -**antitone Galois connection** if for each  $x \in X$  and  $y \in Y$ ,

$$f(x) \leq_Y y \text{ iff } g(y) \leq_X x,$$

- $f$  is said to be a  $(\leq_X, \leq_Y)$ -**left adjoint** of  $g$  if  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection,
- $g$  is said to be a  $(\leq_X, \leq_Y)$ -**right adjoint** of  $f$  if  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.
- $f$  is said to **have a**  $(\leq_X, \leq_Y)$ -**right adjoint** if there exists  $f^* : Y \rightarrow X$  such that  $f^*$  is a  $(\leq_X, \leq_Y)$ -right adjoint of  $f$ .
- $g$  is said to **have a**  $(\leq_X, \leq_Y)$ -**left adjoint** if there exists  $g_* : X \rightarrow Y$  such that  $g_*$  is a  $(\leq_X, \leq_Y)$ -left adjoint of  $g$ .

**Note 7.2.0.2.** When the context is clear, we say

- “monotone (resp. antitone) Galois connection” instead of “ $(\leq_X, \leq_Y)$ -monotone (resp.  $(\leq_X, \leq_Y)$ -antitone) Galois connection”
- “left (resp. right) adjoint” instead of “ $(\leq_X, \leq_Y)$ -left (resp.  $(\leq_X, \leq_Y)$ -right) adjoint”

Question:

There is an interpretation of posets, monotonic maps and galois connections in terms of information.

- Posets are spaces that encode informational direction:  $x_1 \leq_X x_2$  iff  $x_1$  contains less information than  $x_2$ .
- Monotonic maps are those that preserve informational direction: if  $x_1$  has less information than  $x_2$ , then  $f(x_1)$  has less information than  $f(x_2)$ .
- Galois connections encode informational direction in different contexts as well as minimal and maximal changes in information:  $f(x)$  has less information than  $y$  in the context of  $Y$ . More specifically,

- \*  $y$  contains at least as much information as  $x$  in the context of  $Y$  iff  $f(x) \leq_Y y$ .
- \*  $f(x)$  is the element  $y$  of  $Y$  such that

1.  $y$  contains at least as much information as  $x$  in the context of  $Y$ , i.e.  $f(x) \leq_Y y$ .
2. for each  $y' \in Y$ , if  $y$  contains at least as much information as  $x$  in the context of  $Y$ , then  $y'$  contains at least as much information as  $y$  in the context of  $Y$ , i.e.  $f(x) \leq_Y y' \implies y \leq_X y'$ .

So that  $f(x)$  is the object in  $Y$  such that the increase in information (information gain) from  $x$  to  $f(x)$  in the context of  $Y$  is minimal. Equivalently, the change in information  $\Delta_Y I(x, f(x))$  is minimal

- \*  $x$  has less information than  $y$  in the context of  $X$  iff  $x \leq_X g(y)$ .
- \*  $g(y)$  is the element  $x$  of  $X$  such that

1.  $y$  contains at least as much information as  $x$  in the context of  $X$ , i.e.  $x \leq_X g(y)$ .
2. for each  $x' \in X$ , if  $y$  contains at least as much information as  $x'$  in the context of  $X$ , then  $x$  contains at least as much information as  $x'$  in the context of  $X$ , i.e.  $x' \leq_X g(y) \implies x' \leq_X x$ .

So that  $g(y)$  is the object in  $X$  such that the decrease in information (information loss) from  $y$  to  $g(y)$  in the context of  $X$  is minimal. Equivalently, the change in information  $\Delta_X I(y, g(y))$  is maximal ( $-1$  is larger than  $-2$ )

This can be generalized to categories, functors and adjoint functors where

- $x_1 \leq_X x_2$  generalizes to  $\text{Hom}_X(x_2, x_1) \neq \emptyset$ , i.e.  $x_2$  has at least as much information as  $x_1$  generalizes to  $x_1$  is a processed version of  $x_1$ ,
- a monotone map  $f : X \rightarrow Y$  generalizes to a functor  $F : X \rightarrow Y$
- a Galois connection  $(f, g)$  generalizes to an adjunction  $(F, G)$ .

This categorical generalization also generalizes information theory and statistics. However, from most math sources, the typical language of adjunctions is in terms of efficiency instead of information change. For example  $f(x)$  is the most 'efficient' way to obtain an object in  $Y$  from  $x$  which 'preserves  $x$ '. Similarly,  $g(y)$  is the least wasteful way to obtain an object in  $X$  from  $y$  which ?????????? Could we rephrase this to describe dynamical systems, specifically, can we rephrase information or efficiency in terms of energy to capture describe physics? Something like the path in phase space from  $t_1$  to  $t_2$  is a right adjoint functor from the universe (a category) at a time  $t_1$  to the universe at time  $t_2$  so that the information loss is minimal. Maybe show that least action, or newtons equation is a manifestation of this.

**Exercise 7.2.0.3.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Then

1.  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection iff  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection
2.  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection iff  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection
3.  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection iff  $(g, f)$  is a  $(\leq_Y, \leq_X^*)$ -antitone Galois connection

*Proof.*

1. •  $(\implies)$ :  
Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection. Let  $y \in Y$  and  $x \in X$ . Then by assumption,

$$\begin{aligned} g(y) \leq_X^* x &\iff x \leq_X g(y) \\ &\iff f(x) \leq_Y y \\ &\iff y \leq_Y^* f(x). \end{aligned}$$

Since  $y \in Y$  and  $x \in X$  are arbitrary, we have that for each  $y \in Y$  and  $x \in X$ ,  $g(y) \leq_X^* x$  iff  $y \leq_Y^* f(x)$ . Therefore  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection.

- $(\impliedby)$ :  
Suppose that  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection. Let  $x \in X$  and  $y \in Y$ . Then by assumption,

$$\begin{aligned} f(x) \leq_Y y &\iff y \leq_Y^* f(x) \\ &\iff g(y) \leq_X^* x \\ &\iff x \leq_X g(y). \end{aligned}$$

Since  $x \in X$  and  $y \in Y$  are arbitrary, we have that for each  $x \in X$  and  $y \in Y$ ,  $f(x) \leq_Y y$  iff  $x \leq_X g(y)$ . Therefore  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

2. •  $(\implies)$ :  
Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection. Let  $x \in X$  and  $y \in Y$ . Then by assumption,

$$\begin{aligned} f(x) \leq_Y y &\iff g(y) \leq_X x \\ &\iff x \leq_X^* g(y). \end{aligned}$$

Since  $x \in X$  and  $y \in Y$  are arbitrary, we have that for each  $x \in X$  and  $y \in Y$ ,  $f(x) \leq_Y y$  iff  $x \leq_X^* g(y)$ . Therefore  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection.

- $(\impliedby)$ :  
Suppose that  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection. Let  $x \in X$  and  $y \in Y$ . Then by assumption,

$$\begin{aligned} f(x) \leq_Y y &\iff x \leq_X^* g(y) \\ &\iff g(y) \leq_X x. \end{aligned}$$

Since  $x \in X$  and  $y \in Y$  are arbitrary, we have that for each  $x \in X$  and  $y \in Y$ ,  $f(x) \leq_Y y$  iff  $g(y) \leq_X x$ . Therefore  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection.

3. By (1) and (2),

$$\begin{aligned}
 (f, g) \text{ is a } (\leq_X, \leq_Y)\text{-antitone Galois connection} &\iff (f, g) \text{ is a } (\leq_X^*, \leq_Y)\text{-monotone Galois connection} \\
 &\iff (g, f) \text{ is a } (\leq_Y^*, (\leq_X^*)^*)\text{-monotone Galois connection} \\
 &\iff (g, f) \text{ is a } (\leq_Y^*, \leq_X)\text{-monotone Galois connection} \\
 &\iff (g, f) \text{ is a } (\leq_Y, \leq_X)\text{-antitone Galois connection.}
 \end{aligned}$$

□

**Exercise 7.2.0.4.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Then  $f$  is a  $(\leq_X, \leq_Y)$ -left adjoint of  $g$  iff  $g$  is a  $(\leq_X, \leq_Y)$ -right adjoint of  $f$ .

*Proof.* We have that

$$\begin{aligned}
 f \text{ is a } (\leq_X, \leq_Y)\text{-left adjoint of } g &\iff (f, g) \text{ is a } (\leq_X, \leq_Y)\text{-monotone Galois connection} \\
 &\iff g \text{ is a } (\leq_X, \leq_Y)\text{-right adjoint of } f.
 \end{aligned}$$

□

**Exercise 7.2.0.5.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Then

1.  $g$  is a  $(\leq_X, \leq_Y)$ -right adjoint of  $f$  iff  $g$  is a  $(\leq_Y^*, \leq_X^*)$ -left adjoint of  $f$ .
2.  $f$  is a  $(\leq_X, \leq_Y)$ -left adjoint of  $g$  iff  $f$  is a  $(\leq_Y^*, \leq_X^*)$ -right adjoint of  $g$ .

*Proof.*

1. Exercise 7.2.0.3 implies that that

$$\begin{aligned}
 g \text{ is a } (\leq_X, \leq_Y)\text{-right adjoint of } f &\iff (f, g) \text{ is a } (\leq_X, \leq_Y)\text{-monotone Galois connection} \\
 &\iff (g, f) \text{ is a } (\leq_Y^*, \leq_X^*)\text{-monotone Galois connection} \\
 &\iff g \text{ is a } (\leq_Y^*, \leq_X^*)\text{-left adjoint of } f.
 \end{aligned}$$

2. We have that

$$\begin{aligned}
 f \text{ is a } (\leq_X, \leq_Y)\text{-left adjoint of } g &\iff g \text{ is a } (\leq_X, \leq_Y)\text{-right adjoint of } f \quad (\text{by Exercise 7.2.0.4}) \\
 &\iff g \text{ is a } (\leq_Y^*, \leq_X^*)\text{-left adjoint of } f \quad (\text{by prev part}) \\
 &\iff f \text{ is a } (\leq_Y^*, \leq_X^*)\text{-right adjoint of } g \quad (\text{by Exercise 7.2.0.4}).
 \end{aligned}$$

□

**Exercise 7.2.0.6.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Then

1.  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection iff
  - (a) i.  $\text{id}_X \leq_X g \circ f$
  - ii.  $f \circ g \leq_Y \text{id}_Y$
  - (b)  $f$  is  $(\leq_X, \leq_Y)$ -monotone and  $g$  is  $(\leq_Y, \leq_X)$ -monotone
2.  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection iff
  - (a) i.  $g \circ f \leq_X \text{id}_X$
  - ii.  $f \circ g \leq_Y \text{id}_Y$
  - (b)  $f$  is  $(\leq_X, \leq_Y)$ -antitone and  $g$  is  $(\leq_Y, \leq_X)$ -antitone

*Proof.*



1. • ( $\Rightarrow$ ):

Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

- (a) i. Let  $x \in X$ . Define  $y \in Y$  by  $y := f(x)$ . Since  $f(x) \leq_Y f(x)$ , we have that  $f(x) \leq_Y y$ . Therefore

$$\begin{aligned} x &\leq_X g(y) \\ &= g \circ f(x). \end{aligned}$$

Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $x \leq_X g \circ f(x)$ . Hence  $\text{id}_X \leq_X g \circ f$ .

- ii. Let  $y \in Y$ . **the previous ex** implies that  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection. Part 1(a) then implies that for each  $y \in Y$ ,  $y \leq_Y^* f \circ g(y)$ . Therefore for each  $y \in Y$ ,  $f \circ g(y) \leq_Y y$ . Hence  $f \circ g \leq_Y \text{id}_Y$ .
- (b) – Let  $a, b \in X$ . Define  $x \in X$  and  $y \in Y$  by  $x := a$  and  $y := f(b)$ . Suppose that  $a \leq_X b$ . Part 1(a) implies that

$$\begin{aligned} x &= a \\ &\leq_X b \\ &\leq_X g(f(x)) \\ &= g(y). \end{aligned}$$

Since  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, we have that

$$\begin{aligned} f(a) &= f(x) \\ &\leq_Y y \\ &= f(b). \end{aligned}$$

Since  $a, b \in X$  with  $a \leq_X b$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(a) \leq_Y f(b)$ . Thus  $f$  is  $(\leq_X, \leq_Y)$ -monotone.

- Since  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, **a previous ex** implies that  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection. The previous part implies that  $g$  is  $(\leq_Y^*, \leq_X^*)$ -monotone. Exercise 7.1.0.3 then implies that  $g$  is  $(\leq_Y, \leq_X)$ -monotone.

• ( $\Leftarrow$ ):

Suppose that

- (a) i. for each  $x \in X$ ,  $x \leq_X g \circ f(x)$   
 ii. for each  $y \in Y$ ,  $f \circ g(y) \leq_Y y$   
 (b)  $f$  is  $(\leq_X, \leq_Y)$ -monotone and  $g$  is  $(\leq_Y, \leq_X)$ -monotone.

Let  $x \in X$  and  $y \in Y$ .

– ( $\Rightarrow$ ):

Suppose that  $f(x) \leq_Y y$ . Then by (a)i and (a)ii,

$$\begin{aligned} x &\leq_X g \circ f(x) \\ &\leq_X g(y). \end{aligned}$$

– ( $\Leftarrow$ ):

Suppose that  $x \leq_X g(y)$ . Then by (a)i and (a)ii,

$$\begin{aligned} f(x) &\leq_Y f \circ g(y) \\ &\leq_Y y. \end{aligned}$$

Since  $x \in X$  and  $y \in Y$  are arbitrary, we have that for each  $x \in X$  and  $y \in Y$ ,

$$f(x) \leq_Y y \text{ iff } f(x) \leq_Y y.$$

Therefore  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

2. **the previous ex** implies that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection iff  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection. Part (1) implies that  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection iff

- (a) i. for each  $x \in X$ ,  $x \leq_X^* g \circ f(x)$   
 ii. for each  $y \in Y$ ,  $f \circ g(y) \leq_Y y$
- (b)  $f$  is  $(\leq_X^*, \leq_Y)$ -monotone and  $g$  is  $(\leq_Y, \leq_X^*)$ -monotone.

two previous exercise about monotonic functions and the dual order then imply that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection iff

- (a) i. for each  $x \in X$ ,  $g \circ f(x) \leq_X x$   
 ii. for each  $y \in Y$ ,  $f \circ g(y) \leq_Y y$
- (b)  $f$  is  $(\leq_X, \leq_Y)$ -antitone and  $g$  is  $(\leq_Y, \leq_X)$ -antitone.

□

**Exercise 7.2.0.7.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . If  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection or  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection, then  $f \sim_{\leq_Y} f \circ g \circ f$  and  $g \sim_{\leq_X} g \circ f \circ g$ .

*Proof.*

- Suppose that  $(f, g)$  a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

- Let  $x \in X$ .

- \* Define  $a \in X$  and  $b \in Y$  by  $a := g \circ f(x)$  and  $b := f(x)$ . Since

$$\begin{aligned} g \circ f(x) \leq g \circ f(x) &\iff a \leq_X g(b) \\ &\iff f(a) \leq_Y b \\ &\iff f \circ g \circ f(x) \leq_Y f(x), \end{aligned}$$

we have that  $f \circ g \circ f(x) \leq_Y f(x)$ .

- \* **The prev ex** implies that  $x \leq_X g \circ f(x)$  and  $f$  is  $(\leq_X, \leq_Y)$ -monotone. Therefore  $f(x) \leq_Y f \circ g \circ f(x)$ .

Since  $f \circ g \circ f(x) \leq_Y f(x)$  and  $f(x) \leq_Y f \circ g \circ f(x)$ , we have that  $f \circ g \circ f(x) \sim_{\leq_Y} f(x)$ . Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $f \circ g \circ f(x) \sim_{\leq_Y} f(x)$ . Therefore  $f \sim_{\leq_Y} f \circ g \circ f$ .

- Let  $y \in Y$ .

- \* Define  $a \in X$  and  $b \in Y$  by  $a := g(y)$  and  $b := f \circ g(y)$ . Since

$$\begin{aligned} f \circ g(y) \leq_Y f \circ g(y) &\iff f(a) \leq_Y b \\ &\iff a \leq_X g(b) \\ &\iff g(y) \leq_X g \circ f \circ g(y), \end{aligned}$$

we have that  $g(y) \leq_X g \circ f \circ g(y)$ .

- \* **The prev ex** implies that  $f \circ g(y) \leq_Y y$  and  $g$  is  $(\leq_Y, \leq_X)$ -monotone. Therefore  $g \circ f \circ g(y) \leq_X g(y)$ .

Since  $g(y) \leq_X g \circ f \circ g(y)$  and  $g \circ f \circ g(y) \leq_X g(y)$ , we have that  $g(y) \sim_{\leq_X} g \circ f \circ g(y)$ . Since  $y \in Y$  is arbitrary, we have that for each  $y \in Y$ ,  $g(y) \sim_{\leq_X} g \circ f \circ g(y)$ . Therefore  $g \sim_{\leq_X} g \circ f \circ g$ .

- Suppose that  $(f, g)$  a  $(\leq_X, \leq_Y)$ -antitone Galois connection. **The prev ex** then implies that  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection. Part (1) then implies that  $f \sim_{\leq_Y} f \circ g \circ f$  and  $g \sim_{\leq_X^*} g \circ f \circ g$ . **Make previous ex that  $x \sim_{\leq} y$  iff  $x \sim_{\leq^*} y$**  then implies that  $g \sim_{\leq_X} g \circ f \circ g$ .

**FINSIH!!**

□

**Exercise 7.2.0.8.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . If  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection or  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection, then  $f = f \circ g \circ f$  and  $g = g \circ f \circ g$ .

*Proof.* Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection or  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection. Exercise 7.2.0.7 implies that  $f \sim_{\leq_Y} f \circ g \circ f$  and  $g \sim_{\leq_X} g \circ f \circ g$ . Since  $(X, \leq_X)$  and  $(Y, \leq_Y)$  are posets, we have that  $f = f \circ g \circ f$  and  $g = g \circ f \circ g$ . □

**Exercise 7.2.0.9.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ .

1. For each  $g' : Y \rightarrow X$ , if  $g$  and  $g'$  are right adjoints of  $f$ , then  $g = g'$ .
2. For each  $f' : X \rightarrow Y$ , if  $f$  and  $f'$  are left adjoints of  $g$ , then  $f = f'$ .

*Proof.*

1. Let  $g' : Y \rightarrow X$ . Suppose that  $g$  and  $g'$  are right adjoints of  $f$ . Then  $(f, g)$  and  $(f, g')$  are Galois connections. Exercise 7.2.0.8 implies that  $f \circ g' \circ f = f$  and for each  $x \in X$ ,  $g' \circ f(x) \leq_Y g \circ f(g' \circ f(x))$ . Therefore for each  $x \in X$ ,

$$\begin{aligned} g' \circ f(x) &\leq g \circ f(g' \circ f(x)) \\ &= g(f \circ g' \circ f(x)) \\ &= g(f(x)) \\ &= g \circ f(x). \end{aligned}$$

Thus  $g' \circ f \leq g \circ f$ . Similarly,  $g \circ f \leq g' \circ f$ . Hence  $g' \circ f = g \circ f$ . Therefore

$$\begin{aligned} g &= g \circ f \circ g \\ &= g' \circ f \circ g. \end{aligned}$$

and

$$\begin{aligned} g' &= g' \circ f \circ g' \\ &= g \circ f \circ g'. \end{aligned}$$

Let  $y \in Y$ . Since  $f \circ g'(y) \leq y$ , we have that

$$\begin{aligned} g'(y) &= g \circ f \circ g'(y) \\ &= g(f \circ g'(y)) \\ &\leq g(y). \end{aligned}$$

Similarly,  $g(y) \leq g'(y)$ . Hence  $g(y) = g'(y)$ . Since  $y \in Y$  is arbitrary, we have that for each  $y \in Y$ ,  $g(y) = g'(y)$ . Thus  $g = g'$ .

2. Similar to (1).

**TIDY UP!!!!**

□

**Definition 7.2.0.10.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ .

- If  $f$  has a  $(\leq_X, \leq_Y)$ -right adjoint, then we define the  $(\leq_X, \leq_Y)$ -**right adjoint of  $f$** , denoted  $\text{RAdj}(f, (\leq_X, \leq_Y))$ , to be the unique  $g' : Y \rightarrow X$  such that  $g'$  is a  $(\leq_X, \leq_Y)$ -right adjoint of  $f$ .
- If  $g$  has a  $(\leq_X, \leq_Y)$ -left adjoint, then we define the  $(\leq_X, \leq_Y)$ -**left adjoint of  $g$** , denoted  $\text{LAdj}(g, (\leq_X, \leq_Y))$ , to be the unique  $f' : X \rightarrow Y$  such that  $f'$  is a  $(\leq_X, \leq_Y)$ -left adjoint of  $g$ .

**Note 7.2.0.11.** When the context is clear, we write  $f^*$  and  $g_*$  in place of  $\text{RAdj}(f, (\leq_X, \leq_Y))$  and  $\text{LAdj}(g, (\leq_X, \leq_Y))$  respectively.

**Exercise 7.2.0.12.** Let  $(X, \leq_X)$ ,  $(Y, \leq_Y)$ ,  $(Z, \leq_Z)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ .

1. If  $f$  has a right adjoint and  $g$  has a right adjoint, then  $g \circ f$  has a right adjoint and  $(g \circ f)^* = f^* \circ g^*$ .
2. If  $f$  has a left adjoint and  $g$  has a left adjoint, then  $g \circ f$  has a left adjoint and  $(g \circ f)_* = f_* \circ g_*$ .

*Proof.*

1. Suppose that  $f$  has a  $(\leq_X, \leq_Y)$ -right adjoint and  $g$  has a  $(\leq_X, \leq_Y)$ -right adjoint. Since  $(f, f^*)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection and  $(g, g^*)$  is a  $(\leq_Y, \leq_Z)$ -monotone Galois connection, we have that for each  $x \in X$ ,  $y \in Y$  and  $z \in Z$ ,  $x \leq_X f^*(y)$  iff  $f(x) \leq_Y y$  and  $y \leq_Y g^*(z)$  iff  $g(y) \leq_Z z$ . Hence for each  $x \in X$  and  $z \in Z$ ,

$$\begin{aligned} x \leq_X f^* \circ g^*(z) &\iff x \leq_X f^*(g^*(z)) \\ &\iff f(x) \leq_Y g^*(z) \\ &\iff g(f(x)) \leq_Z z \\ &\iff g \circ f(x) \leq_Z z. \end{aligned}$$

Thus  $(g \circ f, f^* \circ g^*)$  is a  $(\leq_X, \leq_Z)$ -monotone Galois connection. Therefore  $g \circ f$  has a  $(\leq_X, \leq_Z)$ -right adjoint and Exercise 7.2.0.9 implies that  $(g \circ f)^* = f^* \circ g^*$ .

2. Suppose that  $f$  has a  $(\leq_Y, \leq_X)$ -left adjoint and  $g$  has a  $(\leq_Z, \leq_Y)$ -left adjoint. Exercise 7.2.0.5 implies that  $\text{LAdj}(f, (\leq_Y, \leq_X)) = \text{RAAdj}(f, (\leq_X^*, \leq_Y^*))$  and  $\text{LAdj}(g, (\leq_Z, \leq_Y)) = \text{RAAdj}(g, (\leq_Y^*, \leq_Z^*))$ . The previous part implies that  $g \circ f$  has a  $(\leq_X^*, \leq_Z^*)$ -right adjoint and  $\text{RAAdj}(g \circ f, (\leq_X^*, \leq_Z^*)) = \text{LAdj}(f, (\leq_Y, \leq_X)) \circ \text{LAdj}(g, (\leq_Z, \leq_Y))$ . Exercise 7.2.0.5 then implies that  $g \circ f$  has a  $(\leq_Z, \leq_X)$ -left adjoint and

$$\begin{aligned} \text{LAdj}(g \circ f, (\leq_Z, \leq_X)) &= \text{RAAdj}(g \circ f, (\leq_X^*, \leq_Z^*)) \\ &= \text{LAdj}(f, (\leq_Y, \leq_X)) \circ \text{LAdj}(g, (\leq_Z, \leq_Y)). \end{aligned}$$

□

**Exercise 7.2.0.13.** Let  $X, Y$  be sets and  $f : X \rightarrow Y$ . We define  $f_* : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$  and  $f^* : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$  by  $f_*(A) := f(A)$  and  $f^*(B) := f^{-1}(B)$ . Then  $(f_*, f^*)$  is a  $(\subset_X, \subset_Y)$ -monotone Galois connection.

*Proof.* We note that for each  $A \in \mathcal{P}(X)$  and  $B \in \mathcal{P}(Y)$ ,

$$\begin{aligned} f_*(A) \subset_Y B &\iff f(A) \subset_Y B \\ &\iff f^{-1}(f(A)) \subset_X f^{-1}(B) \quad \text{by (ref ex in set theory notes)} \\ &\iff A \subset_X f^{-1}(B) \quad \text{by (ref ex in set theory notes)} \\ &\iff A \subset_X f^*(B). \end{aligned}$$

Thus  $(f_*, f^*)$  is a  $(\subset_X, \subset_Y)$ -monotone Galois connection. □

**Definition 7.2.0.14.** maybe get rid of, ask if there is any point in rephrasing/renaming, possibly move right after definition of gallois connections, with discussion on 'information' Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

- Let  $x \in X$ . We define the **right  $(f, g)$ -connection of  $x$** , denoted  $\text{RConn}_Y(x, (f, g)) \in Y$ , by  $\text{RConn}_Y(x, (f, g)) := f(x)$ .
- Let  $y \in Y$ . We define the **left  $(f, g)$ -connection of  $y$** , denoted  $\text{LConn}_X(y, (f, g)) \in X$ , by  $\text{LConn}_X(y, (f, g)) := g(y)$ .

# Chapter 8

## Upper and Lower Sets

### 8.1 Introduction

**Definition 8.1.0.1.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

- $A$  is said to be a  **$\leq$ -upper set** if for each  $a \in A$  and  $x \in X$ ,  $a \leq x$  implies that  $x \in A$ .
- $A$  is said to be an  **$\leq$ -lower set** if for each  $a \in A$  and  $x \in X$ ,  $x \leq a$  implies that  $x \in A$ .

**Note 8.1.0.2.** When the context is clear, we say  $A$  is a

- “upper set” instead of “ $\leq$ -upper set”
- “lower set” instead of “ $\leq$ -lower set”

**Exercise 8.1.0.3.** Let  $(X, \leq)$  be a proset. Then

1.  $X$  is a  $\leq$ -upper set
2.  $X$  is a  $\leq$ -lower set

*Proof.*

1. Let  $a, x \in X$ . Suppose that  $a \leq x$ . By assumption,  $x \in X$ . Since  $a, x \in X$  with  $a \leq x$  are arbitrary, we have that for each  $a, x \in X$ ,  $a \leq x$  implies that  $x \in A$ . Hence  $X$  is a  $\leq$ -upper set.
2. Similar to (1).

□

**Exercise 8.1.0.4.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1.  $A$  is a  $\leq$ -upper set iff  $A$  is a  $\leq^*$ -lower set
2.  $A$  is a  $\leq$ -lower set iff  $A$  is a  $\leq^*$ -upper set

*Proof.*

1. • ( $\implies$ ):  
Suppose that  $A$  is a  $\leq$ -upper set. Let  $a \in A$  and  $x \in X$ . Suppose that  $x \leq^* a$ . Then  $a \leq x$ . Since  $A$  is a  $\leq$ -upper set,  $x \in A$ . Since  $a \in A$  and  $x \in X$  with  $x \leq^* a$  is arbitrary, we have that for each  $a \in A$  and  $x \in X$ ,  $x \leq^* a$  implies that  $x \in A$ . Hence  $A$  is a  $\leq^*$ -lower set.
- ( $\impliedby$ ):  
Suppose that  $A$  is a  $\leq^*$ -lower set. Let  $a \in A$  and  $x \in X$ . Suppose that  $a \leq x$ . Then  $x \leq^* a$ . Since  $A$  is a  $\leq^*$ -lower set,  $x \in A$ . Since  $a \in A$  and  $x \in X$  with  $a \leq x$  is arbitrary, we have that for each  $a \in A$  and  $x \in X$ ,  $a \leq x$  implies that  $x \in A$ . Hence  $A$  is a  $\leq$ -upper set.

2. Similar to (1). **USE DUALITY**

□

**Exercise 8.1.0.5.** Let  $(X, \leq)$  be a poset and  $A \subset X$ . Then

1.  $A$  is an upper set iff  $A^c$  is a lower set
2.  $A$  is a lower set iff  $A^c$  is an upper set

*Proof.*

1. • ( $\Rightarrow$ ):  
Suppose that  $A$  is an upper set. Let  $b \in A^c$  and  $x \in X$ . Suppose that  $x \leq b$ . For the sake of contradiction, suppose that  $x \in A$ . Since  $x \leq b$  and  $A$  is an upper set,  $b \in A$ . This is a contradiction since  $b \in A^c$ . Hence  $x \in A^c$ . Since  $b \in A^c$  and  $x \in X$  with  $x \leq b$  are arbitrary, we have that for each  $b \in A^c$  and  $x \in X$ ,  $x \leq b$  implies that  $x \in A^c$ . Thus  $A^c$  is a lower set.
- ( $\Leftarrow$ ):  
Suppose that  $A^c$  is a  $\leq$ -lower set. Exercise 8.1.0.4 implies that  $A^c$  is a  $\leq^*$ -upper set. The previous part implies that  $A$  is a  $\leq^*$ -lower set. Another application of Exercise 8.1.0.4 implies that  $A$  is a  $\leq$ -upper set.
2. Similar to (1).

□

**Exercise 8.1.0.6.** Let  $(X, \leq)$  be a poset,  $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$ .

1. If for each  $\alpha \in A$ ,  $E_\alpha$  is an upper set, then

- (a)  $\bigcup_{\alpha \in A} E_\alpha$  is an upper set
- (b)  $\bigcap_{\alpha \in A} E_\alpha$  is an upper set

2. If for each  $\alpha \in A$ ,  $E_\alpha$  is a lower set, then

- (a)  $\bigcup_{\alpha \in A} E_\alpha$  is a lower set
- (b)  $\bigcap_{\alpha \in A} E_\alpha$  is a lower set

*Proof.*

1. Suppose that for each  $\alpha \in A$ ,  $E_\alpha$  is an upper set. Set  $E := \bigcup_{\alpha \in A} E_\alpha$ .

- (a) Let  $e \in E$  and  $x \in X$ . Suppose that  $e \leq x$ . Since  $e \in E$ , there exists  $\alpha \in A$  such that  $e \in E_\alpha$ . Since  $E_\alpha$  is an upper set and  $e \leq x$ , we have that

$$\begin{aligned} x &\in E_\alpha \\ &\subset \bigcup_{\alpha \in A} E_\alpha \\ &= E. \end{aligned}$$

Since  $e \in E$  and  $x \in X$  with  $e \leq x$  are arbitrary, we have that for each  $e \in E$  and  $x \in X$ ,  $e \leq x$  implies that  $x \in E$ . Hence  $E$  is an upper set.

- (b) Let  $\alpha \in A$ . Since  $E_\alpha$  is a  $\leq$ -upper set, Exercise 8.1.0.5 implies that  $E_\alpha^c$  is a  $\leq$ -lower set. Exercise 8.1.0.4 then implies that  $E_\alpha^c$  is a  $\leq^*$ -upper set. Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $E_\alpha^c$  is a  $\leq^*$ -upper set. The previous part implies that  $\bigcup_{\alpha \in A} E_\alpha^c$  is a  $\leq^*$ -upper set. Since  $\bigcap_{\alpha \in A} E_\alpha = \left( \bigcup_{\alpha \in A} E_\alpha^c \right)^c$ , Exercise 8.1.0.5 implies that  $\bigcap_{\alpha \in A} E_\alpha$  is a  $\leq^*$ -lower set. Exercise 8.1.0.4 then implies that  $\bigcap_{\alpha \in A} E_\alpha$  is a  $\leq$ -upper set.

2. Similar to (1).

□

**Definition 8.1.0.7.** Let  $(X, \leq)$  be a proset and  $S \subset X$ . (define upper set generated by- as the generated element from a galois connection, then make exercise showing that  $\uparrow(S, \leq) := \bigcap_{U \in \mathcal{U}(S, \leq)} U$ ) We define the

- $\leq$ -upper sets containing  $S$ , denoted  $\mathcal{U}(S, \leq) \subset \mathcal{P}(X)$ , by  $\mathcal{U}(S, \leq) := \{U \subset X : U \text{ is a } \leq\text{-upper set and } S \subset U\}$
- $\leq$ -lower sets containing  $S$ , denoted  $\mathcal{L}(S, \leq) \subset \mathcal{P}(X)$ , by  $\mathcal{L}(S, \leq) := \{L \subset X : L \text{ is a } \leq\text{-lower set and } S \subset L\}$ .
- $\leq$ -upper set generated by  $S$ , denoted  $\uparrow(S, \leq) \subset X$ , by  $\uparrow(S, \leq) := \bigcap_{U \in \mathcal{U}(S, \leq)} U$
- $\leq$ -lower set generated by  $S$ , denoted  $\downarrow(S, \leq) \subset X$ , by  $\downarrow(S, \leq) := \bigcap_{L \in \mathcal{L}(S, \leq)} L$

**Note 8.1.0.8.** • When the context is clear, we write

- $\mathcal{U}(S), \mathcal{L}(S), \uparrow S$  and  $\downarrow S$  in place of  $\mathcal{U}(S, \leq), \mathcal{L}(S, \leq), \uparrow(S, \leq)$  and  $\downarrow(S, \leq)$  respectively.
- $\mathcal{U}(\leq)$  or  $\mathcal{U}_X$  in place of  $\mathcal{U}(\emptyset, \leq)$
- $\mathcal{L}(\leq)$  or  $\mathcal{L}_X$  in place of  $\mathcal{L}(\emptyset, \leq)$ .
- If  $S = \{s\}$ , we write  $\uparrow s$  and  $\downarrow s$  in place of  $\uparrow S$  and  $\downarrow S$  respectively.
- Exercise 8.1.0.3 implies that  $X \in \mathcal{U}(S)$  and  $X \in \mathcal{L}(S)$ .
- Exercise 8.1.0.6 implies that  $\uparrow S \in \mathcal{U}(S)$  and  $\downarrow S \in \mathcal{L}(S)$ .

**Exercise 8.1.0.9.** Let  $(X, \leq)$  be a proset and  $S \subset X$ . Then

1.  $\mathcal{U}(S, \leq^*) = \mathcal{L}(S, \leq)$ ,
2.  $\mathcal{L}(S, \leq^*) = \mathcal{U}(S, \leq)$ ,
3.  $\uparrow(S, \leq^*) = \downarrow(S, \leq)$ ,
4.  $\downarrow(S, \leq^*) = \uparrow(S, \leq)$ .

*Proof.*

1. • Let  $L \in \mathcal{U}(S, \leq^*)$ . Then  $S \subset L$  and  $L$  is a  $\leq^*$ -upper set. Exercise 8.1.0.4 then implies that  $L$  is a  $\leq$ -lower set. Hence  $L \in \mathcal{L}(S, \leq)$ . Since  $L \in \mathcal{U}(S, \leq^*)$  is arbitrary, we have that for each  $L \in \mathcal{U}(S, \leq^*)$ ,  $L \in \mathcal{L}(S, \leq)$ . Thus  $\mathcal{U}(S, \leq^*) \subset \mathcal{L}(S, \leq)$ .
- Let  $L \in \mathcal{L}(S, \leq)$ . Then  $S \subset L$  and  $L$  is a  $\leq$ -lower set. Another application of Exercise 8.1.0.4 implies that  $L$  is a  $\leq^*$ -upper set. Hence  $L \in \mathcal{U}(S, \leq^*)$ . Since  $L \in \mathcal{L}(S, \leq)$  is arbitrary, we have that for each  $L \in \mathcal{L}(S, \leq)$ ,  $L \in \mathcal{U}(S, \leq^*)$ . Thus  $\mathcal{L}(S, \leq) \subset \mathcal{U}(S, \leq^*)$ .

Since  $\mathcal{U}(S, \leq^*) \subset \mathcal{L}(S, \leq)$  and  $\mathcal{L}(S, \leq) \subset \mathcal{U}(S, \leq^*)$ , we have that  $\mathcal{U}(S, \leq^*) = \mathcal{L}(S, \leq)$ .

2. By (1), we have that

$$\begin{aligned} \mathcal{L}(S, \leq^*) &= \mathcal{U}(S, (\leq^*)^*) \\ &= \mathcal{U}(S, \leq). \end{aligned}$$

3. Part (1) implies that

$$\begin{aligned} \uparrow(S, \leq^*) &= \bigcap_{U \in \mathcal{U}(S, \leq^*)} U \\ &= \bigcap_{U \in \mathcal{L}(S, \leq)} U \\ &= \downarrow(S, \leq^*). \end{aligned}$$

4. Similar to (3). □

**Exercise 8.1.0.10.** Let  $(X, \leq)$  be a proset and  $S \subset X$ . Then

1.  $S \in \mathcal{U}(S, \leq)$  iff  $\uparrow S = S$
2.  $S \in \mathcal{L}(S, \leq)$  iff  $\downarrow S = S$

*Proof.*

1. • ( $\implies$ ):  
Suppose that  $S \in \mathcal{U}(S, \leq)$ . Then

$$\begin{aligned}\uparrow S &= \bigcap_{U \in \mathcal{U}(S, \leq)} U \\ &\subset S \\ &\subset \uparrow S.\end{aligned}$$

Hence  $\uparrow S = S$ .

- ( $\impliedby$ ):  
Suppose that  $\uparrow S = S$ . Then

$$\begin{aligned}S &= \uparrow S \\ &\in \mathcal{U}(S, \leq).\end{aligned}$$

2. Similar to (1). □

**Exercise 8.1.0.11.** Let  $(X, \leq)$  be a proset and  $a \in X$ . Then

1.  $\uparrow a = \{x \in X : a \leq x\}$
2.  $\downarrow a = \{x \in X : x \leq a\}$

*Proof.*

1. Define  $U \subset X$  by  $U := \{x \in X : a \leq x\}$ .  
• Let  $u \in U$  and  $x \in X$ . Suppose that  $u \leq x$ . Then

$$\begin{aligned}a &\leq u \\ &\leq x.\end{aligned}$$

Hence  $x \in U$ . Since  $u \in U$  and  $x \in X$  with  $u \leq x$  are arbitrary, we have that for each  $u \in U$  and  $x \in X$ ,  $u \leq x$  implies that  $x \in U$ . Hence  $U$  is an upper set. Since  $a \leq a$ ,  $a \in U$ . By definition,  $U \in \mathcal{U}(a)$ . Therefore

$$\begin{aligned}\uparrow a &= \bigcap_{U' \in \mathcal{U}(a)} U' \\ &\subset U.\end{aligned}$$

- Let  $x \in U$  and  $U' \in \mathcal{U}(a)$ . Since  $x \in U$ ,  $a \leq x$ . Since  $U' \in \mathcal{U}(a)$ ,  $U'$  is an upper set and  $a \in U'$ . Thus  $x \in U'$ . Since  $U' \in \mathcal{U}(a)$  is arbitrary, we have that for each  $U' \in \mathcal{U}(a)$ ,  $x \in U'$ . Thus

$$\begin{aligned}x &\in \bigcap_{U' \in \mathcal{U}(a)} U' \\ &= \uparrow a.\end{aligned}$$

Since  $x \in U$  is arbitrary, we have that for each  $x \in U$ ,  $x \in \uparrow a$ . Hence  $U \subset \uparrow a$ .



Since  $\uparrow a \subset U$  and  $U \subset \uparrow a$ , we have that  $\uparrow a = U$ .

2. Exercise 8.1.0.4 and (1) imply that

$$\begin{aligned}\downarrow(a, \leq) &= \uparrow(a, \leq^*) \\ &= \{x \in X : a \leq^* x\} \\ &= \{x \in X : x \leq a\}.\end{aligned}$$

□

**Exercise 8.1.0.12.** Let  $(X, \leq)$  be a proset and  $a, b \in X$ .

1. If  $\uparrow a = \uparrow b$ , then  $a \sim_{\leq} b$
2. If  $\downarrow a = \downarrow b$ , then  $a \sim_{\leq} b$

follow up later with one about posets

*Proof.*

1. Suppose that  $\uparrow a = \uparrow b$ . Then

$$\begin{aligned}a &\in \uparrow a \\ &= \uparrow b.\end{aligned}$$

Hence  $a \leq b$ . Similarly,  $b \leq a$ . Hence  $a \sim_{\leq} b$ .

- 2.

□

**Exercise 8.1.0.13.** Let  $(X, \leq)$  be a poset and  $a, b \in X$ .

1. If  $\uparrow a = \uparrow b$ , then  $a = b$
2. If  $\downarrow a = \downarrow b$ , then  $a = b$

*Proof.*

1. Suppose that  $\uparrow a = \uparrow b$ . Exercise 8.1.0.12 then implies that  $a \sim_{\leq} b$ . Since  $(X, \leq)$  is a poset, we have that  $a = b$ .
2. Similar to (1).

□

**Exercise 8.1.0.14.** Let  $(X, \leq)$  be a proset and  $S_1, S_2 \subset X$ . If  $S_1 \subset S_2$ , then

1.  $\uparrow S_1 \subset \uparrow S_2$
2.  $\downarrow S_1 \subset \downarrow S_2$

*Proof.* Suppose that  $S_1 \subset S_2$ .

1. Since  $\uparrow S_2 \in \mathcal{U}(S_2)$ ,  $S_2 \subset \uparrow S_2$  and  $\uparrow S_2$  is an upper set. Since

$$\begin{aligned}S_1 &\subset S_2 \\ &\subset \uparrow S_2,\end{aligned}$$

we have that  $\uparrow S_2 \in \mathcal{U}(S_1)$ . Therefore

$$\begin{aligned}\uparrow S_1 &= \bigcap_{U \in \mathcal{U}(S_1)} U \\ &\subset \uparrow S_2.\end{aligned}$$

2. Part (1) and Exercise 8.1.0.9 implies that

$$\begin{aligned}\downarrow(S_1, \leq) &= \uparrow(S_1, \leq^*) \\ &\subset \uparrow(S_2, \leq^*) \\ &= \downarrow(S_2, \leq).\end{aligned}$$

□

**Exercise 8.1.0.15.** Let  $(X, \leq)$  be a proset and  $a, b \in X$ . Then the following are equivalent:

1.  $a \leq b$ ,
2.  $\uparrow b \subset \uparrow a$ ,
3.  $\downarrow a \subset \downarrow b$ .

*Proof.*

1. (1)  $\implies$  (2):

Suppose that  $a \leq b$ . Since  $\uparrow a \in \mathcal{U}(a)$ , we have that  $a \in \uparrow a$  and  $\uparrow a$  is an upper set. Since  $a \leq b$ , we have that  $b \in \uparrow a$ . Hence  $\uparrow a \in \mathcal{U}(b)$ . Therefore

$$\begin{aligned}\uparrow b &= \bigcap_{U \in \mathcal{U}(b)} U \\ &\subset \uparrow a.\end{aligned}$$

2. (2)  $\implies$  (3):

Suppose that  $\uparrow(b, \leq) \subset \uparrow(a, \leq)$ . Exercise 8.1.0.11 then implies that

$$\begin{aligned}b &\in \uparrow(b, \leq) \\ &\subset \uparrow(a, \leq) \\ &= \{x \in X : a \leq x\}.\end{aligned}$$

Hence  $a \leq b$ . Thus  $b \leq^* a$ . Exercise 8.1.0.9 and part (1)  $\implies$  (2) then imply that

$$\begin{aligned}\downarrow(a, \leq) &= \uparrow(a, \leq^*) \\ &\subset \uparrow(b, \leq^*) \\ &= \downarrow(b, \leq).\end{aligned}$$

3. (3)  $\implies$  (1):

Suppose that  $\downarrow a \subset \downarrow b$ . Exercise 8.1.0.11 then implies that

$$\begin{aligned}a &\in \downarrow a \\ &\subset \downarrow b \\ &= \{x \in X : x \leq b\}.\end{aligned}$$

Hence  $a \leq b$ .

□

**Exercise 8.1.0.16.** Let  $(X, \leq)$  be a proset and  $S \subset X$ . Then

1.  $\uparrow S = \bigcup_{s \in S} \uparrow s$
2.  $\downarrow S = \bigcup_{s \in S} \downarrow s$

*Proof.*

1. Define  $U \subset X$  by  $U := \bigcup_{s \in S} \uparrow s$ .

- Since for each  $s \in S$ ,  $\uparrow s$  is an upper set, Exercise 8.1.0.6 implies that  $U$  is an upper set. Let  $s \in S$ . Then

$$\begin{aligned} s &\in \uparrow s \\ &\subset \bigcup_{s \in S} \uparrow s \\ &= U. \end{aligned}$$

Since  $s \in S$  is arbitrary, we have that for each  $s \in S$ ,  $s \in U$ . Hence  $S \subset U$ . Therefore  $U \in \mathcal{U}(S)$  and

$$\begin{aligned} \uparrow S &= \bigcap_{U' \in \mathcal{U}(S)} U' \\ &\subset U. \end{aligned}$$

- Let  $x \in U$  and  $U' \in \mathcal{U}(S)$ . Since  $x \in U$ , there exists  $s \in S$  such that  $x \in \uparrow s$ . Exercise 8.1.0.11 then implies that  $s \leq x$ . Since  $U' \in \mathcal{U}(S)$ ,  $U'$  is an upper set and  $S \subset U'$ . Then

$$\begin{aligned} s &\in S \\ &\subset U'. \end{aligned}$$

Since  $U'$  is an upper set and  $s \leq x$ ,  $x \in U'$ . Since  $U' \in \mathcal{U}(S)$  is arbitrary, we have that for each  $U' \in \mathcal{U}(S)$ ,  $x \in U'$ . Thus

$$\begin{aligned} x &\in \bigcap_{U' \in \mathcal{U}(S)} U' \\ &= \uparrow S. \end{aligned}$$

Since  $x \in U$  is arbitrary, we have that for each  $x \in U$ ,  $x \in \uparrow S$ . Hence  $U \subset \uparrow S$ .

Since  $\uparrow S \subset U$  and  $U \subset \uparrow S$ , we have that  $\uparrow S = U$ .

2. Similar to (1).

□

**Exercise 8.1.0.17.** Let  $(X, \leq)$  be a proset and  $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$ . Then

1.  $\uparrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \uparrow E_\alpha$
2.  $\downarrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \downarrow E_\alpha$

*Proof.*

1. • Let  $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$ . Exercise 8.1.0.16 implies that there exists  $y \in \bigcup_{\alpha \in A} E_\alpha$  such that  $x \in \uparrow y$ . Then there exists  $\alpha_0 \in A$  such that  $y \in E_{\alpha_0}$ . Since  $\{y\} \subset E_{\alpha_0}$ , Exercise 8.1.0.14 implies that  $\uparrow y \subset \uparrow E_{\alpha_0}$ . Therefore

$$\begin{aligned} x &\in \uparrow y \\ &\subset \uparrow E_{\alpha_0} \\ &\subset \bigcup_{\alpha \in A} \uparrow E_\alpha. \end{aligned}$$

Since  $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$  is arbitrary, we have that for each  $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$ ,  $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$ . Hence  $\uparrow \bigcup_{\alpha \in A} E_\alpha \subset \bigcup_{\alpha \in A} \uparrow E_\alpha$ .

- Let  $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$ . Then there exists  $\alpha_0 \in A$  such that  $x \in \uparrow E_{\alpha_0}$ . Exercise 8.1.0.16 implies that there exists  $y \in E_{\alpha_0}$  such that  $x \in \uparrow y$ . Since

$$\begin{aligned} \{y\} &\subset E_{\alpha_0} \\ &\subset \bigcup_{\alpha \in A} E_\alpha, \end{aligned}$$

Exercise 8.1.0.14 implies that

$$\begin{aligned} x &\in \uparrow y \\ &\subset \uparrow \bigcup_{\alpha \in A} E_\alpha. \end{aligned}$$

Since  $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$  is arbitrary, we have that for each  $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$ ,  $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$ . Hence  $\bigcup_{\alpha \in A} \uparrow E_\alpha \subset \uparrow \bigcup_{\alpha \in A} E_\alpha$ .

Since  $\uparrow \bigcup_{\alpha \in A} E_\alpha \subset \bigcup_{\alpha \in A} \uparrow E_\alpha$  and  $\bigcup_{\alpha \in A} \uparrow E_\alpha \subset \uparrow \bigcup_{\alpha \in A} E_\alpha$ , we have that  $\uparrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \uparrow E_\alpha$ .

2. Similar to (1).

□

**Exercise 8.1.0.18.** Let  $(A, \leq)$  be a proset and  $B \subset A$ .

1. If  $B$  is upward directed, then  $\downarrow B$  is upward directed
2. If  $B$  is downward directed, then  $\uparrow B$  is downward directed

*Proof.*

1. Suppose that  $B$  is upward directed.

- Since  $B$  is upward directed,  $(B, \leq|_B)$  is a directed set. Thus  $B \neq \emptyset$ . Thus there exists  $x \in B$ . Since  $B \subset \downarrow B$ , we have that  $x \in \downarrow B$ . Hence  $\downarrow B \neq \emptyset$ .
- Let  $a, b \in \downarrow B$ . Exercise 8.1.0.16 implies that there exists  $x_a, x_b \in B$  such that  $a \in \downarrow x_a$  and  $b \in \downarrow x_b$ . Since  $B$  is upward directed,  $(B, \leq|_B)$  is a directed set. Thus  $\text{ub}(\{x_a, x_b\}, \leq|_B) \neq \emptyset$ . Therefore there exists  $x_0 \in B$  such that  $x_0 \geq x_a$  and  $x_0 \geq x_b$ . Hence

$$\begin{aligned} x_0 &\geq x_a \\ &\geq a. \end{aligned}$$

Similarly,  $x_0 \geq b$ . Since  $B \subset \downarrow B$ ,  $x_0 \in \downarrow B$ . Thus  $\text{ub}(\{a, b\}, \leq|_{\downarrow B}) \neq \emptyset$ . Since  $a, b \in \downarrow B$  are arbitrary, we have that for each  $a, b \in \downarrow B$ ,  $\text{ub}(\{a, b\}, \leq|_{\downarrow B}) \neq \emptyset$ . Then  $\leq|_{\downarrow B}$  is an upward direction on  $\downarrow B$ .

Hence  $B$  is upward directed.

2. Suppose that  $B$  is  $\leq$ -downward directed. **The prev ex** implies that  $B$  is  $\leq^*$ -upward directed. **The prev part** implies that  $\downarrow(B, \leq^*)$  is  $\leq^*$ -upward directed. **A prev ex** implies that  $\downarrow(B, \leq^*) = \uparrow(B, \leq)$ . Since  $\uparrow(B, \leq)$  is  $\leq^*$ -upward directed, **the prev ex** implies that  $\uparrow(B, \leq)$  is  $\leq$ -downward directed.

□

**Exercise 8.1.0.19.** Let  $(X, \leq)$  be a proset and  $x \in X$ . Then

1.  $\uparrow x$  is downward directed,
2.  $\downarrow x$  is upward directed.

*Proof.*

1.
  - Since  $x \in \uparrow x$ ,  $\uparrow x \neq \emptyset$ .
  - Exercise 6.2.0.4 implies that  $\{x\}$  is downward directed. Exercise 8.1.0.18 then implies that  $\uparrow x$  is downward directed.
- 2.

□

**Exercise 8.1.0.20.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ . Then for each  $U \subset Y$ ,  $U$  is a  $\leq_Y$ -upper set implies that  $f^{-1}(U)$  is a  $\leq_X$ -upper set iff for each  $L \subset Y$ ,  $L$  is a  $\leq_Y$ -lower set implies that  $f^{-1}(L)$  is a  $\leq_X$ -lower set.

*Proof.*

- $(\implies)$ :

Suppose that for each  $U \subset Y$ ,  $U$  is a  $\leq_Y$ -upper set implies that  $f^{-1}(U)$  is a  $\leq_X$ -upper set. Let  $L \subset Y$ . Suppose that  $L$  is a  $\leq_Y$ -lower set. Exercise 8.1.0.5 then implies that  $L^c$  is a  $\leq_Y$ -upper set. By assumption,  $f^{-1}(L^c)$  is a  $\leq_X$ -upper set. Another application of Exercise 8.1.0.5 implies that  $f^{-1}(L^c)^c$  is a  $\leq_X$ -lower set. Since

$$\begin{aligned} f^{-1}(L) &= [f^{-1}(L^c)^c]^c \\ &= f^{-1}(L^c)^c, \end{aligned}$$

we have that  $f^{-1}(L)$  is a  $\leq_X$ -lower set. Since  $L \subset Y$  such that  $L$  is a  $\leq_Y$ -lower set is arbitrary, we have that for each  $L \subset Y$ ,  $L$  is a  $\leq_Y$ -lower set implies that  $f^{-1}(L)$  is a  $\leq_X$ -lower set.

- $(\impliedby)$ :

Similar to  $(\implies)$ .

□

**Exercise 8.1.0.21.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ . If  $f$  is  $(\leq_X, \leq_Y)$ -monotone, then for each  $U \subset Y$ ,  $U$  is a  $\leq_Y$ -upper set implies that  $f^{-1}(U)$  is a  $\leq_X$ -upper set.

*Proof.* Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -monotone. Let  $U \subset Y$ . Suppose that  $U$  is a  $\leq_Y$ -upper set. Let  $a \in f^{-1}(U)$  and  $x \in X$ . Suppose that  $a \leq x$ . Since  $a \in f^{-1}(U)$ ,  $f(a) \in U$ . Since  $f$  is monotone,  $f(a) \leq f(x)$ . Since  $U$  is a  $\leq_Y$ -upper set and  $f(a) \in U$ , we have that  $f(x) \in U$ . Hence  $x \in f^{-1}(U)$ . Since  $a \in f^{-1}(U)$  and  $x \in X$  with  $a \leq x$  are arbitrary, we have that for each  $a \in f^{-1}(U)$  and  $x \in X$ ,  $a \leq x$  implies that  $x \in f^{-1}(U)$ . Thus  $f^{-1}(U)$  is a  $\leq_X$ -upper set. Since  $U \subset Y$  such that  $U$  is a  $\leq_Y$ -upper set is arbitrary, we have that for each  $U \subset Y$ ,  $U$  is a  $\leq_Y$ -upper set implies that  $f^{-1}(U)$  is a  $\leq_X$ -upper set. □

**Exercise 8.1.0.22.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ . Then  $f$  has a right adjoint iff for each  $y \in Y$ , there exists  $x \in X$  such that  $f^{-1}(\downarrow y) = \downarrow x$ .

*Proof.*

- $(\implies)$ :

Suppose that  $f$  has a right adjoint. Then there exists  $g : Y \rightarrow X$  such that  $(f, g)$  is a monotone Galois connection. [a prev ex in galois connections on posets section](#) Exercise ?? then implies that  $f$  and  $g$  are monotone,  $\text{id}_X \leq_X g \circ f$  and  $f \circ g \leq_Y \text{id}_Y$ . Let  $y \in Y$ . Define  $x \in X$  by  $x := g(y)$ .

- We note that for each  $a \in X$ ,

$$\begin{aligned} a \in f^{-1}(\downarrow y) &\iff f(a) \in \downarrow y \\ &\iff f(a) \leq_Y y. \end{aligned}$$

Let  $a \in f^{-1}(\downarrow y)$ . Then

$$\begin{aligned} a &\leq_X g \circ f(a) \\ &= g(f(a)) \\ &\leq_X g(y) \\ &= x. \end{aligned}$$

Therefore  $a \in \downarrow x$ . Since  $a \in f^{-1}(\downarrow y)$  is arbitrary, we have that for each  $a \in f^{-1}(\downarrow y)$ ,  $a \in \downarrow x$ . Hence  $a \in f^{-1}(\downarrow y) \subset \downarrow x$ . Then  $f(a) \in \downarrow$ .

– Let  $a \in \downarrow x$ . Then  $a \leq_X x$ . Hence

$$\begin{aligned} f(a) &\leq_Y f(x) \\ &= f(g(y)) \\ &= f \circ g(y) \\ &\leq_Y y. \end{aligned}$$

Thus  $f(a) \in \downarrow y$  and  $a \in f^{-1}(\downarrow y)$ . Since  $a \in \downarrow x$  is arbitrary, we have that for each  $a \in \downarrow x$ ,  $a \in f^{-1}(\downarrow y)$ . Hence  $\downarrow x \subset f^{-1}(\downarrow y)$ .

Since  $f^{-1}(\downarrow y) \subset \downarrow x$  and  $\downarrow x \subset f^{-1}(\downarrow y)$ , we have that  $f^{-1}(\downarrow y) = \downarrow x$ . Since  $y \in Y$  is arbitrary, we have that for each  $y \in Y$ , there exists  $x \in X$  such that  $f^{-1}(\downarrow y) = \downarrow x$ .

• ( $\Leftarrow$ ):

Suppose that for each  $y \in Y$ , there exists  $x \in X$  such that  $f^{-1}(\downarrow y) = \downarrow x$ . The axiom of choice implies that there exists  $g : Y \rightarrow X$  such that for each  $y \in Y$ ,  $f^{-1}(\downarrow y) = \downarrow g(y)$ . Then for each  $x \in X$  and  $y \in Y$ ,

$$\begin{aligned} f(x) \leq_X y &\iff f(x) \in \downarrow y \\ &\iff x \in f^{-1}(\downarrow y) \\ &\iff x \in \downarrow g(y) \\ &\iff x \leq_Y g(y). \end{aligned}$$

Thus  $(f, g)$  is a monotone Galois connection. Hence  $f$  has a right adjoint.

□

**Definition 8.1.0.23.** Let  $(X, \leq_X)$  be a poset.

• Let  $U \in \mathcal{U}(\emptyset, \leq_X)$ . We define the  $(\leq_X, \leq_2)$ -**monotonic function induced by  $U$** , denoted  $m_U : X \rightarrow \mathbf{2}$ , by

$$m_U(x) := \begin{cases} 1, & x \in U \\ 0, & x \in U^c. \end{cases}$$

• Let  $f : X \rightarrow \mathbf{2}$ . Suppose that  $f$  is  $(\leq_X, \leq_2)$ -monotonic. We define the  $\leq_X$ -**upper set induced by  $f$** , denoted  $U_f \subset \mathcal{P}(X)$  by  $U_f := f^{-1}(\{1\})$ .

**Exercise 8.1.0.24.** Let  $(X, \leq_X)$  be a poset.

1. Let  $U \in \mathcal{U}(\leq_X)$ . Then  $m_U \in \text{Mtn}((X, \leq_X), (\mathbf{2}, \leq_2))$ .
2. Let  $f : X \rightarrow \mathbf{2}$ . Suppose that  $f$  is  $(\leq_X, \leq_2)$ -monotonic. Then  $U_f \in \mathcal{U}(\leq_X)$ .

*Proof.*

1. Let  $x, y \in X$ . Suppose that  $x \leq_X y$ .
  - Suppose that  $x \in U$ . Since  $U \in \mathcal{U}(\leq_X)$  and  $x \leq_X y$ , we have that  $y \in U$ . Thus

$$\begin{aligned} m_U(x) &= 1 \\ &= m_U(y). \end{aligned}$$

Hence  $m_U(x) \leq_2 m_U(y)$ .

- Suppose that  $x \notin U$ . Then

$$\begin{aligned} m_U(x) &= 0 \\ &\leq m_U(y). \end{aligned}$$

Since  $x, y \in X$  with  $x \leq_X y$  are arbitrary, we have that for each  $x, y \in X$ ,  $x \leq_X y$  implies that  $m_U(x) \leq_2 m_U(y)$ . Thus  $m_U$  is monotone.

2. Let  $x \in U_f$  and  $y \in X$ . Suppose that  $x \leq_X y$ . Since  $f$  is  $(\leq_X, \leq_2)$ -monotonic, we have that

$$\begin{aligned} 1 &= f(x) \\ &\leq_2 f(y) \\ &\leq_2 1. \end{aligned}$$

Thus  $f(y) = 1$  and

$$\begin{aligned} y &\in f^{-1}(\{1\}) \\ &= U_f. \end{aligned}$$

Since  $x \in U_f$  and  $y \in X$  with  $x \leq_X y$  are arbitrary, we have that for each  $x \in U_f$  and  $y \in X$ ,  $x \leq_X y$  implies that  $y \in U_f$ . Therefore  $U_f \in \mathcal{U}(\leq_X)$ . □

**Definition 8.1.0.25.** Let  $X$  be a poset. We define the **monotone-to-upper-set map**, denoted  $\Phi : \text{Mtn}(X, \mathbf{2}) \rightarrow \mathcal{U}_X$ , by  $\Phi(f) := U_f$ .

**Exercise 8.1.0.26.** Then

1.  $\Phi \in \text{Mtn}(\text{Mtn}(X, \mathbf{2}), \mathcal{U}_X)$ .
2.  $\Phi$  is an isomorphism.

*Proof.*

1. Let  $f, g \in \text{Mtn}(X, \mathbf{2})$ . Suppose that  $f \leq g$ . Let  $x \in U_f$ . Then  $f(x) = 1$ . Since  $f \leq g$ , we have that

$$\begin{aligned} 1 &= f(x) \\ &\leq g(x) \\ &\leq 1. \end{aligned}$$

Hence  $g(x) = 1$  and  $x \in U_g$ . Since  $x \in U_f$  is arbitrary, we have that for each  $x \in U_f$ ,  $x \in U_g$ . Thus

$$\begin{aligned} \Phi(f) &= U_f \\ &\subset U_g \\ &= \Phi(g). \end{aligned}$$

Since  $f, g \in \text{Mtn}(X, \mathbf{2})$  with  $f \leq g$  are arbitrary, we have that for each  $f, g \in \text{Mtn}(X, \mathbf{2})$ ,  $f \leq g$  implies that  $\Phi(f) \subset \Phi(g)$ . Hence  $\Phi \in \text{Mtn}(\text{Mtn}(X, \mathbf{2}), \mathcal{U}_X)$ .

2. • – **injectivity:**

Let  $f, g \in \text{Mtn}(X, \mathbf{2})$ . Suppose that  $\Phi(f) = \Phi(g)$ . Then  $U_f = U_g$  and therefore for each  $x \in X$ ,

- \*  $x \in U_f$  implies that  $x \in U_g$  and  $f(x) = 1 = g(x)$ .
- \*  $x \in U_f^c$  implies that  $x \in U_g^c$  and  $f(x) = 0 = g(x)$ .

Thus for each  $x \in X$ ,  $f(x) = g(x)$ . Hence  $f = g$ . Since  $f, g \in \text{Mtn}(X, \mathbf{2})$  with  $\Phi(f) = \Phi(g)$  are arbitrary, we have that for each  $f, g \in \text{Mtn}(X, \mathbf{2})$ ,  $\Phi(f) = \Phi(g)$  implies that  $f = g$ . Thus  $\Phi$  is injective.

- **surjectivity:**

Let  $U \in \mathcal{U}_X$ . Define  $f \in \text{Mtn}(X, \mathbf{2})$  by  $f := m_U$ . Then

$$\begin{aligned} \Phi(f) &= U_f \\ &= f^{-1}(\{1\}) \\ &= m_U^{-1}(\{1\}) \\ &= U \quad (\text{by Definition 8.1.0.23}). \end{aligned}$$

Since  $U \in \mathcal{U}_X$  is arbitrary, we have that for each  $U \in \mathcal{U}_X$ , there exists  $f \in \text{Mtn}(X, \mathbf{2})$  such that  $\Phi(f) = U$ . Thus  $\Phi$  is surjective.

Therefore  $\Phi$  is a bijection.

- Clearly for each  $U \in \mathcal{U}_X$ ,  $\Phi^{-1}(U) = m_U$ . Let  $U_1, U_2 \in \mathcal{U}_X$ . Suppose that  $U_1 \subset U_2$ . Then for each  $x \in X$ ,
  - $x \in U_2$  implies that

$$\begin{aligned}\Phi^{-1}(U_1)(x) &= m_{U_1}(x) \\ &\leq 1 \\ &= m_{U_2}(x) \\ &= \Phi^{-1}(U_2)(x).\end{aligned}$$

- $x \in U_2^c$  implies that  $x \in U_1^c$  and

$$\begin{aligned}\Phi^{-1}(U_1)(x) &= m_{U_1}(x) \\ &= 0 \\ &= m_{U_2}(x) \\ &= \Phi^{-1}(U_2)(x).\end{aligned}$$

Thus  $\Phi^{-1}(U_1) \leq \Phi^{-1}(U_2)$ . Since  $U_1, U_2 \in \mathcal{U}_X$  with  $U_1 \subset U_2$  are arbitrary, we have that for each  $U_1, U_2 \in \mathcal{U}_X$ ,  $U_1 \subset U_2$  implies that  $\Phi^{-1}(U_1) \leq \Phi^{-1}(U_2)$ . Therefore  $\Phi^{-1} \in \text{Mtn}(\mathcal{U}_X, \text{Mtn}(X, \mathbf{2}))$ .

Thus  $\Phi$  is an isomorphism. □

need to do similar for lower sets  $\mathcal{L}_X$  and antitone? morphisms

## 8.2 Maximal Elements

RELABEL, MAYBE RETITLE

**Definition 8.2.0.1.** Let  $(X, \leq)$  be a proset and  $a \in X$ . Then  $a$  is said to be

- **$\leq$ -maximal** if for each  $x \in X$ ,  $a \leq x$  implies that  $x \sim_{\leq} a$ .
- **$\leq$ -minimal** if for each  $x \in X$ ,  $x \leq a$  implies that  $x \sim_{\leq} a$ .

**Note 8.2.0.2.** When the context is clear, we write “maximal” and “minimal” instead of “ $\leq$ -maximal” and “ $\leq$ -minimal” respectively.

**Exercise 8.2.0.3.** Let  $(X, \leq)$  be a proset and  $a \in X$ . Then

1.  $a$  is maximal iff  $\uparrow a = \pi_{X/\sim_{\leq}}(a)$
2.  $a$  is minimal iff  $\downarrow a = \pi_{X/\sim_{\leq}}(a)$

*Proof.*

1. • ( $\implies$ ):

Suppose that  $a$  is maximal.

- Let  $x \in \uparrow a$ . Exercise 8.1.0.11 implies  $a \leq x$ . Since  $a$  is maximal,  $x \sim_{\leq} a$ . Thus  $x \in \pi_{X/\sim_{\leq}}(a)$ . Since  $x \in \uparrow a$  is arbitrary, we have that for each  $x \in \uparrow a$ ,  $x \in \pi_{X/\sim_{\leq}}(a)$ . Hence  $\uparrow a \subset \pi_{X/\sim_{\leq}}(a)$ .
- Let  $x \in \pi_{X/\sim_{\leq}}(a)$ . Then  $a \sim_{\leq} x$ . Hence  $a \leq x$  and  $x \leq a$ . Since  $a \leq x$ , we have that  $x \in \uparrow a$ . Since  $x \in \pi_{X/\sim_{\leq}}(a)$  is arbitrary, we have that for each  $x \in \pi_{X/\sim_{\leq}}(a)$ ,  $x \in \uparrow a$ . Hence  $\pi_{X/\sim_{\leq}}(a) \subset \uparrow a$ .

Since  $\uparrow a \subset \pi_{X/\sim_{\leq}}(a)$  and  $\pi_{X/\sim_{\leq}}(a) \subset \uparrow a$ , we have that  $\uparrow a = \pi_{X/\sim_{\leq}}(a)$ .

- ( $\impliedby$ ):

Suppose that  $\uparrow a = \pi_{X/\sim_{\leq}}(a)$ . Let  $x \in X$ . Suppose that  $a \leq x$ . Then

$$\begin{aligned}x &\in \uparrow a \\ &= \pi_{X/\sim_{\leq}}(a).\end{aligned}$$

Thus  $x \sim_{\leq} a$ . Since  $x \in X$  with  $a \leq x$  is arbitrary, we have that for each  $x \in X$ ,  $a \leq x$  implies that  $x \sim_{\leq} a$ . Hence  $a$  is maximal.



2. Similar to (1).

□

**Exercise 8.2.0.4.** Let  $(X, \leq)$  be a poset and  $a \in X$ . Then

1.  $a$  is maximal iff  $\uparrow a = \{a\}$
2.  $a$  is minimal iff  $\downarrow a = \{a\}$

*Proof.* Since  $(X, \leq)$  is a poset,  $\pi_{\sim_{\leq}}(a) = \{a\}$ . Exercise 8.2.0.3 then implies the result.

□



# Chapter 9

## Suprema and Infima

### 9.1 Upper and Lower Bounds

**Definition 9.1.0.1.** maybe move to suprema infima section, or make a section on upper and lower bounds Let  $(X, \leq)$  be a proset and  $A \subset X$ .

- Let  $x \in X$ . Then  $x$  is said to be a
  - **$\leq$ -upper bound of  $A$**  if for each  $a \in A$ ,  $a \leq x$
  - **$\leq$ -lower bound of  $A$**  if for each  $a \in A$ ,  $x \leq a$
- We define
  - $\text{ub}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-upper bound of } A\}$
  - $\text{lb}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-lower bound of } A\}$
- Then  $A$  is said to be
  - **$\leq$ -bounded above** if  $\text{ub}(A, \leq) \neq \emptyset$
  - **$\leq$ -bounded below** if  $\text{lb}(A, \leq) \neq \emptyset$

**Note 9.1.0.2.** When the context is clear, we write

- “upper bound” and “lower bound” instead of “ $\leq$ -upper bound” and “ $\leq$ -lower bound” respectively
- $\text{ub } A$  and  $\text{lb } A$  in place of  $\text{ub}(A, \leq)$  and  $\text{lb}(A, \leq)$  respectively
- “bounded above” and “bounded below” instead of “ $\leq$ -bounded above” and “ $\leq$ -bounded below” respectively

**Exercise 9.1.0.3.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1.  $\text{ub}(A, \leq) = \text{lb}(A, \leq^*)$
2.  $\text{ub}(A, \leq^*) = \text{lb}(A, \leq)$

*Proof.*

1. We note that for each  $x \in X$ ,

$$\begin{aligned} x \in \text{ub}(A, \leq) &\iff \text{for each } a \in A, a \leq x \\ &\iff \text{for each } a \in A, x \leq^* a \\ &\iff x \in \text{lb}(A, \leq^*). \end{aligned}$$

Hence  $\text{ub}(A, \leq) = \text{lb}(A, \leq^*)$ .

2. Similar to (1) **ref**.

□

**Exercise 9.1.0.4.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1.  $\text{ub } A = \bigcap_{x \in A} \uparrow x$
2.  $\text{lb } A = \bigcap_{x \in A} \downarrow x$

*Proof.*

1.
  - Let  $a \in \text{ub } A$  and  $x \in A$ . Then  $x \leq a$ . Hence  $a \in \uparrow x$ . Since  $x \in A$  is arbitrary, we have that for each  $x \in A$ ,  $a \in \uparrow x$ . Thus  $a \in \bigcap_{x \in A} \uparrow x$ . Since  $a \in \text{ub } A$  is arbitrary, we have that for each  $a \in \text{ub } A$ ,  $a \in \bigcap_{x \in A} \uparrow x$ . Hence  $\text{ub } A \subset \bigcap_{x \in A} \uparrow x$ .
  - Let  $a \in \bigcap_{x \in A} \uparrow x$  and  $x_0 \in A$ . Then  $a \in \uparrow x_0$ . Hence  $x_0 \leq a$ . Since  $x_0 \in A$  is arbitrary, we have that for each  $x_0 \in A$ ,  $x_0 \leq a$ . Hence  $a \in \text{ub } A$ . Since  $a \in \bigcap_{x \in A} \uparrow x$  is arbitrary, we have that for each  $a \in \bigcap_{x \in A} \uparrow x$ ,  $a \in \text{ub } A$ . Thus  $\bigcap_{x \in A} \uparrow x \subset \text{ub } A$ .

Since  $\text{ub } A \subset \bigcap_{x \in A} \uparrow x$  and  $\bigcap_{x \in A} \uparrow x \subset \text{ub } A$ , we have that  $\text{ub } A = \bigcap_{x \in A} \uparrow x$ .

2. Similar to (1).

□

**Exercise 9.1.0.5.** Let  $(X, \leq)$  be a proset and  $A, B \subset X$ . If  $A \subset B$ , then

1.  $\text{ub } B \subset \text{ub } A$
2.  $\text{lb } B \subset \text{lb } A$

*Proof.* Suppose that  $A \subset B$ .

1. Exercise 9.1.0.4 implies that

$$\begin{aligned} \text{ub } B &= \bigcap_{x \in B} \uparrow x \\ &\subset \bigcap_{x \in A} \uparrow x \\ &= \text{ub } A. \end{aligned}$$

2. Similar to (1). **ref**

□

**Exercise 9.1.0.6.** Let  $(X, \leq)$  be a proset and  $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$ . Then  $\text{ub } \bigcup_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \text{ub } E_\alpha$ .

*Proof.* Define  $E_\vee \subset X$  by  $E_\vee := \bigcup_{\alpha \in A} E_\alpha$ . Exercise 9.1.0.4 then implies that

$$\begin{aligned} \text{ub } \bigcup_{\alpha \in A} E_\alpha &= \bigcap_{x \in E_\vee} \uparrow x \\ &= \bigcap_{\alpha \in A} \left[ \bigcap_{x \in E_\alpha} \uparrow x \right] \\ &= \bigcap_{\alpha \in A} \text{ub } E_\alpha. \end{aligned}$$

□

**Exercise 9.1.0.7.** Let  $(X, \leq)$  be a proset. For each  $A \subset X$ ,

1.  $A \subset \text{lb}(\text{ub } A)$
2.  $A \subset \text{ub}(\text{lb } A)$

*Proof.*

1. Let  $a \in A$ . Then for each  $x \in \text{ub } A$ ,  $a \leq x$ . Hence  $a \in \text{lb}(\text{ub } A)$ . Since  $a \in A$  is arbitrary, we have that for each  $a \in A$ ,  $a \in \text{lb}(\text{ub } A)$ . Thus  $A \subset \text{lb}(\text{ub } A)$ .
2. Let  $a \in A$ . Then for each  $x \in \text{lb } A$ ,  $x \leq a$ . Hence  $a \in \text{ub}(\text{lb } A)$ . Since  $a \in A$  is arbitrary, we have that for each  $a \in A$ ,  $a \in \text{ub}(\text{lb } A)$ . Thus  $A \subset \text{ub}(\text{lb } A)$ .

□

**Exercise 9.1.0.8.** Let  $(X, \leq)$  be a proset and  $A, B \subset X$ . Suppose that  $B \subset A$ . Then

1.  $\text{ub}(B, \leq|_A) = \text{ub}(B, \leq) \cap A$ .
2.  $\text{lb}(B, \leq|_A) = \text{lb}(B, \leq) \cap A$ .

*Proof.*

1. We note that for each  $x \in X$ ,

$$\begin{aligned} x \in \text{ub}(B, \leq|_A) &\iff x \in A \text{ and for each } y \in B, y \leq|_A x \\ &\iff x \in A \text{ and for each } y \in B, y \leq x \\ &\iff x \in \text{ub}(B, \leq) \cap A. \end{aligned}$$

2. We have that

$$\begin{aligned} \text{lb}(B, \leq|_A) &= \text{ub}(B, \leq|_A^*) \quad (\text{by Exercise 9.1.0.3}) \\ &= \text{ub}(B, \leq^*|_A) \quad (\text{by ref ex in set theory notes in binary hetero relations}) \\ &= \text{ub}(B, \leq^*) \cap A \quad (\text{by ref prev part}) \\ &= \text{lb}(B, \leq) \cap A \quad (\text{by Exercise 9.1.0.3}). \end{aligned}$$

□

**Exercise 9.1.0.9.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be prosets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ .

1. If  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, then
  - (a) for each  $A \subset X$ ,  $\text{ub}(f(A), \leq_Y) = g^{-1}(\text{ub}(A, \leq_X))$ ,
  - (b) for each  $B \subset Y$ ,  $\text{lb}(g(B), \leq_X) = f^{-1}(\text{lb}(B, \leq_Y))$ .
2. If  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection, then
  - (a) for each  $A \subset X$ ,  $\text{ub}(f(A), \leq_Y) = g^{-1}(\text{lb}(A, \leq_X))$ ,
  - (b) for each  $B \subset Y$ ,  $\text{ub}(g(B), \leq_X) = f^{-1}(\text{lb}(B, \leq_Y))$ .

*Proof.* Since  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, a prev ex implies that  $f$  is  $(\leq_X, \leq_Y)$ -monotonoe,  $g$  is  $(\leq_Y, \leq_X)$ -monotone,  $\text{id}_X \leq_X g \circ f$  and  $f \circ g \leq_Y \text{id}_Y$ .

1. Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.
  - (a) Let  $A \subset X$ .

- Let  $y \in \text{ub}(f(A), \leq_Y)$  and  $a \in A$ . Then  $f(a) \leq_Y y$ . Hence

$$\begin{aligned} a &\leq_X g \circ f(a) \\ &= g(f(a)) \\ &\leq_X g(y). \end{aligned}$$

Since  $a \in A$  is arbitrary, we have that for each  $a \in A$ ,  $a \leq_X g(y)$ . Thus  $g(y) \in \text{ub}(A, \leq_X)$ . Hence  $y \in g^{-1}(\text{ub}(A, \leq_X))$ . Since  $y \in \text{ub}(f(A), \leq_Y)$  is arbitrary, we have that for each  $y \in \text{ub}(f(A), \leq_Y)$ ,  $y \in g^{-1}(\text{ub}(A, \leq_X))$ . Thus  $\text{ub}(f(A), \leq_Y) \subset g^{-1}(\text{ub}(A, \leq_X))$ .

- Let  $y \in g^{-1}(\text{ub}(A, \leq_X))$ . Then  $g(y) \in \text{ub}(A, \leq_X)$ . Let  $b \in f(A)$ . Then there exists  $a \in A$  such that  $b = f(a)$ . Then  $a \leq_X g(y)$ . Therefore

$$\begin{aligned} b &= f(a) \\ &\leq_Y f(g(y)) \\ &= f \circ g(y) \\ &= \leq_Y y. \end{aligned}$$

Since  $b \in f(A)$  is arbitrary, we have that for each  $b \in f(A)$ ,  $b \leq_Y y$ . Thus  $y \in \text{ub}(f(A), \leq_Y)$ . Since  $y \in g^{-1}(\text{ub}(A, \leq_X))$  is arbitrary, we have that for each  $y \in g^{-1}(\text{ub}(A, \leq_X))$ ,  $y \in \text{ub}(f(A), \leq_Y)$ . Hence  $g^{-1}(\text{ub}(A, \leq_X)) \subset \text{ub}(f(A), \leq_Y)$ .

Since  $\text{ub}(f(A), \leq_Y) \subset g^{-1}(\text{ub}(A, \leq_X))$  and  $g^{-1}(\text{ub}(A, \leq_X)) \subset \text{ub}(f(A), \leq_Y)$ , we have that

$$\text{ub}(f(A), \leq_Y) = g^{-1}(\text{ub}(A, \leq_X)).$$

- (b) Since  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, [a prev rex](#) implies that  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection. [The previous part \(ref\)](#) and Exercise 9.1.0.3 then imply that for each  $B \subset Y$ ,

$$\begin{aligned} \text{lb}(g(B), \leq_X) &= \text{ub}(g(B), \leq_X^*) \\ &= f^{-1}(\text{ub}(B, \leq_Y^*)) \\ &= f^{-1}(\text{lb}(B, \leq_Y)). \end{aligned}$$

2. Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -antitone Galois connection. [A prev ex](#) then implies that  $(f, g)$  is a  $(\leq_X^*, \leq_Y)$ -monotone Galois connection and  $(g, f)$  is a  $(\leq_Y^*, \leq_X)$ -monotone Galois connection. [The prev part](#) implies that

- (a) for each  $A \subset X$ ,

$$\begin{aligned} \text{ub}(f(A), \leq_Y) &= g^{-1}(\text{ub}(A, \leq_X^*)) \\ &= g^{-1}(\text{lb}(A, \leq_X)). \end{aligned}$$

- (b) for each  $B \subset Y$ ,

$$\begin{aligned} \text{ub}(g(B), \leq_X) &= f^{-1}(\text{ub}(B, \leq_Y^*)) \\ &= f^{-1}(\text{lb}(B, \leq_Y)). \end{aligned}$$

□

**Definition 9.1.0.10.** Let  $(X, \leq)$  be a proset. We define the

- **$\leq$ -upper bound map**, denoted  $\text{ub}(\cdot, \leq) : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ , by  $\text{ub}(\cdot, \leq)(A) := \text{ub}(A, \leq)$ .
- **$\leq$ -lower bound map**, denoted  $\text{lb}(\cdot, \leq) : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ , by  $\text{lb}(\cdot, \leq)(A) := \text{lb}(A, \leq)$ .

**Note 9.1.0.11.** When the context is clear, we may write  $\text{ub} : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  and  $\text{lb} : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  in place of  $\text{ub}(\cdot, \leq) : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  and  $\text{lb}(\cdot, \leq) : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ .

**Exercise 9.1.0.12.** Let  $(X, \leq)$  be a proset. Then  $(\text{ub}(\cdot, \leq), \text{lb}(\cdot, \leq))$  is a  $(\subset^*, \subset^*)$ -antitone Galois connection.

*Proof.* Define  $f, g : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  by  $f := \text{ub}(\cdot, \leq)$  and  $g := \text{lb}(\cdot, \leq)$ .

- Exercise 9.1.0.7 then implies that for each  $A \in \mathcal{P}(X)$ ,  $\text{id}_{\mathcal{P}(X)}(A) \subset g \circ f(A)$  and  $\text{id}_{\mathcal{P}(X)}(A) \subset f \circ g(A)$ . Therefore for each  $A \in \mathcal{P}(X)$ ,  $g \circ f(A) \subset^* \text{id}_{\mathcal{P}(X)}(A)$  and  $f \circ g(A) \subset^* \text{id}_{\mathcal{P}(X)}(A)$ .
- Exercise 9.1.0.5 implies that  $f, g$  are  $(\subset, \subset)$ -antitone. Exercise 7.1.0.3 implies that  $f, g$  are  $(\subset^*, \subset^*)$ -antitone.

Exercise 7.2.0.6 then implies that  $(f, g)$  is a  $(\subset^*, \subset^*)$ -antitone Galois connection. □

**Exercise 9.1.0.13.** Let  $(X, \leq_X)$   $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $A \subset X$ . If  $f$  is  $(\leq_X, \leq_Y)$ -monotone, then

1.  $f(\text{ub}(A, \leq_X)) \subset \text{ub}(f(A), \leq_Y)$ ,
2.  $f(\text{lb}(A, \leq_X)) \subset \text{lb}(f(A), \leq_Y)$

*Proof.* Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -monotone.

1. Let  $x \in \text{ub}(A, \leq_X)$  and  $y_0 \in f(A)$ . Then there exists  $x_0 \in A$  such that  $f(x_0) = y_0$ . Since  $x \in \text{ub}(A, \leq_X)$ ,  $x_0 \leq x$ . Since  $f$  is  $(\leq_X, \leq_Y)$ -monotone,

$$\begin{aligned} y_0 &= f(x_0) \\ &\leq f(x). \end{aligned}$$

Since  $y_0 \in f(A)$  is arbitrary, we have that for each  $y_0 \in f(A)$ ,  $y_0 \leq f(x)$ . Therefore  $f(x) \in \text{ub}(f(A), \leq_Y)$ . Since  $x \in \text{ub}(A, \leq_X)$  is arbitrary, we have that for each  $x \in \text{ub}(A, \leq_X)$ ,  $f(x) \in \text{ub}(f(A), \leq_Y)$ . Hence  $f(\text{ub}(A, \leq_X)) \subset \text{ub}(f(A), \leq_Y)$ .

2. Exercise 9.1.0.3 and the prev part imply that

$$\begin{aligned} f(\text{lb}(A, \leq_X)) &= f(\text{ub}(A, \leq_X^*)) \\ &\subset \text{ub}(f(A), \leq_Y^*) \\ &= \text{lb}(f(A), \leq_Y). \end{aligned}$$

□

## 9.2 SupSets and InfSets

**Definition 9.2.0.1.** Let  $(X, \leq)$  be a proset and  $A \subset X$ .

- Let  $x \in X$ . Then  $x$  is said to be a
  - **$\leq$ -supremum of  $A$  or  $\leq$ -least upper bound of  $A$** , if
    1.  $x \in \text{ub}(A, \leq)$
    2. for each  $y \in \text{ub}(A, \leq)$ ,  $x \leq y$
  - **$\leq$ -infimum of  $A$  or  $\leq$ -greatest lower bound of  $A$** , if
    1.  $x \in \text{lb}(A, \leq)$
    2. for each  $y \in \text{lb}(A, \leq)$ ,  $y \leq x$ .
- – We define  $\text{supSet}(A, \leq) \subset X$  by  $\text{supSet}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-supremum of } A\}$
- We define  $\text{infSet}(A, \leq) \subset X$  by  $\text{infSet}(A, \leq) := \{x \in X : x \text{ is an } \leq\text{-infimum of } A\}$

**Note 9.2.0.2.** When the context is clear,

- we say “supremum” and “infimum” in place of “ $\leq$ -supremum” and “ $\leq$ -infimum” respectively,
- we write  $\text{supSet } A$  and  $\text{infSet } A$  in place of  $\text{supSet}(A, \leq)$  and  $\text{infSet}(A, \leq)$  respectively.

**Exercise 9.2.0.3.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1. (a)  $\text{supSet } A = \text{ub } A \cap \text{lb}(\text{ub } A)$   
 (b)  $\text{supSet } A = \text{infSet}(\text{ub } A)$
2. (a)  $\text{infSet } A = \text{lb } A \cap \text{ub}(\text{lb } A)$   
 (b)  $\text{infSet } A = \text{supSet}(\text{lb } A)$

*Proof.*

1. (a) We note that for each  $x \in X$ ,

$$\begin{aligned}
 x \in \text{supSet } A &\iff x \in \text{ub } A \text{ and for each } y \in \text{ub } A, x \leq y \\
 &\iff x \in \text{ub } A \text{ and } x \in \text{lb}(\text{ub } A) \\
 &\iff x \in \text{ub } A \cap \text{lb}(\text{ub } A).
 \end{aligned}$$

Therefore  $\text{supSet } A = \text{ub } A \cap \text{lb}(\text{ub } A)$ .

- (b) Exercise 9.1.0.12 implies that  $(\text{ub}(\cdot, \leq), \text{lb}(\cdot, \leq))$  is a  $(\subset^*, \subset^*)$ -antitone Galois connection. Since  $(\mathcal{P}(X), \subset)$  is a poset Exercise 7.2.0.8 implies that  $\text{ub}(\text{lb}(\text{ub } A)) = \text{ub } A$ . Therefore

$$\begin{aligned}
 \text{supSet } A &= \text{ub } A \cap \text{lb}(\text{ub } A) \quad (\text{by prev part (ref)}) \\
 &= \text{ub}(\text{lb}(\text{ub } A)) \cap \text{lb}(\text{ub } A) \quad (\text{by prev part (ref)}) \\
 &= \text{lb}(\text{ub } A) \cap \text{ub}(\text{lb}(\text{ub } A)) \\
 &= \text{infSet}(\text{ub } A).
 \end{aligned}$$

2. Similar to (1) **USE DUALITY???**.

□

**Exercise 9.2.0.4.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1.  $\text{supSet}(A, \leq^*) = \text{infSet}(A, \leq)$
2.  $\text{supSet}(A, \leq) = \text{infSet}(A, \leq^*)$



*Proof.*

1. We have that

$$\begin{aligned}\text{supSet}(A, \leq^*) &= \text{ub}(A, \leq^*) \cap \text{lb}(\text{ub}(A, \leq^*), \leq^*) \quad (\text{by Exercise 9.2.0.3}) \\ &= \text{lb}(A, \leq) \cap \text{ub}(\text{lb}(A, \leq), \leq) \quad (\text{by Exercise 9.1.0.3}) \\ &= \text{infSet}(A, \leq) \quad (\text{by Exercise 9.2.0.3}).\end{aligned}$$

2. We have that

$$\begin{aligned}\text{supSet}(A, \leq) &= \text{supSet}(A, (\leq^*)^*) \\ &= \text{infSet}(A, \leq^*) \quad (\text{by prev part (ref prev part)}).\end{aligned}$$

Exercise 9.2.0.3 □

**Exercise 9.2.0.5.** Let  $(X, \leq)$  be a proset and  $A \subset X$ . Then

1. for each  $x, y \in \text{supSet}(A, \leq)$ ,  $x \sim_{\leq} y$ .
2. for each  $x, y \in \text{infSet}(A, \leq)$ ,  $x \sim_{\leq} y$ .

*Proof.*

1. Let  $x, y \in \text{supSet}(A, \leq)$ . Then  $x, y \in \text{ub}(A, \leq)$ . Since  $x \in \text{supSet}(A, \leq)$  and  $y \in \text{ub}(A, \leq)$ ,  $x \leq y$ . Since  $y \in \text{supSet}(A, \leq)$  and  $x \in \text{ub}(A, \leq)$ ,  $y \leq x$ . Hence  $x \sim_{\leq} y$ . Since  $x, y \in \text{supSet}(A, \leq)$  are arbitrary, we have that for each  $x, y \in \text{supSet}(A, \leq)$ ,  $x \sim_{\leq} y$ .
2. We note that for each  $x, y \in X$ ,

$$\begin{aligned}x, y \in \text{infSet}(A, \leq) &\iff x, y \in \text{supSet}(A, \leq^*) \quad (\text{by Exercise 9.2.0.4}) \\ &\implies x \sim_{\leq^*} y \quad (\text{by prev part (ref)}) \\ &\iff x \sim_{\leq} y \quad (\text{by Exercise 4.1.0.6}).\end{aligned}$$

□

**Exercise 9.2.0.6.** Let  $(X, \leq)$  be a poset and  $A \subset X$ . Then

1. for each  $x, y \in \text{supSet } A$ ,  $x = y$ .
2. for each  $x, y \in \text{infSet } A$ ,  $x = y$ .

*Proof.*

1. Let  $x, y \in \text{supSet } A$ . Exercise 9.2.0.5 implies that  $x \sim_{\leq} y$ . Exercise 5.1.0.2 then implies that  $x = y$ .
2. Similar to (1).

□

**Exercise 9.2.0.7.** Let  $(X, \leq)$  be a proset and  $A, B \subset X$ . If  $A \subset B$ , then

1. for each  $s_A \in \text{supSet } A$  and  $s_B \in \text{supSet } B$ ,  $s_A \leq s_B$ ,
2. for each  $l_A \in \text{infSet } A$  and  $l_B \in \text{infSet } B$ ,  $l_B \leq l_A$ .

*Proof.* Suppose that  $A \subset B$ .

1. Let  $s_A \in \text{supSet } A$  and  $s_B \in \text{supSet } B$ . Then Exercise 9.1.0.5 implies that

$$\begin{aligned} s_B &\in \text{supSet } B \\ &\subset \text{ub } B \\ &\subset \text{ub } A. \end{aligned}$$

Since  $s_A \in \text{lb}(\text{ub } A)$ , we have that  $s_A \leq s_B$ . Since  $s_A \in \text{supSet } A$  and  $s_B \in \text{supSet } B$  are arbitrary, we have that for each  $s_A \in \text{supSet } A$  and  $s_B \in \text{supSet } B$ ,  $s_A \leq s_B$ .

2. **similar, use duality**

□

**Exercise 9.2.0.8.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$ ,  $g : Y \rightarrow X$  and  $A \subset X$ . If  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, then

1.  $f(\text{supSet}(A, \leq_X)) \subset \text{supSet}(f(A), \leq_Y)$ .
2. **can we say something about  $\text{infSet}(f(A), \leq_Y)$  or  $\text{infSet}(g(B), \leq_X)$ ?**

*Proof.* Suppose that  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection.

1. **A prev ex in Galois connection subsection** implies that  $f$  is  $(\leq_X, \leq_Y)$ -monotone,  $g$  is  $(\leq_Y, \leq_X)$ -monotone,  $g \circ f \leq_X \text{id}_X$  and  $\text{id}_Y \leq_Y f \circ g$ . Let  $s \in \text{supSet}(A, \leq_X)$ . Since  $\text{supSet}(A, \leq_X) \subset \text{ub}(A, \leq_X)$ , we have that  $s \in \text{ub}(A, \leq_X)$ . Exercise 9.1.0.13 then implies that

$$\begin{aligned} f(s) &\in f(\text{ub}(A, \leq_X)) \\ &\subset \text{ub}(f(A), \leq_Y). \end{aligned}$$

Let  $y \in \text{ub}(f(A), \leq_Y)$ . **A prev ex in galois connection subsection** imply that

$$\begin{aligned} y &\in \text{ub}(f(A), \leq_Y) \\ &= g^{-1}(\text{ub}(A, \leq_X)). \end{aligned}$$

Then  $g(y) \in \text{ub}(A, \leq_X)$ . Since  $s \in \text{supSet}(A, \leq_X)$ , we have that  $s \leq_X g(y)$ . Hence

$$\begin{aligned} f(s) &\leq_Y f(g(y)) \\ &= f \circ g(y) \\ &\leq_Y y. \end{aligned}$$

Since  $y \in \text{ub}(f(A), \leq_Y)$  is arbitrary, we have that for each  $y \in \text{ub}(f(A), \leq_Y)$ ,  $f(s) \leq_Y y$ . Thus  $f(s) \in \text{supSet}(f(A), \leq_Y)$ . Since  $s \in \text{supSet}(A, \leq_X)$  is arbitrary, we have that for each  $s \in \text{supSet}(A, \leq_X)$ ,  $f(s) \in \text{supSet}(f(A), \leq_Y)$ . Hence  $f(\text{supSet}(A, \leq_X)) \subset \text{supSet}(f(A), \leq_Y)$ .

2. **FINISH!!!**

□

## 9.3 Suprema and Infima

relabel ex/def references

**Definition 9.3.0.1.** Let  $(X, \leq)$  be a poset and  $A \subset X$ .

- If  $\text{supSet } A \neq \emptyset$ , we define the **supremum of**  $A$ , denoted  $\sup A$ , to be the unique  $x \in X$  such that  $x \in \text{supSet } A$ .
- If  $\text{infSet } A \neq \emptyset$ , we define the **infimum of**  $A$ , denoted  $\inf A$ , to be the unique  $x \in X$  such that  $x \in \text{infSet } A$ .
- We say that  $\sup A$  (**resp.**  $\inf A$ ) **exists** if  $\text{supSet } A \neq \emptyset$  (**resp.**  $\text{infSet } A \neq \emptyset$ ).

**Exercise 9.3.0.2.** Let  $(X, \leq)$  be a poset and  $A \subset X$ . Then

- (a)  $\sup A$  exists iff  $\inf(\text{ub } A)$  exists  
(b)  $\sup A$  exists implies that  $\sup A = \inf(\text{ub } A)$
- (a)  $\inf A$  exists iff  $\sup(\text{lb } A)$  exists  
(b)  $\inf A$  exists implies that  $\inf A = \sup(\text{lb } A)$

*Proof.*

- (a) Exercise 9.2.0.3 implies that

$$\begin{aligned} \sup A \text{ exists} &\iff \text{supSet } A \neq \emptyset \\ &\iff \text{infSet}(\text{ub } A) \neq \emptyset \\ &\iff \inf(\text{ub } A) \text{ exists.} \end{aligned}$$

- (b) Suppose that  $\sup A$  exists. **the previous part (a)** then implies that  $\inf(\text{ub } A)$  exists. **a previous ex** implies that  $\text{supSet } A = \{\sup A\}$  and  $\text{infSet}(\text{ub } A) = \{\inf(\text{ub } A)\}$ . **The previous parts (a)** then implies that

$$\begin{aligned} \{\sup A\} &= \text{supSet } A \\ &= \text{infSet}(\text{ub } A) \\ &= \{\inf(\text{ub } A)\}. \end{aligned}$$

Hence  $\sup A = \inf(\text{ub } A)$ .

- Similar to (1).

□

**Exercise 9.3.0.3.** Let  $(X, \leq)$  be a poset and  $A \subset X$ . Then

- (a)  $\sup(A, \leq)$  exists iff  $\inf(A, \leq^*)$  exists  
(b)  $\sup A$  exists implies that  $\sup(A, \leq) = \inf(A, \leq^*)$
- (a)  $\inf(A, \leq)$  exists iff  $\sup(A, \leq^*)$  exists  
(b)  $\inf A$  exists implies that  $\inf(A, \leq) = \sup(A, \leq^*)$

*Proof.*

- (a) We have that

$$\begin{aligned} \sup(A, \leq) \text{ exists} &\iff \text{supSet}(A, \leq) \neq \emptyset \\ &\iff \text{infSet}(A, \leq^*) \neq \emptyset \quad (\text{by Exercise 9.2.0.4}) \\ &\iff \inf(A, \leq^*) \text{ exists.} \end{aligned}$$

- (b) Suppose that  $\sup(A, \leq)$  exists. **The prev part** implies that  $\inf(A, \leq^*)$  exists. Since  $\sup\text{Set}(A, \leq) = \{\sup(A, \leq)\}$  and  $\inf\text{Set}(A, \leq^*) = \{\inf(A, \leq^*)\}$ , Exercise 9.2.0.4 implies that

$$\begin{aligned}\{\sup(A, \leq)\} &= \sup\text{Set}(A, \leq) \\ &= \inf\text{Set}(A, \leq^*) \\ &= \{\inf(A, \leq^*)\}.\end{aligned}$$

Thus  $\sup(A, \leq) = \inf(A, \leq^*)$ .

2. **Similar to (1).**

□

**Exercise 9.3.0.4.** Let  $(X, \leq)$  be a poset and  $A, B \subset X$ . If  $A \subset B$ , then

1.  $\sup A$  and  $\sup B$  exist implies that  $\sup A \leq \sup B$ .
2.  $\inf A$  and  $\inf B$  exist implies that  $\inf B \leq \inf A$ .

*Proof.* 1. Since  $\sup A \in \sup\text{Set } A$  and  $\sup B \in \sup\text{Set } B$ , Exercise 9.2.0.7 implies that  $\sup A \leq \sup B$ .

2. **FINISH!!!**

□

**Exercise 9.3.0.5. Associativity of Supremum and Infima:**

Let  $(X, \leq)$  be a poset and  $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$ .

1. Suppose that

- for each  $\alpha \in A$ ,  $\sup E_\alpha$  exists
- $\sup_{\alpha \in A} \left[ \sup E_\alpha \right]$  exists.

Then  $\sup \bigcup_{\alpha \in A} E_\alpha$  exists and

$$\sup \bigcup_{\alpha \in A} E_\alpha = \sup_{\alpha \in A} \left[ \sup E_\alpha \right]$$

2. Suppose that

- for each  $\alpha \in A$ ,  $\inf E_\alpha$  exists
- $\inf_{\alpha \in A} \left[ \inf E_\alpha \right]$  exists.

Then  $\inf \bigcup_{\alpha \in A} E_\alpha$  exists

$$\inf \bigcup_{\alpha \in A} E_\alpha = \inf_{\alpha \in A} \left[ \inf E_\alpha \right]$$

*Proof.*

1. Define  $s \in X$  by  $s := \sup_{\alpha \in A} \left[ \sup E_\alpha \right]$ .

- Let  $x \in \bigcup_{\alpha \in A} E_\alpha$ . Then there exists  $\alpha_0 \in A$  such that  $x \in E_{\alpha_0}$ . We note by definition,  $s \in \text{ub}\{\sup E_\alpha : \alpha \in A\}$ . Hence for each  $\alpha \in A$ ,  $s \geq \sup E_\alpha$ . Since  $\sup E_{\alpha_0} \in \text{ub } E_{\alpha_0}$ , we have that

$$\begin{aligned}x &\leq \sup E_{\alpha_0} \\ &\leq s.\end{aligned}$$

Since  $x \in \bigcup_{\alpha \in A} E_\alpha$  is arbitrary, we have that for each  $x \in \bigcup_{\alpha \in A} E_\alpha$ ,  $x \leq s$ . Hence  $s \in \text{ub} \bigcup_{\alpha \in A} E_\alpha$ .

- Let  $x \in \text{ub} \bigcup_{\alpha \in A} E_\alpha$  and  $\alpha_0 \in A$ . Since  $E_{\alpha_0} \subset \bigcup_{\alpha \in A} E_\alpha$ , Exercise 9.1.0.5 implies that

$$\begin{aligned} x &\in \text{ub} \bigcup_{\alpha \in A} E_\alpha \\ &\subset \text{ub} E_{\alpha_0}. \end{aligned}$$

Therefore  $\sup E_{\alpha_0} \leq x$ . Since  $\alpha_0 \in A$  is arbitrary, we have that for each  $\alpha_0 \in A$ ,  $\sup E_{\alpha_0} \leq x$ . Hence  $x \in \text{ub}\{\sup E_\alpha : \alpha \in A\}$ . Therefore  $s \leq x$ . Since  $x \in \text{ub} \bigcup_{\alpha \in A} E_\alpha$  is arbitrary, we have that for each  $x \in \text{ub} \bigcup_{\alpha \in A} E_\alpha$ ,  $s \leq x$ .

Therefore  $s \in \text{supSet} \bigcup_{\alpha \in A} E_\alpha$  and  $\sup \bigcup_{\alpha \in A} E_\alpha = s$ .

2. Similar to (1). **Maybe fill out, use duality**

□

**Exercise 9.3.0.6.** Let  $(X, \leq)$  be a poset and  $a, b \in X$ . Then

1. (a)  $\sup\{a\}$  exists and  $\sup\{a\} = a$ ,  
(b)  $a \leq b$  iff  $\sup\{a, b\} = b$ ,
2. (a)  $\inf\{a\}$  exists and  $\inf\{a\} = a$ ,  
(b)  $a \leq b$  iff  $\inf\{a, b\} = a$ ,
3.  $\sup \downarrow a = a$ ,
4.  $\inf \uparrow a = a$ .

*Proof.*

1. (a) Since  $a \leq a$ ,  $a \in \text{ub}\{a\}$ . By definition, for each  $x \in \text{ub}\{a\}$ ,  $a \leq x$ . Hence  $a \in \text{supSet}\{a\}$ . Thus  $\sup\{a\}$  exists and  $a = \sup\{a\}$ .  
(b) • ( $\implies$ ):  
Suppose that  $a \leq b$ . Since  $b \leq b$ , we have that for each  $c \in \{a, b\}$ ,  $c \leq b$ . Hence  $b \in \text{lb}\{a, b\}$ . Let  $c \in \text{ub}\{a, b\}$ . Then  $b \leq c$ . Since  $c \in \text{ub}\{a, b\}$  is arbitrary, we have that for each  $c \in \text{ub}\{a, b\}$ ,  $b \leq c$ . Hence  $b \in \text{supSet}\{a, b\}$  and  $b = \sup\{a, b\}$ .  
• ( $\impliedby$ ):  
Suppose that  $\sup\{a, b\}$  exists and  $b = \sup\{a, b\}$ . Then  $b \in \text{ub}\{a, b\}$ . Therefore  $a \leq b$ .
2. (a) **FINISH!!!**  
(b) **FINISH!!!**
3. Since for each  $x \in \downarrow a$ ,  $x \leq a$ , we have that  $a \in \text{ub} \downarrow a$ . Since  $a \in \downarrow a$ , we have that for each  $x \in \text{ub} \downarrow a$ ,  $a \leq x$ . Hence  $a = \sup \downarrow a$ .
4. Similar to (3).

□

**Exercise 9.3.0.7.** Let  $(X, \leq)$  be a poset. Then for each  $a, b \in X$ ,  $\sup\{a, b\}$  exists iff for each  $n \in \mathbb{N}$  and  $(a_k)_{k=1}^n \in X^n$ ,  $\sup\{a_1, \dots, a_n\}$  exists and

$$\sup\{a_1, \dots, a_n\} = \begin{cases} a_1, & n = 1 \\ \sup\left\{\sup\{a_1, \dots, a_{n-1}\}, a_n\right\}, & n \geq 2. \end{cases}$$

*Proof.*

- ( $\implies$ ):  
Suppose that for each  $a, b \in X$ ,  $\sup\{a, b\}$  exists. Let  $n \in \mathbb{N}$  and  $(a_k)_{k=1}^n \in X^n$ .

– **Base Case:**

Suppose that  $n = 1$ . Exercise 9.3.0.6 implies that  $\sup\{a_1\}$  exists and  $\sup\{a_1\} = a_1$ .

– **Induction Step:**

Suppose that  $n \geq 2$  and  $\sup\{a_1, \dots, a_{n-1}\}$  exists. Then by assumption,  $\sup\left\{\sup\{a_1, \dots, a_{n-1}\}, a_n\right\}$  exists. Exercise 9.3.0.5 then implies that  $\sup\{a_1, \dots, a_n\}$  exists and

$$\begin{aligned}\sup\{a_1, \dots, a_n\} &= \sup\left\{\{a_1, \dots, a_{n-1}\} \cup \{a_n\}\right\} \\ &= \sup\left\{\sup\{a_1, \dots, a_{n-1}\}, \sup\{a_n\}\right\} \\ &= \sup\left\{\sup\{a_1, \dots, a_{n-1}\}, a_n\right\}.\end{aligned}$$

By induction, for each  $n \in \mathbb{N}$  and  $(a_k)_{k=1}^n \in X^n$ ,  $\sup\{a_1, \dots, a_n\}$  exists and

$$\sup\{a_1, \dots, a_n\} = \begin{cases} a_1, & n = 1 \\ \sup\left\{\sup\{a_1, \dots, a_{n-1}\}, a_n\right\}, & n \geq 2. \end{cases}$$

- ( $\Leftarrow$ ):  
Clear.

□

**Exercise 9.3.0.8.** Let  $(X, \leq)$  be a poset. Then

1. for each  $a, b \in \text{ub } X$ ,  $a = b$ ,
2. for each  $a \in \text{ub } X$ ,  $\sup X = a$ ,
3. for each  $a, b \in \text{lb } X$ ,  $a = b$ ,
4. for each  $a \in \text{lb } X$ ,  $\inf X = a$

*Proof.*

1. Let  $a, b \in \text{ub } X$ . Since  $a \in X$  and  $b \in \text{ub } X$ ,  $a \leq b$ . Similarly,  $b \leq a$ . Hence  $a = b$ .
2. Let  $a, b \in \text{ub } X$ . **the previous part** implies that  $a = b$ . Since  $b \in \text{ub } X$  is arbitrary, we have that for each  $b \in \text{ub } X$ ,  $a \leq b$ . Thus  $a = \sup X$ .
3. Similar to (1) **ref.**
4. Similar to (2) **ref.**

□

**Exercise 9.3.0.9.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Suppose that  $(f, g)$  is a monotone Galois connection. Then

1. for each  $A \subset X$ , if  $\sup(A, \leq_X)$  exists, then  $\sup(f(A), \leq_Y)$  exists and  $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$ .
2. for each  $B \subset Y$ , if  $\inf(B, \leq_Y)$  exists, then  $g(\inf(B, \leq_Y)) = \inf(g(B), \leq_X)$ . **FIX!!!**

*Proof.*

1. Suppose that  $\sup(A, \leq_X)$  exists. Then  $\sup\text{Set}(A, \leq_X) \neq \emptyset$ . Exercise 9.2.0.8 then implies that

$$\begin{aligned} \emptyset &\neq f(\sup\text{Set}(A, \leq_X)) \\ &\subset \sup\text{Set}(f(A), \leq_Y). \end{aligned}$$

Hence  $\sup(f(A), \leq_Y)$  exists and  $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$ .

2. ???

□

**Exercise 9.3.0.10.** Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Suppose that  $(f, g)$  is a monotone Galois connection. Then

1. for each  $x \in X$ ,  $f(x) = \inf(g^{-1}(\uparrow(x, \leq_X)), \leq_Y)$
2. for each  $y \in Y$ ,  $g(y) = \sup(f^{-1}(\downarrow(y, \leq_Y)), \leq_X)$

*Proof.*

1. Let  $x \in X$ . We note that for each  $y \in Y$ ,

$$\begin{aligned} y \in g^{-1}(\uparrow(x, \leq_X)) &\iff g(y) \in \uparrow(x, \leq_X) \\ &\iff x \leq_X g(y) \\ &\iff f(x) \leq_Y y \quad (\text{since } (f, g) \text{ is a Galois connection}) \\ &\iff y \in \uparrow(f(x), \leq_Y). \end{aligned}$$

Thus  $g^{-1}(\uparrow(x, \leq_X)) = \uparrow(f(x), \leq_Y)$ . Exercise 9.3.0.6 then implies that

$$\begin{aligned} \inf(g^{-1}(\uparrow(x, \leq_X)), \leq_Y) &= \inf(\uparrow(f(x), \leq_Y), \leq_Y) \\ &= f(x). \end{aligned}$$

Since  $x \in X$  is arbitrary, we have that for each  $x \in X$ ,  $f(x) = \inf(g^{-1}(\uparrow(x, \leq_X)), \leq_Y)$ .

2. Let  $y \in Y$ . Since  $(f, g)$  is a  $(\leq_X, \leq_Y)$ -monotone Galois connection, Exercise ?? ref prev ex in poset galois connection section implies that  $(g, f)$  is a  $(\leq_Y^*, \leq_X^*)$ -monotone Galois connection. The previous part and exercises in previous section on poset upper bounds implies that

$$\begin{aligned} g(y) &= \inf(f^{-1}(\uparrow(y, \leq_Y^*)), \leq_X^*) \\ &= \sup(f^{-1}(\downarrow(y, \leq_Y)), \leq_X). \end{aligned}$$

Since  $y \in Y$  is arbitrary, we have that for each  $y \in Y$ ,  $g(y) = \sup(f^{-1}(\downarrow(y, \leq_Y)), \leq_X)$ .

□

**Definition 9.3.0.11.** Let  $(X, \leq)$  be a poset.

- Then  $(X, \leq)$  is said to
  - \* **have a one** or **have a top** if  $\sup(X, \leq)$  exists,
  - \* **have a zero** or **have a bottom** if  $\inf(X, \leq)$  exists.
- be a
  - \* **poset with one** or **poset with top** if  $(X, \leq)$  has a one,
  - \* **poset with zero** or **poset with bottom** if  $(X, \leq)$  has a zero.
- – If  $(X, \leq)$  has a one, we define  $1_\leq \in X$  by  $1_\leq := \sup(X, \leq)$ .
- If  $(X, \leq)$  has a zero, we define  $0_\leq \in X$  by  $0_\leq := \inf(X, \leq)$ .

**Note 9.3.0.12.**

- We the context is clear, we may write 1 and 0 in place of  $1_{\leq}$  and  $0_{\leq}$  respectively.
- Another common notation for 1 and 0 is  $\top$  and  $\perp$  respectively.

**Exercise 9.3.0.13.** Let  $X$  be a poset. Then

1.  $X$  has a one iff there exists  $a \in X$  such that for each  $x \in X$ ,  $\inf\{x, a\} = x$ .
2.  $X$  has a zero iff there exists  $a \in X$  such that for each  $x \in X$ ,  $\sup\{x, a\} = x$ .

*Proof.*

1.
  - ( $\implies$ ):  
If  $X$  has a one, then there exists  $1 \in X$  such that  $1 = \sup X$ . Since for each  $x \in X$ ,  $x \leq 1$ , Exercise 9.3.0.6 implies that for each  $x \in X$ ,  $\inf\{x, 1\} = x$ .
  - ( $\impliedby$ ):  
Suppose that there exists  $a \in X$  such that for each  $x \in X$ ,  $\inf\{x, a\} = x$ . Exercise 9.3.0.6 implies that for each  $x \in X$ ,  $x \leq a$ . Therefore  $a \in \text{ub}(X)$ . Exercise 9.3.0.8 implies that  $a = \sup X$ . Hence  $X$  has a one.
2. **Similar to prev part FINISH!!!**

□

**Exercise 9.3.0.14.** Let  $(X, \leq)$  be a poset.

1. If  $(X, \leq)$  has a one, then for each  $U \in \mathcal{U}(X)$ ,  $U \neq \emptyset$  implies that  $1 \in U$ .
2. If  $(X, \leq)$  has a zero, then for each  $L \in \mathcal{L}(X)$ ,  $L \neq \emptyset$  implies that  $0 \in L$ .

*Proof.* 1. Suppose that  $(X, \leq)$  has a one. Let  $U \in \mathcal{U}(X)$ . Suppose that  $U \neq \emptyset$ . Then there exists  $x \in X$  such that  $x \in U$ . Since  $x \leq 1$  and  $U \in \mathcal{U}(X)$ , we have that  $1 \in U$ . Since  $U \in \mathcal{U}(X)$  with  $U \neq \emptyset$  is arbitrary, we have that for each  $U \in \mathcal{U}(X)$ ,  $U \neq \emptyset$  implies that  $1 \in U$ .

2. **Use duality**

□



# Chapter 10

## Structures on Ordered Sets

### 10.1 Ideals and Filters

**Definition 10.1.0.1.** Let  $(X, \leq)$  be a poset.

- Let  $F, I \subset X$ . Then
  - $I$  is said to be a  $\leq$ -**ideal** if
    1.  $I \neq \emptyset$ ,
    2.  $I \in \mathcal{L}(\leq)$ ,
    3.  $I$  is  $\leq$ -upward directed,
  - item  $F$  is said to be a  $\leq$ -**filter** if
    1.  $F \neq \emptyset$ ,
    2.  $F \in \mathcal{U}(\leq)$ ,
    3.  $F$  is  $\leq$ -downward directed.
- We define
  - $\mathcal{I}(X, \leq) := \{I \subset L : I \text{ is an } \leq\text{-ideal}\}$
  - $\mathcal{F}(X, \leq) := \{F \subset L : F \text{ is a } \leq\text{-filter}\}$

**Note 10.1.0.2.** When the context is clear,

- we may say “ideal” and “filter” instead of “ $\leq$ -ideal” and “ $\leq$ -filter” respectively,
- we may write  $\mathcal{I}(X)$  and  $\mathcal{F}(X)$  in place of  $\mathcal{I}(X, \leq)$  and  $\mathcal{F}(X, \leq)$  respectively.

**Exercise 10.1.0.3.** Let  $(X, \leq)$  be a poset. Suppose that  $X \neq \emptyset$ .

1. If  $X$  has a one, then  $X \in \mathcal{I}(X)$ ,
2. If  $X$  has a zero, then  $X \in \mathcal{F}(X)$ .

*Proof.*

1.
  - By assumption,  $X \neq \emptyset$ .
  - Exercise 8.1.0.3 implies that  $X \in \mathcal{L}(X, \leq)$ .
  - Since  $X$  has a one, for each  $x, y \in X$ ,  $x \leq 1$  and  $y \leq 1$ . Hence  $X$  is  $\leq$ -upward directed.
2. **FINISH!!! use duality?.**

□

**Exercise 10.1.0.4.** Let  $(X, \leq)$  be a poset. Then for each  $x \in X$ ,

1.  $\downarrow x \in \mathcal{I}(X)$
2.  $\uparrow x \in \mathcal{F}(X)$

*Proof.* Let  $x \in X$ .

1. (a) Since  $x \in \downarrow x$ ,  $\downarrow x \neq \emptyset$ .  
 (b) By Note 8.1.0.8,  $\downarrow x \in \mathcal{L}(X)$  change earlier notation to agree.  
 (c) Exercise 8.1.0.19 implies that  $\downarrow x$  is upward directed.

Thus  $\downarrow x \in \mathcal{I}(X)$

2. use duality

□

**Exercise 10.1.0.5.** Let  $(X, \leq)$  be a poset and  $I, F \subset X$ . Then

1.  $I \in \mathcal{I}(X, \leq)$  iff  $I \in \mathcal{F}(X, \leq^*)$ ,
2.  $F \in \mathcal{F}(X, \leq)$  iff  $F \in \mathcal{I}(X, \leq^*)$ .

*Proof.*

1. • ( $\implies$ ):  
 Suppose that  $I \in \mathcal{I}(X, \leq)$ .  
 – Since  $I \in \mathcal{I}(X, \leq)$ ,  $I \neq \emptyset$ .  
 – Since  $I \in \mathcal{I}(X, \leq)$ ,  $I \in \mathcal{L}(X, \leq)$ . Exercise 8.1.0.4 implies that  $I \in \mathcal{U}(X, \leq^*)$ .  
 – Since  $I \in \mathcal{I}(X, \leq)$ ,  $I$  is  $\leq$ -upward directed. Exercise 6.2.0.3 implies that  $I$  is  $\leq^*$ -downward directed. Thus  $I \in \mathcal{F}(X, \leq^*)$ .  
 • ( $\impliedby$ ) Suppose that  $I \in \mathcal{F}(X, \leq^*)$ . The prev part implies that Thus

$$\begin{aligned} I &\in \mathcal{I}(X, (\leq^*)^*) \\ &= \mathcal{I}(X, \leq) \quad (\text{by ref ex on binary converse relations}) \end{aligned}$$

2. Similar to (1) use duality, prev part FINISH!!!

□

**Exercise 10.1.0.6.** Let  $(X, \leq)$  be a poset.

1. If  $(X, \leq)$  has a zero, then for each  $I \in \mathcal{I}(X, \leq)$ ,  $0 \in I$ .
2. If  $(X, \leq)$  has a one, then for each  $F \in \mathcal{F}(X, \leq)$ ,  $1 \in F$ .

*Proof.*

1. Suppose that  $(X, \leq)$  has a zero. Since for each  $I \in \mathcal{I}(X, \leq)$ ,  $I \in \mathcal{L}(X, \leq)$  and  $I \neq \emptyset$ , Exercise 9.3.0.14 implies that for  $I \in \mathcal{L}(X, \leq)$ ,  $0 \in I$ .
2. use duality

□

**Exercise 10.1.0.7.** move or fix!! Let  $(X, \leq)$  be a poset.

1. For each  $I_1, I_2 \in \mathcal{I}(X, \leq)$ .  
 (a) If  $\bigcap_{\alpha \in A} I_\alpha \neq \emptyset$ , then  $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(X, \leq)$ .

- (b) If  $A = \{1, 2\}$ , then  $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(X, \leq)$ .
- (c) If  $(X, \leq)$  has a zero, then  $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(X, \leq)$ .

2. Let  $(F_\alpha)_{\alpha \in A} \subset \mathcal{F}(X, \leq)$ .

- (a) If  $\bigcap_{\alpha \in A} F_\alpha \neq \emptyset$ , then  $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(X, \leq)$ .
- (b) If  $A = \{1, 2\}$ , then  $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(X, \leq)$ .
- (c) If  $(X, \leq)$  has a one, then  $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(X, \leq)$ .

*Proof.*

1. Set  $I := \bigcap_{\alpha \in A} I_\alpha$ .

- (a) Suppose that  $I \neq \emptyset$ .
- i. By assumption  $I \neq \emptyset$ .
  - ii. Let  $a, b \in I$  and  $\alpha_0 \in A$ . Then  $a, b \in I_{\alpha_0}$  and  $I_{\alpha_0} \in \mathcal{I}(X, \leq)$ . Therefore  $I_{\alpha_0}$  is upward directed. Exercise ?? implies that  $a \vee b \in I_{\alpha_0}$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $a \vee b \in I_\alpha$ . Hence  $a \vee b \in I$ .
  - iii. Let  $x \in L$ ,  $a \in I$  and  $\alpha \in A$ . Then  $a \in I_\alpha$ . Since  $I_\alpha \in \mathcal{I}(L)$ ,  $x \in L$  and  $a \in I_\alpha$ , we have that  $x \wedge a \in I_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $x \wedge a \in I_\alpha$ . Hence  $x \wedge a \in I$ . Since  $x \in L$  and  $a \in I$  are arbitrary, we have that for each  $x \in L$  and  $a \in I$ ,  $x \wedge a \in I$ .

Thus  $I \in \mathcal{I}(L)$ .

- (b) Suppose that  $A = \{1, 2\}$ .
- i. Since  $I_1, I_2 \in \mathcal{I}(L)$ ,  $I_1 \neq \emptyset$  and  $I_2 \neq \emptyset$ . Thus there exist  $x_1 \in I_1$  and  $x_2 \in I_2$ . Since  $I_1 \in \mathcal{I}(L)$ ,  $x_2 \in L$  and  $x_1 \in I_1$ , we have that  $x_1 \wedge x_2 \in I_1$ . Similarly,  $x_1 \wedge x_2 \in I_2$ . Hence  $x_1 \wedge x_2 \in I_1 \cap I_2$  and  $I_1 \cap I_2 \neq \emptyset$ . Part 1(a) implies that  $I_1 \cap I_2 \in \mathcal{I}(L)$ .
  - (c) Suppose that  $L$  has a zero. Let  $\alpha \in A$ . Since  $I_\alpha \in \mathcal{I}(L)$ , a previous exercise implies that  $0 \in I_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $0 \in I_\alpha$ . Hence  $0 \in I$ . Thus  $I \neq \emptyset$ . The previous part implies that  $I \in \mathcal{I}(L)$ .

2. Similar to (1).

□

**Definition 10.1.0.8.** *move or fix!! (consider changing notation)* Let  $L$  be a lattice and  $A \subset L$ .

• We define

- $\text{Ideals}(A, L) := \{I \in \mathcal{I}(L) : A \subset I\}$
- $\text{Filters}(A, L) := \{F \in \mathcal{F}(L) : A \subset F\}$

• We define

- the  $L$ -ideal generated by  $A$ , denoted  $\text{IdealGen}(A, L)$ , by  $\text{IdealGen}(A, L) := \bigcap_{I \in \text{Ideals}(A, L)} I$
- the  $L$ -filter generated by  $A$ , denoted  $\text{FilterGen}(A, L)$ , by  $\text{FilterGen}(A, L) := \bigcap_{F \in \text{Filters}(A, L)} F$

**Note 10.1.0.9.** *move or fix!! the prev ex* implies that  $\text{IdealGen}(A, L) \in \text{Ideals}(A, L)$  and  $\text{FilterGen}(A, L) \in \text{Filters}(A, L)$ .

**Exercise 10.1.0.10.** *move or fix!!* Let  $L$  be a lattice and  $B \subset L$ .

1. If  $B$  is upward directed, then

- (a)  $\downarrow B \in \mathcal{I}(L)$

$$(b) \text{ IdealGen}(B) = \downarrow(B, \leq_L).$$

2. If  $B$  is downward directed, then

$$(a) \uparrow B \in \mathcal{F}(L)$$

$$(b) \text{ FilterGen}(B) = \uparrow(B, \leq_L).$$

*Proof.*

1. Suppose that  $B$  is upward directed.

(a) Exercise ??(ref ex) implies that  $\downarrow B$  is upward directed and Exercise ?? (ref ex) implies that  $\downarrow B$  is an lower set. Exercise ?? then implies that  $\downarrow B \in \mathcal{I}(L)$ .

(b) The prev part implies that  $\downarrow B \in \text{Ideals}(B, L)$ . Therefore

$$\begin{aligned} \text{IdealGen}(B, L) &= \bigcap_{I \in \text{Ideals}(B, L)} I \\ &\subset \uparrow B. \end{aligned}$$

We note that for each  $I \in \mathcal{I}(L)$ ,  $I$  is a lower set. Thus  $\text{Ideals}(B, L) \subset \mathcal{L}(B, \leq_L)$ . Hence

$$\begin{aligned} \downarrow B &= \bigcap_{I \in \text{Ideals}(B, L)} I \\ &\subset \uparrow B. \end{aligned}$$

2. Suppose that  $B$  is  $\leq$ -downward directed. Exercise ?? implies that  $B$  is  $\leq^*$ -upward directed. the prev part implies that  $\uparrow(B, \leq^*) \in \mathcal{I}(L)$ . Exercise ?? then implies that  $\uparrow(B, \leq^*) \in \mathcal{I}(L)$

make new notation  $\leq_L :=$  the unique partial order on  $L$  compatible with  $\vee, \wedge$ . Show  $\leq_{L^*} = \leq_L^*$ , make note that we write  $\leq$  in place

□

**Definition 10.1.0.11.** move or fix!! define prefilter or filter basis, preideal or ideal basis as a downward directed/upward directed set, maybe don't need to, but look into what we need for topology.

# Part III

## Lattices



pulling from [20]





# Chapter 11

## Lattices

### 11.1 Semilattices

#### 11.1.1 Skew Semilattices

*cite Leech, skew lattices in rings*

**Definition 11.1.1.1.** Let  $X$  be a set and  $\diamond : X \times X \rightarrow X$  a binary operator on  $X$ . Then

- $\diamond$  is said to be a **skew semilattice operator** on  $X$  if  $\diamond$  is associative and idempotent.
- $(X, \diamond)$  is said to be a **skew semilattice** if

1.  $X \neq \emptyset$
2.  $\diamond$  is a skew semilattice operator on  $X$

**Definition 11.1.1.2.** Let  $(X, \diamond)$  be an skew semilattice. We define the

- **join preorder on  $X$  induced by  $\diamond$** , denoted  $\leq_{\diamond}^{\vee} \subset X \times X$ , by  $a \leq_{\diamond}^{\vee} b$  iff  $(b \diamond a) \diamond b = b$
- **meet preorder on  $X$  induced by  $\diamond$** , denoted  $\leq_{\diamond}^{\wedge} \subset X \times X$ , by  $a \leq_{\diamond}^{\wedge} b$  iff  $(a \diamond b) \diamond a = a$ .

**Exercise 11.1.1.3.** Let  $(X, \diamond)$  be an skew semilattice. Then

1.  $\leq_{\diamond}^{\vee}$  is a preorder on  $X$ .  
**Hint:** If  $cbc = c$  and  $bab = b$ , then
  - (a)  $c = (cb)(abc)(abc)$  and  $c = cab$
  - (b)  $cac = (ca)(ca)(bc)$
2.  $\leq_{\diamond}^{\wedge}$  is a preorder on  $X$ .

*Proof.*

1. Let  $a, b, c \in X$ .
  - (a) Since  $\diamond$  is idempotent, we have that

$$\begin{aligned}(a \diamond a) \diamond a &= a \diamond a \\ &= a\end{aligned}$$

and therefore  $a \leq_{\diamond}^{\vee} a$ .

(b) Suppose that  $a \leq_{\diamond}^{\vee} b$  and  $b \leq_{\diamond}^{\vee} c$ . Then  $(b \diamond a) \diamond b = b$  and  $(c \diamond b) \diamond c = c$ . Since  $\diamond$  is associative and idempotent,

$$\begin{aligned}
 c &= cbc \\
 &= cbabc \\
 &= (cb)(abc) \\
 &= (cb)(abc)(abc) \\
 &= c(bab)cabc \\
 &= cbcabc \\
 &= (cbc)abc \\
 &= cabc.
 \end{aligned}$$

Therefore

$$\begin{aligned}
 cac &= (ca)(c) \\
 &= (ca)(cab) \\
 &= (ca)(ca)(bc) \\
 &= (ca)(bc) \\
 &= cabc \\
 &= c.
 \end{aligned}$$

Hence  $a \leq_{\diamond}^{\vee} c$ .

So  $\leq_{\diamond}^{\vee}$  is a preorder on  $X$ .

2. Similar to (1).

□

**Exercise 11.1.1.4.** Let  $(X, \diamond)$  be an skew semilattice. Then

1.  $(\leq_{\diamond}^{\vee})^* = \leq_{\diamond}^{\wedge}$
2.  $(\leq_{\diamond}^{\wedge})^* = \leq_{\diamond}^{\vee}$

*Proof.*

1. Let  $a, b \in X$ . Then

$$\begin{aligned}
 a(\leq_{\diamond}^{\vee})^* b &\iff b \leq_{\diamond}^{\vee} a \\
 &\iff a \diamond b \diamond a = a \\
 &\iff a \leq_{\diamond}^{\wedge} b.
 \end{aligned}$$

2. Similar to (1).

□

**Exercise 11.1.1.5.** Let  $(X, \diamond)$  be an skew semilattice. Then for each  $a, b \in X$ ,

1.  $a \diamond b \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$
2.  $a \diamond b \in \text{lb}(\{a, b\}, \leq_{\diamond}^{\wedge})$

*Proof.* Let  $a, b \in X$ .

1. We note that

$$\begin{aligned}(a \diamond b) \diamond [a \diamond (a \diamond b)] &= (a \diamond b) \diamond [(a \diamond a) \diamond b] \\ &= (a \diamond b) \diamond (a \diamond b) \\ &= a \diamond b.\end{aligned}$$

Similarly,  $[(a \diamond b) \diamond b] \diamond (a \diamond b) = a \diamond b$ . Therefore  $a \leq_{\diamond}^{\vee} a \diamond b$  and  $b \leq_{\diamond}^{\vee} a \diamond b$ . Hence  $a \diamond b \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$ .

2. **FINISH!!!**

□

### 11.1.2 Skew Semilattice Homomorphisms

**Definition 11.1.2.1.** Let  $(X, \diamond_X)$  and  $(Y, \diamond_Y)$  be skew semilattices and  $f : X \rightarrow Y$ . Then  $f$  is said to be a  $(\diamond_X, \diamond_Y)$ -**homomorphism** if for each  $a, b \in X$ ,  $f(a \diamond_X b) = f(a) \diamond_Y f(b)$ .

**Note 11.1.2.2.** When the context is clear, we say homomorphism instead of  $(\diamond_X, \diamond_Y)$ -homomorphism.

**Exercise 11.1.2.3.** Let  $(X, \diamond_X)$  and  $(Y, \diamond_Y)$  be skew semilattices and  $f : X \rightarrow Y$ . If  $f$  is a  $(\diamond_X, \diamond_Y)$ -homomorphism, then

1.  $f$  is  $(\leq_{\diamond_X}^{\vee}, \leq_{\diamond_Y}^{\vee})$ -monotone
2.  $f$  is  $(\leq_{\diamond_X}^{\wedge}, \leq_{\diamond_Y}^{\wedge})$ -monotone

*Proof.* Suppose that  $f$  is a  $(\diamond_X, \diamond_Y)$ -homomorphism.

1. Let  $a, b \in X$ . Suppose that  $a \leq_{\diamond_X}^{\vee} b$ . Then  $(b \diamond_X a) \diamond_X b = b$ . Therefore

$$\begin{aligned}f(b) &= (b \diamond_X a) \diamond_X b \\ &= f(b \diamond_X a) \diamond_Y f(b) \\ &= (f(b) \diamond_Y f(a)) \diamond_Y f(b)\end{aligned}$$

Thus  $f(a) \leq_{\diamond_Y}^{\vee} f(b)$ . Since  $a, b \in X$  with  $a \leq_{\diamond_X}^{\vee} b$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_{\diamond_X}^{\vee} b$  implies that  $f(a) \leq_{\diamond_Y}^{\vee} f(b)$ . Hence  $f$  is  $(\leq_{\diamond_X}^{\vee}, \leq_{\diamond_Y}^{\vee})$ -monotone.

2. **use duality?, FINISH!!!**

□

### 11.1.3 Semilattices

**Definition 11.1.3.1.** Let  $X$  be a set and  $\diamond : X \times X \rightarrow X$ . Then

- $\diamond$  is said to be a **semilattice operator on  $X$**  if  $\diamond$  associative, idempotent and commutative.
- $(X, \diamond)$  is said to be a **semilattice** if
  1.  $X \neq \emptyset$
  2.  $\diamond$  is a semilattice operator on  $X$ .

**Exercise 11.1.3.2.** Let  $X$  be a set and  $\diamond : X \times X \rightarrow X$ . Then  $\diamond$  is a semilattice operator on  $X$  iff

1.  $\diamond$  is a skew semilattice operator on  $X$ ,
2.  $\diamond$  is commutative.

*Proof.* Clear by definition. □

**Exercise 11.1.3.3.** Let  $(X, \diamond)$  be a semilattice. Then

1. for each  $a, b \in X$ ,
  - (a)  $a \leq_{\diamond}^{\vee} b$  iff  $a \diamond b = b$
  - (b)  $a \leq_{\diamond}^{\wedge} b$  iff  $a \diamond b = a$
2. (a)  $\leq_{\diamond}^{\vee}$  is a partial order on  $X$
- (b)  $\leq_{\diamond}^{\wedge}$  is a partial order on  $X$

*Proof.*

1. Let  $a, b \in X$ .
  - (a) Since  $\diamond$  is commutative, associative and idempotent,

$$\begin{aligned}
 a \leq_{\diamond}^{\vee} b &\iff (b \diamond a) \diamond b = b \\
 &\iff (a \diamond b) \diamond b = b \\
 &\iff a \diamond (b \diamond b) = b \\
 &\iff a \diamond b = b.
 \end{aligned}$$

- (b) Similar to 1(a).

2. (a) Exercise 11.1.1.3 implies that  $\leq_{\diamond}^{\vee}$  is a preorder on  $X$ . Let  $a, b \in X$ . Suppose that  $a \leq_{\diamond}^{\vee} b$  and  $b \leq_{\diamond}^{\vee} a$ . Part (1) then implies that  $a \diamond b = b$  and  $b \diamond a = a$ . Since  $\diamond$  is commutative,

$$\begin{aligned}
 a &= b \diamond a \\
 &= a \diamond b \\
 &= b.
 \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ , if  $a \leq_{\diamond}^{\vee} b$  and  $b \leq_{\diamond}^{\vee} a$ , then  $a = b$ . Thus  $\leq_{\diamond}^{\vee}$  is a partial order on  $X$ .

- (b) Similar to 2(a). □

**Exercise 11.1.3.4.** Let  $(X, \diamond)$  be a semilattice. Then for each  $a, b \in X$ ,

1.  $a \diamond b = \sup(\{a, b\}, \leq_{\diamond}^{\vee})$
2.  $a \diamond b = \inf(\{a, b\}, \leq_{\diamond}^{\wedge})$

*Proof.* Let  $a, b \in X$ .

- Exercise 11.1.1.5 implies that  $a \diamond b \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$ . Let  $c \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$ . Since  $a \leq_{\diamond}^{\vee} c$  and  $b \leq_{\diamond}^{\vee} c$ , Exercise 11.1.3.3 implies that  $a \diamond c = c$  and  $b \diamond c = c$ . Therefore

$$\begin{aligned}(a \diamond b) \diamond c &= a \diamond (b \diamond c) \\ &= a \diamond c \\ &= c.\end{aligned}$$

Another application of Exercise 11.1.3.3 implies that  $a \diamond b \leq_{\diamond}^{\vee} c$ . Since  $c \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$  is arbitrary, we have that for each  $c \in \text{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$ ,  $a \diamond b \leq_{\diamond}^{\vee} c$ . Therefore  $a \diamond b = \sup(\{a, b\}, \leq_{\diamond}^{\vee})$ .

- FINISH!!!**

□

**Definition 11.1.3.5.** Let  $(X, \leq)$  be a poset. Then  $(X, \leq)$  is said to be a

- **join semilattice** if

1.  $X \neq \emptyset$
2. for each  $a, b \in X$ ,  $\sup\{a, b\}$  exists.

- **meet semilattice** if

1.  $X \neq \emptyset$
2. for each  $a, b \in X$ ,  $\inf\{a, b\}$  exists.

**Exercise 11.1.3.6.** Let  $(X, \leq)$  be a poset. Then

1.  $(X, \leq)$  is a join semilattice iff  $(X, \leq^*)$  is a meet semilattice
2.  $(X, \leq^*)$  is a join semilattice iff  $(X, \leq)$  is a meet semilattice

*Proof.*

- UPDATE ONCE SUPREMA/INFIMA EXERCISES ARE REORGANIZED**

$$\begin{aligned}(X, \leq) \text{ is a join semilattice} &\iff X \neq \emptyset \text{ and for each } a, b \in X, \sup(\{a, b\}, \leq) \text{ exists} \\ &\iff X \neq \emptyset \text{ and for each } a, b \in X, \inf(\{a, b\}, \leq^*) \text{ exists} \quad (\text{by Exercise ?? ADD EXERCISE REF}) \\ &\iff (X, \leq) \text{ is a meet semilattice}\end{aligned}$$

- 2.

□

**Exercise 11.1.3.7.** Let  $(X, \diamond)$  be a semilattice. Then

1.  $(X, \leq_{\diamond}^{\vee})$  is a join semilattice
2.  $(X, \leq_{\diamond}^{\wedge})$  is a meet semilattice

*Proof.* Exercise 11.1.3.4 implies that

1. For each  $a, b \in X$ ,  $\sup(\{a, b\}, \leq_{\diamond}^{\vee}) = a \diamond b$ . Thus  $(X, \leq_{\diamond}^{\vee})$  is a join semilattice.
2. for each  $a, b \in X$ ,  $\inf(\{a, b\}, \leq_{\diamond}^{\wedge}) = a \diamond b$ . Thus  $(X, \leq_{\diamond}^{\wedge})$  is a meet semilattice.

□

**Definition 11.1.3.8.** Let  $(X, \leq)$  be a poset.

- If  $(X, \leq)$  is a join semilattice, we define the **join operator on  $X$  induced by  $\leq$** , denoted  $\vee_{\leq} : X \times X \rightarrow X$ , by  $a \vee_{\leq} b := \sup\{a, b\}$ .
- If  $(X, \leq)$  is a meet semilattice, we define the **meet operator on  $X$  induced by  $\leq$** , denoted  $\wedge_{\leq} : X \times X \rightarrow X$ , by  $a \wedge_{\leq} b := \inf\{a, b\}$ .

**Exercise 11.1.3.9.** Let  $(X, \leq)$  be a poset.

1. If  $(X, \leq)$  is a join semilattice, then  $\vee_{\leq}$  is a semilattice operator on  $X$ .
2. If  $(X, \leq)$  is a meet semilattice, then  $\wedge_{\leq}$  is a semilattice operator on  $X$ .

*Proof.*

1. Suppose that  $(X, \leq)$  is a join semilattice.

- (a) • Let  $a, b, c \in X$ . Exercise 9.3.0.5 implies that

$$\begin{aligned}
 (a \vee_{\leq} b) \vee_{\leq} c &= \sup\{\sup\{a, b\}, c\} \\
 &= \sup\{\sup\{a, b\}, \sup\{c\}\} \\
 &= \sup\{a, b, c\} \\
 &= \sup\{\sup\{a\}, \sup\{b, c\}\} \\
 &= \sup\{a, \sup\{b, c\}\} \\
 &= a \vee_{\leq} (b \vee_{\leq} c).
 \end{aligned}$$

Hence  $\vee_{\leq}$  is associative.

- Let  $a \in X$ . Then

$$\begin{aligned}
 a \vee_{\leq} a &= \sup\{a, a\} \\
 &= \sup\{a\} \\
 &= a.
 \end{aligned}$$

Hence  $\vee_{\leq}$  is idempotent.

- Let  $a, b \in X$ . Then

$$\begin{aligned}
 a \vee_{\leq} b &= \sup\{a, b\} \\
 &= \sup\{b, a\} \\
 &= b \vee_{\leq} a.
 \end{aligned}$$

Hence  $\vee_{\leq}$  is commutative.

Since  $\vee_{\leq}$  is associative, idempotent and commutative,  $\vee_{\leq}$  is a semilattice operator on  $X$ .

2. Similar to (1).

□

**Exercise 11.1.3.10.** Let  $(X, \leq)$  be a poset.

1. If  $(X, \leq)$  is a join semilattice, then

- (a)  $\leq_{\vee_{\leq}}^{\vee} = \leq$ ,  
 (b)  $\leq_{\vee_{\leq}}^{\wedge} = \leq^*$ .

2. If  $(X, \leq)$  is a meet semilattice, then

- (a)  $\leq_{\wedge_{\leq}}^{\wedge} = \leq$ ,  
 (b)  $\leq_{\wedge_{\leq}}^{\vee} = \leq^*$ .

*Proof.*

1. Suppose that  $(X, \leq)$  is a join semilattice.

(a) Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq_{\vee}^{\vee} b &\iff a \vee_{\leq} b = b \quad (\text{Exercise 11.1.3.3}) \\ &\iff \sup(\{a, b\}, \leq) = b \quad (\text{Definition 11.1.3.8}) \\ &\iff a \leq b \quad (\text{Exercise 9.3.0.6}). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that  $\leq_{\vee}^{\vee} = \leq$ .

(b) Exercise 11.1.1.4 implies that

$$\begin{aligned} \leq_{\vee}^{\wedge} &= (\leq_{\vee}^{\vee})^* \\ &= \leq^*. \end{aligned}$$

2. Similar to (1).

□

**Exercise 11.1.3.11.** Let  $(X, \diamond)$  be a semilattice. Then

1.  $\leq_{\diamond}^{\vee}$  is the unique partial order  $\leq$  on  $X$  such that  $(X, \leq)$  is a join semilattice and for each  $a, b \in X$ ,  $a \diamond b = \sup(\{a, b\}, \leq)$ .
2.  $\leq_{\diamond}^{\wedge}$  is the unique partial order  $\leq$  on  $X$  such that  $(X, \leq)$  is a meet semilattice and for each  $a, b \in X$ ,  $a \diamond b = \inf(\{a, b\}, \leq)$ .

*Proof.*

1. Let  $\leq$  be partial order on  $X$ . Suppose that  $(X, \leq)$  is a join semilattice and for each  $a, b \in X$ ,  $a \diamond b = \sup(\{a, b\}, \leq)$ . Let  $a, b \in X$ . Then

$$\begin{aligned} a \leq b &\iff b = \sup(\{a, b\}, \leq) \quad (\text{Exercise 9.3.0.6}) \\ &\iff a \diamond b = b \quad (\text{by assumption}) \\ &\iff a \leq_{\diamond}^{\vee} b \quad (\text{Exercise 11.1.3.3}). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq b$  iff  $a \leq_{\diamond}^{\vee} b$ . Hence  $\leq = \leq_{\diamond}^{\vee}$ .

2. Similar to (1).

□

**Exercise 11.1.3.12.** Let  $(X, \leq)$  be a poset.

1. If  $(X, \leq)$  is a join semilattice, then  $\vee_{\leq}$  is the unique semilattice operator  $\diamond$  on  $X$  such that for each  $a, b \in X$ ,  $a \diamond b = \sup(\{a, b\}, \leq)$ .
2. If  $(X, \leq)$  is a meet semilattice, then  $\wedge_{\leq}$  is the unique semilattice operator  $\diamond$  on  $X$  such that for each  $a, b \in X$ ,  $a \diamond b = \inf(\{a, b\}, \leq)$ .

*Proof.*

1. Let  $\diamond : X \times X \rightarrow X$  be a semilattice operator on  $X$ . Suppose that  $a, b \in X$ ,  $a \diamond b = \sup(\{a, b\}, \leq)$ . Then for each  $a, b \in X$ ,

$$\begin{aligned} a \diamond b &= \sup(\{a, b\}, \leq) \quad (\text{by assumption}) \\ &= a \vee_{\leq} b \quad (\text{Definition 11.1.3.8}). \end{aligned}$$

Hence  $\diamond = \vee_{\leq}$ .

2. Similar to (1).

□

**Exercise 11.1.3.13.** Let  $(L, \leq)$  be a poset.

1. Suppose that  $(L, \leq)$  is a join semilattice. Then  $(L, \leq)$  has a zero iff for each  $A \subset L$ ,  $A$  is finite implies that  $\sup A$  exists.
2. Suppose that  $(L, \leq)$  is a meet semilattice. Then  $(L, \leq)$  has a one iff for each  $A \subset L$ ,  $A$  is finite implies that  $\inf A$  exists.

*Proof.*

1. •  $(\implies)$ :  
Suppose that  $(L, \leq)$  has a zero. Let  $A \subset L$ . Suppose that  $A$  is finite.  
– Suppose that  $\#A = 0$ . Then  $A = \emptyset$ . Exercise 9.3.0.2 implies that

$$\begin{aligned} \sup A \text{ exists} &\iff \inf(\text{ub } A) \text{ exists} \\ &\iff \inf(\text{ub } \emptyset) \text{ exists} \\ &\iff \inf L \text{ exists.} \end{aligned}$$

Since  $L$  has a zero, Definition 9.3.0.11 implies that  $\inf L$  exists. Therefore  $\sup A$  exists.

- Suppose that  $\#A \geq 1$ . Since  $(L, \leq)$  is an join semilattice, we have that for each  $a, b \in L$ ,  $\sup\{a, b\}$  exists. Exercise 9.3.0.7 then implies that  $\sup A$  exists.

- $(\impliedby)$ :  
Suppose that for each  $A \subset L$ ,  $A$  is finite implies that  $\sup A$  exists. Since  $\emptyset$  is finite,  $\sup \emptyset$  exists. By Definition 9.3.0.11,  $(L, \leq)$  has a zero.

2. **FINISH!!!!**

□

### 11.1.4 Semilattice Homomorphisms

**Definition 11.1.4.1.** Let  $(X, \leq_X), (Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ .

1. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are join semilattices. Then  $f$  is said to be  $(\leq_X, \leq_Y)$ -**binary join preserving** if for each  $a, b \in X$ ,  $f(\sup(\{a, b\}, \leq_X)) = \sup(\{f(a), f(b)\}, \leq_Y)$ .
2. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are meet semilattices. Then  $f$  is said to be  $(\leq_X, \leq_Y)$ -**binary meet preserving** if for each  $a, b \in X$ ,  $f(\inf(\{a, b\}, \leq_X)) = \inf(\{f(a), f(b)\}, \leq_Y)$ .

**Exercise 11.1.4.2.** Let  $(X, \leq_X), (Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ .

1. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are join semilattices. Then  $f$  is finite  $(\leq_X, \leq_Y)$ -binary join preserving iff  $f$  is a  $(\vee_{\leq_X}, \vee_{\leq_Y})$ -homomorphism.
2. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are meet semilattices. Then  $f$  is finite  $(\leq_X, \leq_Y)$ -binary meet preserving iff  $f$  is a  $(\wedge_{\leq_X}, \wedge_{\leq_Y})$ -homomorphism.

*Proof.* Clear.

□

**Exercise 11.1.4.3.** Let  $(X, \diamond_X)$  and  $(Y, \diamond_Y)$  be semilattices and  $f : X \rightarrow Y$ . Then the following are equivalent:

1.  $f$  is a  $(\diamond_X, \diamond_Y)$ -homomorphism
2.  $f$  is  $(\leq_{\diamond_X}^\vee, \leq_{\diamond_Y}^\vee)$ -binary join preserving
3.  $f$  is  $(\leq_{\diamond_X}^\wedge, \leq_{\diamond_Y}^\wedge)$ -binary meet preserving

*Proof.*

- $(1) \implies (2)$ :  
Suppose that  $f$  is a  $(\diamond_X, \diamond_Y)$ -homomorphism. **make/ref previous ex here** implies that  $\vee_{\leq_{\diamond_X}^\vee} = \diamond_X$  and  $\vee_{\leq_{\diamond_Y}^\vee} = \diamond_Y$ . Therefore  $f$  is a  $(\vee_{\leq_{\diamond_X}^\vee}, \vee_{\leq_{\diamond_Y}^\vee})$ -homomorphism. **the previous ex** implies that  $f$  is  $(\leq_{\diamond_X}^\vee, \leq_{\diamond_Y}^\vee)$ -binary join preserving.



- (2)  $\implies$  (3):  
Suppose that **FINISH!!!**
- (3)  $\implies$  (1):

□

**Exercise 11.1.4.4.** Let  $(X, \leq_X), (Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ .

1. Suppose that  $(X, \leq_X)$  and  $(Y, \leq_Y)$  are join semilattices. Then  $f$  is  $(\leq_X, \leq_Y)$ -monotone iff for each  $a, b \in X$ ,  $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$ .
2. Suppose that  $(X, \leq_X)$  and  $(Y, \leq_Y)$  are meet semilattices. Then  $f$  is  $(\leq_X, \leq_Y)$ -monotone iff for each  $a, b \in X$ ,  $f(a \vee_{\leq_X} b) \leq_Y f(a) \wedge_{\leq_Y} f(b)$ .

*Proof.*

1. • ( $\implies$ ):  
Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -monotone. Let  $a, b \in X$ . Since  $a \vee_{\leq_X} b \in \text{ub}(\{a, b\}, \leq_X)$ , Exercise 9.1.0.13 implies that

$$\begin{aligned} f(a \vee_{\leq_X} b) &\in f(\text{ub}(\{a, b\}, \leq_X)) \\ &\subset \text{ub}(f(\{a, b\}), \leq_Y) \\ &= \text{ub}(\{f(a), f(b)\}, \leq_Y) \end{aligned}$$

Therefore

$$\begin{aligned} f(a) \vee_{\leq_Y} f(b) &= \sup(\{f(a), f(b)\}, \leq_Y) \\ &\leq f(a \vee_{\leq_X} b). \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$ .

- ( $\impliedby$ ):  
Suppose that for each  $a, b \in X$ ,  $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$ . Let  $a, b \in X$ . Suppose that  $a \leq_X b$ . Exercise 9.3.0.6 then implies that  $a \vee_{\leq_X} b = b$ . By assumption,

$$\begin{aligned} f(a) \vee_{\leq_Y} f(b) &\leq_Y f(a \vee_{\leq_X} b) \\ &= f(b) \\ &\leq f(a) \vee_{\leq_Y} f(b). \end{aligned}$$

Therefore  $f(a) \vee_{\leq_Y} f(b) = f(b)$ . Another application of Exercise 9.3.0.6 implies that  $f(a) \leq_Y f(b)$ . Since  $a, b \in X$  with  $a \leq b$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_X b$  implies that  $f(a) \leq_Y f(b)$ . Hence  $f$  is  $(\leq_X, \leq_Y)$ -monotone.

2. **use duality FINISH!!!**

□

**Exercise 11.1.4.5.** Let  $(X, \leq_X), (Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ .

1. Suppose that  $(X, \leq_X)$  and  $(Y, \leq_Y)$  are join semilattices. If  $f$  binary join preserving, then  $f$  is  $(\leq_X, \leq_Y)$ -monotone.
2. Suppose that  $(X, \leq_X)$  and  $(Y, \leq_Y)$  are meet semilattices. If  $f$  binary meet preserving, then  $f$  is  $(\leq_X, \leq_Y)$ -monotone.

*Proof.*

1. Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -binary join preserving. Then **the previous exercise** implies that

$$\begin{aligned} \text{for each } a, b \in X, f(a) \vee_{\leq_Y} f(b) &= f(a \vee_{\leq_X} b) \implies \text{for each } a, b \in X, f(a) \vee_{\leq_Y} f(b) \leq f(a \vee_{\leq_X} b) \\ &\implies f \text{ is } (\leq_X, \leq_Y)\text{-monotone} \end{aligned}$$

2. **FINISH!!!** , use duality or just similar

□

**Definition 11.1.4.6.** Let  $(X, \leq_X), (Y, \leq_Y)$  be posets and  $f : X \rightarrow Y$ .

1. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are join semilattices with zero. Then  $f$  is said to be  $(\leq_X, \leq_Y)$ -**finite join preserving** if for each  $A \subset X$ ,  $A$  is finite implies that  $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$ .
2. Suppose that  $(X, \leq_X), (Y, \leq_Y)$  are meet semilattices with one. Then  $f$  is said to be  $(\leq_X, \leq_Y)$ -**finite meet preserving** if for each  $A \subset X$ ,  $A$  is finite implies that  $f(\inf(A, \leq_X)) = \inf(f(A), \leq_Y)$ .

**Exercise 11.1.4.7.** Let  $(L, \leq_L)$  and  $(M, \leq_M)$  be posets and  $f : L \rightarrow M$ .

1. Suppose that  $(L, \leq_L)$  and  $(M, \leq_M)$  are join semilattices with zero. Then  $f$  is  $(\leq_L, \leq_M)$ -finite join preserving iff  $f$  is  $(\leq_L, \leq_M)$ -binary join preserving and  $f(0_L) = 0_M$ .
2. Suppose that  $(L, \leq_L)$  and  $(M, \leq_M)$  are meet semilattices with one. Then  $f$  is  $(\leq_L, \leq_M)$ -finite meet preserving iff  $f$  is  $(\leq_L, \leq_M)$ -binary meet preserving and  $f(1_L) = 1_M$ .

*Proof.*

1. • ( $\implies$ ):  
Suppose that  $f$  is  $(\leq_L, \leq_M)$ -finite join preserving. Then clearly  $f$  is  $(\leq_L, \leq_M)$ -binary join preserving. Since  $\emptyset$  is finite, **a prev ex about  $\sup \emptyset = 0$**  implies that

$$\begin{aligned} f(0_L) &= f(\sup(\emptyset, \leq_L)) \\ &= \sup(f(\emptyset), \leq_M) \\ &= \sup(\emptyset, \leq_M) \\ &= 0_M. \end{aligned}$$

- ( $\impliedby$ ):  
Suppose that  $f$  is  $(\leq_L, \leq_M)$ -binary join preserving and  $f(0_L) = 0_M$ . Let  $A \subset L$ . Suppose that  $A$  is finite.
  - Suppose that  $\#A = 0$ . Then  $A = \emptyset$  and

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\sup(\emptyset, \leq_L)) \\ &= f(0_L) \\ &= 0_M \\ &= \sup(\emptyset, \leq_M) \\ &= \sup(f(\emptyset), \leq_M) \\ &= \sup(f(A), \leq_M). \end{aligned}$$

- Suppose that  $\#A = 1$ . Then there exists  $a \in L$  such that  $A = \{a\}$ . Hence

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\sup(\{a\}, \leq_L)) \\ &= f(a) \\ &= \sup(\{f(a)\}, \leq_M) \\ &= \sup(f(A), \leq_M). \end{aligned}$$

- Suppose that  $\#A \geq 2$ . Then there exists  $n \in \mathbb{N}$  and  $a_1, \dots, a_n \in L$  such that  $n \geq 2$  and  $A = \{a_1, \dots, a_n\}$ . **ref prev ex** implies that  $\sup(A, \leq_L) = (\dots((a_1 \vee_{\leq_L} a_2) \vee_{\leq_L} a_3) \dots a_n)$ . Then

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\dots((a_1 \vee_{\leq_L} a_2) \vee_{\leq_L} a_3) \vee_{\leq_L} \dots \vee_{\leq_L} a_n) \\ &= \end{aligned}$$

do by induction on  $\#A$  so it is less messy.

2.

**FINISH!!!**

□

### 11.1.5 Semilattice Structures

**Exercise 11.1.5.1.** Let  $(X, \leq)$  be a poset and  $E \subset X$ .

1. Suppose that  $(X, \leq)$  is a join semilattice. Then
  - (a)  $E \in \mathcal{U}_X$  iff  $\vee(X \times E) \subset E$ ,
  - (b)  $E \neq \emptyset$  and  $\vee(E \times E) \subset E$  implies that  $E$  is upward directed.
2. Suppose that  $(X, \leq)$  is a meet semilattice. Then
  - (a)  $E \in \mathcal{L}_X$  iff  $\wedge(X \times E) \subset E$ .
  - (b)  $E \neq \emptyset$  and  $\wedge(E \times E) \subset E$  implies that  $E$  is downward directed.

*Proof.*

1. (a) •  $(\implies)$ :  
 Suppose that  $E \in \mathcal{U}_X$ . Then for each  $x \in X$  and  $a \in E$ ,  $a \leq x$  implies that  $x \in E$ . Let  $x \in X$  and  $a \in E$ . Since  $a \leq x \vee a$ , we have that  $x \vee a \in E$ . Since  $x \in X$  and  $a \in E$  are arbitrary, we have that for each  $x \in X$  and  $a \in E$ ,  $x \vee a \in E$ .  
 •  $(\impliedby)$ :  
 Suppose that for each  $x \in X$  and  $a \in E$ ,  $x \vee a \in E$ . Let  $x \in X$  and  $a \in E$ . Suppose that  $a \leq x$ . Then

$$\begin{aligned} x &= x \vee a \\ &\in E. \end{aligned}$$

Since  $x \in X$  and  $a \in E$  with  $a \leq x$  are arbitrary, we have that for each  $x \in X$  and  $a \in E$ ,  $a \leq x$  implies that  $x \in E$ . Hence  $E \in \mathcal{U}_X$ .

- (b) Suppose that  $E \neq \emptyset$  and for each  $\vee(E \times E) \subset E$ . Let  $x, y \in E$ . Then

$$\begin{aligned} x \vee y &\in \text{ub}(\{x, y\}, \leq) \cap E \\ &= \text{ub}(\{x, y\}, \leq|_E). \end{aligned}$$

Hence  $\text{ub}(\{x, y\}, \leq|_E) \neq \emptyset$ . Since  $x, y \in E$  are arbitrary, we have that for each  $x, y \in E$ ,  $\text{ub}(\{x, y\}, \leq|_E) \neq \emptyset$ . [ref prev](#)  
[ex in section on directed subsets](#) implies that  $E$  is  $\leq$ -upward directed.

2. **FINISH!!! use duality**

□

**Exercise 11.1.5.2.** Let  $(X, \leq)$  be a poset and  $I, F \subset X$ .

1. Suppose that  $(X, \leq)$  is a join semilattice. Then  $I \in \mathcal{I}(X)$  iff
  - $I \neq \emptyset$
  - $I \in \mathcal{L}_X$
  - $\vee(I \times I) \subset I$ .
2. Suppose that  $(X, \leq)$  is a meet semilattice. Then  $F \in \mathcal{F}(X)$  iff
  - $F \neq \emptyset$
  - $F \in \mathcal{U}_X$
  - $\wedge(F \times F) \subset F$ .

*Proof.*

1. • ( $\Rightarrow$ ):  
 Suppose that  $I \in \mathcal{I}(X)$ . Then  $I \neq \emptyset$ ,  $I \in \mathcal{L}_X$  and  $I$  is upward directed. Let  $x, y \in I$ . Since  $I$  is upward directed, there exists  $z \in I$  such that  $z \in \text{ub}\{x, y\}$ . Then

$$\begin{aligned} x \vee y &= \sup\{x, y\} \\ &= \inf(\text{ub}\{x, y\}) \\ &\leq z. \end{aligned}$$

Since  $I \in \mathcal{L}_X$ ,  $z \in I$  and  $x \vee y \leq z$ , we have that  $x \vee y \in I$ . Since  $x, y \in I$  are arbitrary, we have that for each  $x, y \in I$ ,  $x \vee y \in I$ . Thus  $\vee(I \times I) \subset I$ .

- ( $\Leftarrow$ ):  
 Suppose that  $I \neq \emptyset$ ,  $I \in \mathcal{L}(X)$  and  $\vee(I \times I) \subset I$ . Then
- by assumption,  $I \neq \emptyset$ ,
- by assumption,  $I \in \mathcal{L}(X)$ ,
- Exercise 11.1.5.1 then implies that  $I$  is upward directed.

Thus  $I \in \mathcal{I}(X)$ .

2. **FINISH!!! use duality**

□

**Exercise 11.1.5.3.** Let  $(X, \leq)$  be a poset. Suppose that  $X \neq \emptyset$ .

1. If  $(X, \leq)$  is a join semilattice, then  $X \in \mathcal{I}(X)$ .
2. If  $(X, \leq)$  is a meet semilattice, then  $X \in \mathcal{F}(X)$ .

*Proof.*

1. Suppose that  $(X, \leq)$  is a join semilattice. By assumption,  $X \neq \emptyset$ . Exercise 8.1.0.3 implies that  $X \in \mathcal{L}(X, \leq)$ . Since  $\vee(X \times X) \subset X$ , Exercise 11.1.5.2 implies that  $X \in \mathcal{I}(X, \leq)$ .

2. **FINISH!!! use duality?**

□

**Exercise 11.1.5.4.** Let  $(X, \leq)$  be a poset and  $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$ . Define  $E \in \mathcal{P}(X)$  by  $E := \bigcap_{\alpha \in A} E_\alpha$ .

1. Suppose that  $(X, \leq)$  is a join semilattice. If for each  $\alpha \in A$ ,  $\vee(E_\alpha \times E_\alpha) \subset E_\alpha$ , then  $\vee(E_\times E) \subset E$ .
2. Suppose that  $(X, \leq)$  is a meet semilattice. If for each  $\alpha \in A$ ,  $\wedge(E_\alpha \times E_\alpha) \subset E_\alpha$ , then  $\wedge(E_\times E) \subset E$ .

*Proof.*

1. Suppose that for each  $\alpha \in A$ ,  $\vee(E_\alpha \times E_\alpha) \subset E_\alpha$ . Let  $x, y \in E$  and  $\alpha \in A$ . Then  $x, y \in E_\alpha$ . By assumption,  $x \vee y \in E_\alpha$ . Since  $\alpha \in A$  is arbitrary, we have that for each  $\alpha \in A$ ,  $x \vee y \in E_\alpha$ . Thus  $x \vee y \in E$ . Since  $x, y \in E$  are arbitrary, we have that for each  $x, y \in E$ ,  $x \vee y \in E$ . Therefore  $\vee(E \times E) \subset E$ .

2. **use duality**

□

**Exercise 11.1.5.5.** Let  $(X, \leq)$  be a poset.

1. Suppose that  $(X, \leq)$  is a join semilattice. Let  $(I_\alpha)_{\alpha \in A} \subset \mathcal{I}(X)$ .

(a) If  $\bigcap_{\alpha \in A} I_\alpha \neq \emptyset$ , then  $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(X)$ .

(b) If  $(X, \leq)$  has a zero, then  $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(X, \leq)$ .

2. Suppose that  $(X, \leq)$  is a meet semilattice. Let  $(F_\alpha)_{\alpha \in A} \subset \mathcal{F}(X)$ .

- (a) If  $\bigcap_{\alpha \in A} F_\alpha \neq \emptyset$ , then  $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(X, \leq)$ .
- (b) If  $(X, \leq)$  has a one, then  $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(X, \leq)$ .

*Proof.*

1. Define  $I \subset X$  by  $I := \bigcap_{\alpha \in A} I_\alpha$ . Since  $(I_\alpha)_{\alpha \in A} \subset \mathcal{I}(X)$ , we have that for each  $\alpha \in A$ ,  $I_\alpha \neq \emptyset$ ,  $I_\alpha \in \mathcal{L}(X)$  and  $I_\alpha$  is upward directed.

- (a) Suppose that  $I \neq \emptyset$ .
  - By assumption  $I \neq \emptyset$ .
  - Exercise 8.1.0.6 implies that  $I \in \mathcal{L}(X)$ .
  - Exercise 11.1.5.4 implies that  $\vee(I \times I) \subset I$ .

Exercise 11.1.5.2 then implies that  $I \in \mathcal{I}(X)$ .

- (b) Suppose that  $(X, \leq)$  has a zero. Exercise 10.1.0.6 implies that for each  $\alpha \in A$ ,  $0 \in I_\alpha$ . Thus  $0 \in I$ . Hence  $I \neq \emptyset$ . **The prev part** then implies that  $I \in \mathcal{I}(X)$ .

2. **use duality**

□

## 11.2 Lattices

### 11.2.1 Skew Lattices

**Definition 11.2.1.1.** Let  $X$  be a set and  $\vee, \wedge$  skew semilattice operators on  $X$ . Then  $\vee$  and  $\wedge$  are said to **satisfy the skew lattice absorption identities** if for each  $a, b \in X$ ,

1.  $a \wedge (a \vee b) = a$
2.  $a \vee (a \wedge b) = a$
3.  $(b \vee a) \wedge a = a$
4.  $(b \wedge a) \vee a = a$

**Exercise 11.2.1.2.** Let  $X$  be a set and  $\vee, \wedge$  skew semilattice operators on  $X$ . Then  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities iff for each  $a, b \in X$ ,

1.  $a \vee b = b$  iff  $a \wedge b = a$
2.  $a \vee b = a$  iff  $a \wedge b = b$

*Proof.*

- $(\implies)$ :

Suppose that  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities. Let  $a, b \in X$ .

1.  $-(\implies)$ :  
If  $a \vee b = b$ , then

$$\begin{aligned} a \wedge b &= a \wedge (a \vee b) \\ &= a. \end{aligned}$$

- ( $\Leftarrow$ ):  
If  $a \wedge b = a$ , then

$$\begin{aligned} a \vee b &= (a \wedge b) \vee b \\ &= b. \end{aligned}$$

Thus  $a \vee b = b$  iff  $a \wedge b = a$ .

2. Similar to (1)

- ( $\Leftarrow$ ):

Suppose that for each  $a, b \in X$ ,

1.  $a \vee b = b$  iff  $a \wedge b = a$ ,
2.  $a \vee b = a$  iff  $a \wedge b = b$ .

Let  $a, b \in X$ .

1. Set  $c := a \vee b$ . By assumption,

$$\begin{aligned} a \wedge (a \vee b) &= a \iff a \wedge c = a \\ &\iff a \vee c = c \\ &\iff a \vee (a \vee b) = c \\ &\iff (a \vee a) \vee b = c \\ &\iff a \vee b = c \\ &\iff c = c. \end{aligned}$$

Since  $c = c$ , we have that  $a \wedge (a \vee b) = a$ .

2. Similar to (1)
3. Similar to (1)
4. Similar to (1)

□

**Exercise 11.2.1.3.** Let  $X$  be a set and  $\vee, \wedge$  skew semilattice operators on  $X$ . If  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities, then for each  $a, b \in X$ ,  $b \vee a \vee b = b$  iff  $a \wedge b \wedge a = a$ .

**Hint:** Exercise 11.2.1.2 implies that  $b \vee (a \vee b) = b$  iff  $b \wedge (a \vee b) = a \vee b$ . Use, skew lattice absorption identities and associativity.

*Proof.* Suppose that  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities. Let  $a, b \in X$ .

- ( $\Rightarrow$ ):

Suppose that  $b \vee a \vee b = b$ . Then Exercise 11.2.1.2 implies that  $b \wedge (a \vee b) = (a \vee b)$ . Therefore

$$\begin{aligned} a \wedge b &= (a \wedge b) \vee [(a \wedge b) \wedge (a \vee b)] \quad (\text{absorption}) \\ &= (a \wedge b) \vee [a \wedge (b \wedge [a \vee b])] \quad (\text{associativity}) \\ &= (a \wedge b) \vee [a \wedge (a \vee b)] \quad (\text{from earlier}) \\ &= (a \wedge b) \vee a \quad (\text{absorption}). \end{aligned}$$

Another application of Exercise 11.2.1.2 implies that  $(a \wedge b) \wedge a = a$ .

- ( $\Leftarrow$ ):

Similar to ( $\Rightarrow$ ).

□

**Definition 11.2.1.4.** Let  $X$  be a set and  $\vee, \wedge$  skew semilattice operators on  $X$ . Then  $(X, \vee, \wedge)$  is said to be a **skew lattice** if

1.  $X \neq \emptyset$
2.  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities

**Exercise 11.2.1.5.** Let  $(X, \vee, \wedge)$  be an skew lattice. Then

1.  $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$ .
2.  $\leq_{\vee}^{\wedge} = \leq_{\wedge}^{\vee} = (\leq_{\vee}^{\vee})^*$ .

*Proof.*

1. Let  $a, b \in X$ . Since  $(X, \vee, \wedge)$  is an skew lattice,  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities. Exercise 11.2.1.3 then implies that

$$\begin{aligned} a \leq_{\vee}^{\vee} b &\iff b \vee a \vee b = b \\ &\iff a \wedge b \wedge a = a \\ &\iff a \leq_{\wedge}^{\wedge} b. \end{aligned}$$

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,  $a \leq_{\vee}^{\vee} b$  iff  $a \leq_{\wedge}^{\wedge} b$ . Hence  $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$ .

2. Exercise 11.1.1.4 and part (1) imply that

$$\begin{aligned} \leq_{\vee}^{\wedge} &= (\leq_{\vee}^{\vee})^* \\ &= (\leq_{\wedge}^{\wedge})^* \\ &= \leq_{\wedge}^{\vee}. \end{aligned}$$

□

**Note 11.2.1.6.** Let  $(X, \vee, \wedge)$  be an skew lattice. When the context is clear, we write  $\leq$  in place of  $\leq_{\vee}^{\vee}$ .

## 11.2.2 Lattices

**Definition 11.2.2.1.** Let  $X$  be a set and  $\vee, \wedge$  semilattice operators on  $X$ . Then  $\vee$  and  $\wedge$  are said to **satisfy the lattice absorption identities** if for each  $a, b \in X$ ,

1.  $a \wedge (a \vee b) = a$
2.  $a \vee (a \wedge b) = a$

**Exercise 11.2.2.2.** Let  $X$  be a set and  $\vee, \wedge$  semilattice operators on  $X$ . Then  $\vee$  and  $\wedge$  satisfy the lattice absorption identities iff  $\vee$  and  $\wedge$  satisfy the skew lattice absorption identities.

*Proof.*

- $(\implies)$ :  
Clear by commutativity.
- $(\impliedby)$ :  
Immediate.

□

**Definition 11.2.2.3.** Let  $X$  be a set and  $\vee, \wedge$  semilattice operators on  $X$ . Then  $(X, \vee, \wedge)$  is said to be an **lattice** if

1.  $X \neq \emptyset$
2.  $\vee$  and  $\wedge$  satisfy the lattice absorption identities.

**Definition 11.2.2.4.** Let  $(X, \leq)$  be a poset. Then  $(X, \leq)$  is said to be an **ordered lattice** if  $(X, \leq)$  is a join semilattice and  $(X, \leq)$  is a meet semilattice.

**Exercise 11.2.2.5.** Let  $(X, \vee, \wedge)$  be a lattice. Then  $(X, \leq_{\vee}^{\vee})$  is an ordered lattice.

*Proof.* Exercise 11.1.3.7 implies that  $(X, \leq_{\vee}^{\vee})$  is a join semilattice and  $(X, \leq_{\wedge}^{\wedge})$  is a meet semilattice. Exercise 11.2.1.5 implies that  $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$ . Therefore  $(X, \leq_{\vee}^{\vee})$  is a join semilattice and  $(X, \leq_{\vee}^{\vee})$  is a meet semilattice. Hence  $(X, \leq_{\vee}^{\vee})$  is an ordered lattice.  $\square$

**Exercise 11.2.2.6.** Let  $(X, \leq)$  be an ordered lattice. Then  $(X, \vee_{\leq}, \wedge_{\leq})$  is a lattice.

*Proof.* Let  $a, b \in X$ .

1.
  - Since  $\sup\{a, b\} \in \text{ub}\{a, b\}$  is an upper bound of  $\{a, b\}$ , we have that  $a \leq \sup\{a, b\}$ . Since  $a \leq a$ , we have that  $a \in \text{lb}\{a, \sup\{a, b\}\}$ . Hence  $a \leq \inf\{a, \sup\{a, b\}\}$ .
  - Since  $\inf\{a, \sup\{a, b\}\} \in \text{lb}\{a, \sup\{a, b\}\}$ , we have that  $\inf\{a, \sup\{a, b\}\} \leq a$ .

Since  $a \leq \inf\{a, \sup\{a, b\}\}$  and  $\inf\{a, \sup\{a, b\}\} \leq a$ , we have that

$$\begin{aligned} a \wedge_{\leq} (a \vee_{\leq} b) &= \inf\{a, \sup\{a, b\}\} \\ &= a. \end{aligned}$$

2. Similarly,  $a \vee_{\leq} (a \wedge_{\leq} b) = a$ .

Since  $a, b \in X$  are arbitrary, we have that for each  $a, b \in X$ ,

1.  $a \wedge_{\leq} (a \vee_{\leq} b) = a$
2.  $a \vee_{\leq} (a \wedge_{\leq} b) = a$

Hence  $\vee_{\leq}$  and  $\wedge_{\leq}$  satisfy the lattice absorption identities. Therefore  $(X, \vee_{\leq}, \wedge_{\leq})$  is an lattice.  $\square$

**Exercise 11.2.2.7.** Let  $(X, \leq)$  be an ordered lattice. Then

1.  $\leq_{\vee_{\leq}}^{\vee} = \leq_{\wedge_{\leq}}^{\wedge} = \leq$
2.  $\leq_{\vee_{\leq}}^{\wedge} = \leq_{\wedge_{\leq}}^{\vee} = \leq^*$

*Proof.* Since  $(X, \leq)$  is an ordered lattice,  $(X, \leq)$  is a join semilattice  $(X, \leq)$  is a meet semilattice. Exercise 11.1.3.10 then implies that

- 1.

$$\begin{aligned} \leq_{\vee_{\leq}}^{\vee} &= \leq \\ &= \leq_{\wedge_{\leq}}^{\wedge} \end{aligned}$$

- 2.

$$\begin{aligned} \leq_{\vee_{\leq}}^{\wedge} &= \leq^* \\ &= \leq_{\wedge_{\leq}}^{\vee} \end{aligned}$$

$\square$

**Exercise 11.2.2.8.** Let  $(X, \vee, \wedge)$  be a lattice. Then  $\leq_{\vee}^{\vee}$  is the unique partial order  $\leq$  on  $X$  such that  $(X, \leq)$  is an ordered lattice and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq)$ .

*Proof.* Let  $\leq$  be a partial order on  $X$ . Suppose that  $(X, \leq)$  is an ordered lattice and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq)$ . Since  $(X, \vee)$  is a semilattice,  $(X, \leq_{\vee}^{\vee})$  and  $(X, \leq_{\wedge}^{\wedge})$  are join semilattices and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq_{\vee}^{\vee})$  and  $a \vee b = \sup(\{a, b\}, \leq)$ , Exercise 11.1.3.11 implies that  $\leq = \leq_{\vee}^{\vee}$ .  $\square$

**Exercise 11.2.2.9.** Let  $(X, \leq)$  be an ordered lattice. Then  $\vee_{\leq}, \wedge_{\leq}$  are the unique semilattice operators  $\vee, \wedge$  on  $X$  such that  $(X, \vee, \wedge)$  is an lattice and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq)$



*Proof.* Let  $\vee, \wedge$  be semilattice operators on  $X$ . Suppose that  $(X, \vee, \wedge)$  is a lattice and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq)$ . Then for each  $a, b \in X$ ,

$$\begin{aligned} a \vee_{\leq} b &= \sup(\{a, b\}, \leq) \quad (\text{Definition 11.1.3.8}) \\ &= a \vee b \quad (\text{by assumption}). \end{aligned}$$

Thus  $\vee = \vee_{\leq}$ . Since  $(X, \vee, \wedge)$  is a lattice and for each  $a, b \in X$ ,  $a \vee b = \sup(\{a, b\}, \leq)$ , Exercise 11.2.2.8 implies that  $\leq_{\vee}^{\vee} = \leq$ . Exercise 11.2.1.5 then implies that

$$\begin{aligned} \leq_{\wedge}^{\wedge} &= \leq_{\vee}^{\vee} \\ &= \leq. \end{aligned}$$

Therefore for each  $a, b \in X$ ,

$$\begin{aligned} a \wedge b &= \inf(\{a, b\}, \leq_{\wedge}^{\wedge}) \quad (\text{Exercise 11.1.3.4}) \\ &= \inf(\{a, b\}, \leq) \\ &= a \wedge_{\leq} b \quad (\text{Definition 11.1.3.8}). \end{aligned}$$

Hence  $\wedge = \wedge_{\leq}$ . □

**Definition 11.2.2.10.** Let  $(X, \vee, \wedge)$  be a lattice. We define the

- **$(\vee, \wedge)$ -dual join operator on  $X$** , denoted  $\vee^* : X \times X \rightarrow X$ , by  $\vee^* := \wedge$
- **$(\vee, \wedge)$ -dual meet operator on  $X$** , denoted  $\wedge^* : X \times X \rightarrow X$ , by  $\wedge^* := \vee$
- **dual lattice of  $(X, \vee, \wedge)$** , denoted  $(X, \vee, \wedge)^*$ , by  $(X, \vee, \wedge)^* := (X, \vee^*, \wedge^*)$ .

**Note 11.2.2.11.** When the context is clear, we may say “dual join operator on  $X$ ” and “dual meet operator on  $X$ ” in place of “ $(\vee, \wedge)$ -dual join operator on  $X$ ” and “ $(\vee, \wedge)$ -dual meet operator on  $X$ ” respectively.

**Exercise 11.2.2.12.** Let  $(X, \leq)$  be an ordered lattice. Then

1.  $\vee_{\leq}^* = \vee_{\leq^*}$  and  $\wedge_{\leq}^* = \wedge_{\leq^*}$
2.  $(\vee^*)^* = \vee$  and  $(\wedge^*)^* = \wedge$

*Proof.*

1. For each  $a, b \in X$ ,

$$\begin{aligned} a \vee_{\leq}^* b &= a \wedge_{\leq} b \\ &= \inf(\{a, b\}, \leq) \\ &= \sup(\{a, b\}, \leq^*) \quad (\text{by ref ex}) \\ &= a \vee_{\leq^*} b \end{aligned}$$

Hence  $\vee_{\leq}^* = \vee_{\leq^*}$ . Similarly,  $\wedge_{\leq}^* = \wedge_{\leq^*}$ .

2. By Definition 11.2.2.10,

$$\begin{aligned} (\vee^*)^* &= \wedge^* \\ &= \vee. \end{aligned}$$

Similarly,  $(\wedge^*)^* = \wedge$ . □

**Exercise 11.2.2.13.** Let  $(X, \vee, \wedge)$  be a lattice. Then  $\leq_{\vee^*}^{\vee} = (\leq_{\vee}^{\vee})^*$ .

*Proof.* We have that

$$\begin{aligned} \leq_{\vee^*}^{\vee} &= \leq_{\wedge}^{\vee} \quad (\text{by Definition ?? ref prev definition}) \\ &= (\leq_{\vee}^{\vee})^* \quad (\text{by Exercise ?? ref prev ex}) \end{aligned}$$
□

**Note 11.2.2.14.** When the context is clear, we write  $\leq_X$  and  $\leq_{X^*}$  in place of  $\leq_{\vee}^{\vee}$  and  $\leq_{\vee^*}^{\vee}$  respectively.

11.2.3 Lattice Homomorphisms

???

## 11.3 Fundamental Examples

### 11.3.1 Bounded, Modular and Distributive Lattices

**Definition 11.3.1.1.** Let  $(L, \leq)$  be a lattice. Then  $(L, \leq, 0, 1)$  is said to be a **bounded lattice** if 0 is a  $\leq$ -zero and 1 is a  $\leq$ -one  
 put some exercises?

**Definition 11.3.1.2.** Let  $L$  be a lattice. Then  $L$  is said to be a **distributive lattice** if for each  $a, b, c \in L$ ,  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ .

**Exercise 11.3.1.3.** Let  $L$  be a lattice. Then for each  $a, b, c \in L$ ,

1.  $(a \wedge b) \vee (a \wedge c) \leq a \wedge (b \vee c)$ .
2.  $a \wedge (b \vee c) \leq (a \vee b) \wedge (a \vee c)$

*Proof.* Let  $a, b, c \in L$ .

1. • Since  $a \wedge b \leq a$  and  $a \wedge c \leq a$ , we have that  $a \in \text{ub}\{a \wedge b, a \wedge c\}$ . Therefore

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &= \sup\{a \wedge b, a \wedge c\} \\ &\leq a. \end{aligned}$$

- Since

$$\begin{aligned} a \wedge b &\leq b \\ &\leq b \vee c \end{aligned}$$

and

$$\begin{aligned} a \wedge c &\leq c \\ &\leq b \vee c, \end{aligned}$$

we have that  $b \vee c \in \text{ub}\{a \wedge b, a \wedge c\}$ . Therefore

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &= \sup\{a \wedge b, a \wedge c\} \\ &\leq b \vee c. \end{aligned}$$

Then  $(a \wedge b) \vee (a \wedge c) \in \text{lb}\{a, b \vee c\}$ . Hence

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &\leq \inf\{a, b \vee c\} \\ &= a \wedge (b \vee c). \end{aligned}$$

2. **FINISH!!! (use duality)**

□

**Exercise 11.3.1.4.** Let  $L$  be a lattice. Then for each  $a, b, c \in L$ ,

1.  $c \leq a$  implies that  $(a \wedge b) \vee c \leq a \wedge (b \vee c)$
2.  $a \leq c$  implies that  $a \vee (b \wedge c) \leq (a \vee b) \wedge c$

*Proof.* Let  $a, b, c \in L$ .

1. Suppose that  $c \leq a$ . Then  $a \wedge c = c$ . The previous exercise then implies that

$$\begin{aligned} (a \wedge b) \vee c &= (a \wedge b) \vee (a \wedge c) \\ &\leq a \wedge (b \vee c). \end{aligned}$$

2. **FINISH!!!** use duality

□

**Exercise 11.3.1.5.** Let  $L$  be a lattice. Then for each  $a, b, c \in L$ ,

$$(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) \leq (a \vee b) \wedge (b \vee c) \wedge (c \vee a).$$

*Proof.* Let  $a, b, c \in L$ .

- We first note that

$$a \wedge b \leq a \leq a \vee b,$$

$$b \wedge c \leq b \leq a \vee b,$$

$$c \wedge a \leq a \leq a \vee b.$$

Therefore  $a \vee b \in \text{ub}\{a \wedge b, b \wedge c, c \wedge a\}$  and

$$\begin{aligned} (a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq a \vee b. \end{aligned}$$

- Similarly,

$$a \wedge b \leq b \leq b \vee c,$$

$$b \wedge c \leq c \leq b \vee c,$$

$$c \wedge a \leq c \leq b \vee c.$$

Therefore  $b \vee c \in \text{ub}\{a \wedge b, b \wedge c, c \wedge a\}$  and

$$\begin{aligned} (a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq b \vee c. \end{aligned}$$

- Finally, we have that

$$a \wedge b \leq a \leq c \vee a,$$

$$b \wedge c \leq c \leq c \vee a,$$

$$c \wedge a \leq c \leq c \vee a.$$

Therefore  $c \vee a \in \text{ub}\{a \wedge b, b \wedge c, c \wedge a\}$  and

$$\begin{aligned} (a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq c \vee a. \end{aligned}$$

Hence  $(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) \in \text{lb}\{a \vee b, b \vee c, c \vee a\}$  and

$$\begin{aligned} (a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &\leq \inf\{a \vee b, b \vee c, c \vee a\} \\ &= (a \vee b) \wedge (b \vee c) \wedge (c \vee a). \end{aligned}$$

□

**Exercise 11.3.1.6.** Let  $L$  be a lattice. Then the following are equivalent:

1. For each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee c$ .
2. For each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ .
3. For each  $p, q, r \in L$ ,  $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$ .

*Proof.*

1. (1)  $\implies$  (2):

Suppose that for each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee c$ . Let  $a, b, c \in L$ . Suppose that  $c \leq a$ . Then  $a \wedge c = c$ . By assumption

$$\begin{aligned} a \wedge (b \vee c) &= (a \wedge b) \vee c \\ &= (a \wedge b) \vee (a \wedge c) \end{aligned}$$

Since  $a, b, c \in L$  with  $c \leq a$  are arbitrary, we have that for each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ .

2. (2)  $\implies$  (3):

Suppose that for each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ . Let  $p, q, r \in L$ . Define  $a, b, c \in L$  by  $a := p$ ,  $b := q$  and  $c := p \wedge r$ . Then

$$\begin{aligned} a \wedge c &= p \wedge (p \wedge r) \\ &= (p \wedge p) \wedge r \\ &= p \wedge r. \end{aligned}$$

By assumption,

$$\begin{aligned} p \wedge (q \vee (p \wedge r)) &= a \wedge (b \vee c) \\ &= (a \wedge b) \vee (a \wedge c) \\ &= (p \wedge q) \vee (p \wedge r). \end{aligned}$$

Since  $p, q, r \in L$  are arbitrary, we have that for each  $p, q, r \in L$ ,  $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$ .

3. (3)  $\implies$  (1):

Suppose that for each  $p, q, r \in L$ ,  $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$ . Let  $a, b, c \in L$ . Suppose that  $c \leq a$ . Define  $p, q, r \in L$  by  $p := a$ ,  $q := b$  and  $r := c$ . Since  $c \leq a$ , we have that

$$\begin{aligned} p \wedge r &= a \wedge c \\ &= c. \end{aligned}$$

By assumption,

$$\begin{aligned} a \wedge (b \vee c) &= p \wedge (q \vee (p \wedge r)) \\ &= (p \wedge q) \vee (p \wedge r) \\ &= (a \wedge b) \vee c. \end{aligned}$$

Since  $a, b, c \in L$  with  $c \leq a$  are arbitrary, we have that for each  $a, b, c \in L$ ,  $c \leq a$  implies that  $a \wedge (b \vee c) = (a \wedge b) \vee c$ .

□

**Definition 11.3.1.7.** Let  $L$  be a lattice. Then  $L$  is said to be **modular** if for each  $a, b, c \in L$ ,

$$c \leq a \text{ implies that } a \wedge (b \vee c) = (a \wedge b) \vee c.$$

**Exercise 11.3.1.8.** Let  $L$  be a lattice. Then the following are equivalent:

1. For each  $a, b, c \in L$ ,  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ .
2. For each  $p, q, r \in L$ ,  $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$ .

*Proof.*

1. (1)  $\implies$  (2):

Suppose that for each  $a, b, c \in L$ ,  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ . Let  $p, q, r \in L$ . Define  $a, b, c \in L$  by  $a := p \vee q$ ,  $b := p$  and  $c := r$ . By absorption,

$$\begin{aligned} a \wedge b &= (p \vee q) \wedge p \\ &= p \wedge (p \vee q) \\ &= p. \end{aligned}$$

By assumption,

$$\begin{aligned} a \wedge c &= (p \vee q) \wedge r \\ &= r \wedge (p \vee q) \\ &= (r \wedge p) \vee (r \wedge q). \end{aligned}$$

Then by assumption and absorption,

$$\begin{aligned} (p \vee q) \wedge (p \vee r) &= a \wedge (b \vee c) \\ &= (a \wedge b) \vee (a \wedge c) \\ &= p \vee [(r \wedge p) \vee (r \wedge q)] \\ &= [p \vee (r \wedge p)] \vee (r \wedge q) \\ &= [p \vee (p \wedge r)] \vee (q \wedge r) \\ &= p \vee (q \wedge r). \end{aligned}$$

Since  $p, q, r \in L$  are arbitrary, we have that for each  $p, q, r \in L$ ,  $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$ .

2. (2)  $\implies$  (1):

Suppose that for each  $p, q, r \in L$ ,  $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$ . Let  $a, b, c \in L$ . Define  $p, q, r \in L$  by  $p := a \wedge b$ ,  $q := a$  and  $r := c$ . By absorption,

$$\begin{aligned} p \vee q &= (a \wedge b) \vee a \\ &= a \vee (a \wedge b) \\ &= a. \end{aligned}$$

By assumption,

$$\begin{aligned} p \vee r &= (a \wedge b) \vee c \\ &= c \vee (a \wedge b) \\ &= (c \vee a) \wedge (c \vee b). \end{aligned}$$

Then by assumption and absorption,

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &= p \vee (q \wedge r) \\ &= (p \vee q) \wedge (p \vee r) \\ &= a \wedge [(c \vee a) \wedge (c \vee b)] \\ &= [a \wedge (c \vee a)] \wedge (c \vee b) \\ &= [a \wedge (a \vee c)] \wedge (b \vee c) \\ &= a \wedge (b \vee c). \end{aligned}$$

Since  $a, b, c \in L$  are arbitrary, we have that for each  $a, b, c \in L$ ,  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ .

□

**Definition 11.3.1.9.** Let  $L$  be a lattice. Then  $L$  is said to be **distributive** if for each  $a, b, c \in L$ ,

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

**Exercise 11.3.1.10.** Let  $L$  be a lattice. If  $L$  is distributive, then  $L$  is modular.

*Proof.* Suppose that  $L$  is distributive. **FINISH!!!**

□

**Exercise 11.3.1.11.** Let  $L$  be a lattice. Then for each  $a, b, c \in L$ ,  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$  iff  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ .

*Proof.* Let  $a, b, c \in L$ .

- $(\implies)$ :  
Suppose that  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ .
- $(\impliedby)$ :

□

### 11.3.2 Heyting Algebras

**Definition 11.3.2.1.** Let  $(L, \vee, \wedge)$  be a lattice and  $\rightarrow: L \times L \rightarrow L$ . Then

- $\rightarrow$  is said to be a  $(\vee, \wedge)$ -**Heyting implication operator** if for each  $a, x, y \in L$ ,  $(x \wedge a) \leq_L y$  iff  $x \leq_L (a \rightarrow y)$ ,
- $(L, \vee, \wedge, \rightarrow)$  is said to be a **Heyting algebra** if
  1.  $(L, \vee, \wedge)$  is a lattice with zero,
  2.  $\rightarrow: L \times L \rightarrow L$  is a  $(L, \vee, \wedge)$ -Heyting implication operator.

(ask if there is any way to show that  $L$  has a zero instead of showing it. Maybe there is a result saying that this is impossible in general.)

**Note 11.3.2.2.** When the context is clear, we say “Heyting implication operator” instead of “ $(L, \vee, \wedge)$ -Heyting implication operator” and we write  $L$  in place of  $(L, \vee, \wedge, \rightarrow)$ .

**Exercise 11.3.2.3.** Let  $L$  be a Heyting algebra. Then

1.  $L$  is a lattice with one
2. for each  $a \in L$ ,  $a \rightarrow a = 1_L$ .

(ask if there is any way to show that  $L$  has a zero instead of showing it. Maybe there is a result saying that this is impossible in general.)

*Proof.* Let  $a \in L$ . Define  $y \in L$  by  $y := a$ . Then

$$\begin{aligned} (x \wedge a) &\leq a \\ &= y. \end{aligned}$$

Then by **definition**, we have that  $x \leq a \rightarrow y$ . Since  $x \in L$  is arbitrary, we have that for each  $x \in L$ ,

$$\begin{aligned} x &\leq (a \rightarrow y) \\ &= a \rightarrow a. \end{aligned}$$

Thus  $a \rightarrow a$  is a one of  $L$ . Since  $a \in L$  is arbitrary, we have that for each  $a \in L$ ,  $a \rightarrow a$  is a one of  $L$ .

1. Since  $L$  is a lattice,  $L \neq \emptyset$ . Thus there exists  $a \in L$ . Since  $a \rightarrow a \in L$  and  $a \rightarrow a$  is a one of  $L$ ,  $L$  is a lattice with one.
2. **the part above** implies that for each  $a \in L$ ,  $a \rightarrow a = 1_L$ .

□

**Exercise 11.3.2.4.** Let  $(L, \vee, \wedge)$  be a lattice with zero and  $\rightarrow: L \times L \rightarrow L$ . Then  $(L, \vee, \wedge, \rightarrow)$  is a Heyting algebra iff for each  $a \in L$ ,  $((\cdot \wedge a), (a \rightarrow \cdot))$  is a  $(\leq_L, \leq_L)$ -monotone Galois connection.

Can we get rid of assumption about having 0?

*Proof.* For each  $a \in L$ , define  $f_a, g_a: L \rightarrow L$  by  $f_a := \cdot \wedge a$  and  $g_a := a \rightarrow \cdot$ .

•  $(\implies)$ :

Suppose that  $(L, \vee, \wedge, \rightarrow)$  is a Heyting algebra. By definition,  $\rightarrow: L \times L \rightarrow L$  is a  $(L, \vee, \wedge)$ -Heyting implication operator. Therefore, for each  $a, x, y \in L$ ,

$$\begin{aligned} f_a(x) \leq_L y &\iff (x \wedge a) \leq_L y \\ &\iff x \leq_L (a \rightarrow y) \\ &\iff x \leq_L g_a(y). \end{aligned}$$

Thus for each  $a \in L$ ,  $(f_a, g_a)$  is a  $(\leq_L, \leq_L)$ -monotone Galois connection.

•  $(\impliedby)$ :

Suppose that for each  $a \in L$ ,  $(f_a, g_a)$  is a  $(\leq_L, \leq_L)$ -monotone Galois connection.

1. By assumption,  $(L, \vee, \wedge)$  is a lattice with zero.
2. We have that for each  $a, x, y \in L$ ,

$$\begin{aligned} (x \wedge a) \leq_L y &\iff f_a(x) \leq_L y \\ &\iff x \leq_L g_a(y) \quad (\text{by assumption}) \\ &\iff x \leq_L (a \rightarrow y). \end{aligned}$$

Hence  $\rightarrow: L \times L \rightarrow L$  is a  $(\vee, \wedge)$ -Heyting implication operator.

Thus  $(L, \vee, \wedge, \rightarrow)$  is a Heyting algebra. □

**Exercise 11.3.2.5.** Let  $L$  be a Heyting algebra. Then for each  $a, b \in L$ ,  $a \rightarrow b = \sup\{x \in L : x \wedge a \leq b\}$  (maybe write  $\max A$ ? need to define  $\max A$ ).

*Proof.* Let  $a, b \in L$ . Define  $A \subset L$  by  $A := \{x \in L : x \wedge a \leq b\}$ . Define  $y \in L$  by  $y := b$ . We note that

$$\begin{aligned} x \in A &\iff x \wedge a \leq b \\ &\iff x \wedge a \leq y \\ &\iff x \leq a \rightarrow y \quad (\text{by def}) \\ &\iff x \leq a \rightarrow b. \end{aligned}$$

1. Define  $x_0 \in L$  by  $x_0 := (a \rightarrow b)$ . Since

$$\begin{aligned} x_0 &= a \rightarrow b \\ &\leq a \rightarrow b, \end{aligned}$$

we have that  $x_0 \in A$ .

2. Let  $x \in A$ . Then

$$\begin{aligned} x &\leq a \rightarrow b \\ &= x_0. \end{aligned}$$

Since  $x \in A$  is arbitrary, we have that for each  $x \in A$ ,  $x \leq x_0$ .

Exercise ?? ref prev ex implies that  $x_0 = \sup A$  (maybe write  $\max A$ ? need to define  $\max A$ ). □



**Exercise 11.3.2.6.** Let  $L$  be a Heyting algebra. Then for each  $x, y \in L$ ,  $x \leq y$  iff  $x \rightarrow y = 1$ .

*Proof.* Let  $x, y \in L$ . We note that  $1 \wedge x = x$  and for each  $a \in L$ ,  $a = 1$  iff  $1 \leq a$ . Therefore

$$\begin{aligned} x \rightarrow y = 1 &\iff 1 \leq x \rightarrow y \\ &\iff 1 \wedge x \leq y \\ &\iff x \leq y. \end{aligned}$$

Since  $x, y \in L$  are arbitrary, we have that for each  $x, y \in L$ ,  $x \leq y$  iff  $x \rightarrow y = 1$ . □

**Definition 11.3.2.7.** Let  $(L, \vee, \wedge, \rightarrow)$  be a Heyting algebra. We define the  $\rightarrow$ -**negation operator**, denoted  $\neg_{\rightarrow} : L \rightarrow L$ , by  $\neg_{\rightarrow} x := x \rightarrow 0$ .

**Note 11.3.2.8.** When the context is clear, we write  $\neg$  in place of  $\neg_{\rightarrow}$ .

**Exercise 11.3.2.9.** Let  $L$  be a Heyting algebra. Then  $\neg : L \rightarrow L$  is a pseudocomplement.

*Proof.* **FINISH!!!!**, define pseudocomplement (and complement and orthocomplemented lattice, etc) □

### 11.3.3 Boolean Algebras

## 11.4 Lattice Structures

### 11.4.1 Irreducibility and Primality

**Definition 11.4.1.1.** Let  $(L, \leq)$  be a poset.

- Suppose that  $(L, \leq)$  is a join semilattice.
  - Let  $x \in L$ . Then  $x$  is said to be
    - \*  **$\leq$ -join irreducible** if
      1.  $x \neq 0$ ,
      2. for each  $a, b \in L$ ,  $x = a \vee b$  implies that  $x = a$  or  $x = b$ ,
    - \*  **$\leq$ -join prime**
      1.  $x \neq 0$ ,
      2. if for each  $a, b \in L$ ,  $x \leq a \vee b$  implies that  $x \leq a$  or  $x \leq b$ .
  - We define
    - \*  $\text{irred}_{\vee}(L, \leq) := \{x \in L : x \text{ is } \leq\text{-join irreducible}\}$ ,
    - \*  $\text{prime}_{\vee}(L, \leq) := \{x \in L : x \text{ is } \leq\text{-join prime}\}$ .
- Suppose that  $(L, \leq)$  is a meet semilattice.
  - Let  $x \in L$ . Then  $x$  is said to be
    - \*  **$\leq$ -meet irreducible**
      1.  $x \neq 1$ ,
      2. if for each  $a, b \in L$ ,  $x = a \wedge b$  implies that  $x = a$  or  $x = b$ ,
    - \*  **$\leq$ -meet prime** if
      1.  $x \neq 1$ ,
      2. for each  $a, b \in L$ ,  $a \wedge b \leq x$  implies that  $a \leq x$  or  $b \leq x$ .
  - We define
    - \*  $\text{irred}_{\wedge}(L, \leq) := \{x \in L : x \text{ is } \leq\text{-meet irreducible}\}$ ,
    - \*  $\text{prime}_{\wedge}(L, \leq) := \{x \in L : x \text{ is } \leq\text{-meet prime}\}$ .

**Note 11.4.1.2.** When the context is clear,

- we say “join irreducible”, “join prime”, “meet irreducible” and “meet prime” in place of “ $\leq$ -join irreducible”, “ $\leq$ -join prime”, “ $\leq$ -meet irreducible” and “ $\leq$ -meet prime” respectively,
- we write  $\text{irred}_{\vee}(L)$ ,  $\text{prime}_{\vee}(L)$ ,  $\text{irred}_{\wedge}(L)$  and  $\text{prime}_{\wedge}(L)$  in place of  $\text{irred}_{\vee}(L, \leq)$ ,  $\text{prime}_{\vee}(L, \leq)$ ,  $\text{irred}_{\wedge}(L, \leq)$  and  $\text{prime}_{\wedge}(L, \leq)$  respectively

**Exercise 11.4.1.3.** Let  $(L, \leq)$  be a poset.

1. Suppose that  $(L, \leq)$  is a join semilattice. Then  $\text{prime}_{\vee}(L) \subset \text{irred}_{\vee}(L)$ .
2. Suppose that  $(L, \leq)$  is a meet semilattice. Then  $\text{prime}_{\wedge}(L) \subset \text{irred}_{\wedge}(L)$ .

*Proof.*

1. Let  $x \in \text{prime}_{\vee}(L)$ .
  - (a) Since  $x$  is join prime,  $x$  is not at zero of  $L$ .
  - (b) Let  $a, b \in L$ . Suppose that  $x = a \vee b$ . Then  $x \leq a \vee b$ . Since  $x$  is join prime,  $x \leq a$  or  $x \leq b$ .

- Suppose that  $x \leq a$ . Then

$$\begin{aligned} a \vee b &= x \\ &\leq a \\ &\leq a \vee b \end{aligned}$$

and therefore  $x = a$ . Thus  $x \leq a$  implies that  $x = a$ .

- Similarly,  $x \leq b$  implies that  $x = b$ .

Since  $x \leq a$  or  $x \leq b$ , we have that  $x = a$  or  $x = b$ . Since  $a, b \in L$  with  $x = a \vee b$  are arbitrary, we have that for each  $a, b \in L$ ,  $x = a \vee b$  implies that  $x = a$  or  $x = b$ .

Thus  $x \in \text{irred}_\vee(L)$ . Since  $x \in \text{prime}_\vee(L)$  is arbitrary, we have that for each  $x \in \text{prime}_\vee(L)$ ,  $x \in \text{irred}_\vee(L)$ . Hence  $\text{prime}_\vee(L) \subset \text{irred}_\vee(L)$ .

2. Similar to (1) **use duality**.

□

### 11.4.2 Ideals and Filters

**Exercise 11.4.2.1.** Let  $L$  be a lattice and  $I, F \subset L$ . Then

1.  $I \in \mathcal{I}(L)$  iff
  - $I \neq \emptyset$ ,
  - $\wedge(I \times L) \subset I$ ,
  - $\vee(I \times I) \subset I$ .
2.  $F \in \mathcal{I}(L)$  iff
  - $I \neq \emptyset$ ,
  - $\vee(F \times L) \subset F$ ,
  - $\wedge(F \times F) \subset F$ .

*Proof.*

1. • ( $\implies$ ):  
 Suppose that  $I \in \mathcal{I}(L)$ . Exercise 11.1.5.2 implies that  $I \neq \emptyset$ ,  $I \in \mathcal{L}(L)$  and  $\vee(I \times I) \subset I$ .
  - As noted above,  $I \neq \emptyset$ .
  - Exercise 11.1.5.1 implies that  $\wedge(I \times L) \subset I$ .
  - As noted above,  $\vee(I \times I) \subset I$ .
- ( $\impliedby$ ):  
 Suppose that
  - $I \neq \emptyset$ ,
  - $\wedge(I \times L) \subset I$ ,
  - $\vee(I \times I) \subset I$ .
 Exercise 11.1.5.1 implies that  $I \in \mathcal{L}(L)$ . Exercise 11.1.5.2 then implies that  $I \in \mathcal{I}(L)$ .

- 2.

□

**Exercise 11.4.2.2.** Let  $L$  be a lattice. Then

1. for each  $I_1, I_2 \in \mathcal{I}(L)$ ,  $\{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\} \in \mathcal{I}(L)$ .

2. for each  $F_1, F_2 \in \mathcal{F}(L)$ ,  $\{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } a \vee b \leq x\} \in \mathcal{F}(L)$  (maybe FIX!!!).

*Proof.*

1. Let  $I_1, I_2 \in \mathcal{I}(L)$ . Set  $I := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$ .

- (a) Since  $I_1, I_2 \in \mathcal{I}(L)$ , we have that  $I_1 \neq \emptyset$  and  $I_2 \neq \emptyset$ . Hence there exist  $x_1 \in I_1$  and  $x_2 \in I_2$ . Since  $x_1 \leq x_1 \vee x_2$ , we have that  $x_1 \in I$ . Thus  $I \neq \emptyset$ .
- (b) Let  $a, b \in I$ . Then there exist  $x_a, x_b \in I_1$  and  $y_a, y_b \in I_2$  such that  $a \leq x_a \vee y_a$  and  $b \leq x_b \vee y_b$ . Define  $z_1, z_2 \in L$  by  $z_1 := x_a \vee x_b$  and  $z_2 := y_a \vee y_b$ . Since  $I_1, I_2 \in \mathcal{I}(L)$ , we have that  $z_1 \in I_1$  and  $z_2 \in I_2$ . We note that

$$\begin{aligned} a &\leq x_a \vee y_a \\ &\leq (x_a \vee y_a) \vee (x_b \vee y_b) \\ &= (x_a \vee x_b) \vee (y_a \vee y_b) \\ &= z_1 \vee z_2. \end{aligned}$$

and similarly  $b \leq z_1 \vee z_2$ . Therefore  $z_1 \vee z_2 \in \text{ub}(\{a, b\}, \leq)$ . Hence

$$\begin{aligned} a \vee b &= \sup(\{a, b\}, \leq) \\ &\leq z_1 \vee z_2. \end{aligned}$$

Thus  $a \vee b \in I$ . Since  $a, b \in I$  are arbitrary, we have that for each  $a, b \in I$ ,  $a \vee b \in I$ .

- (c) Let  $x \in L$  and  $y \in I$ . Then there exists  $a \in I_1$  and  $b \in I_2$  such that  $y \leq a \vee b$ . Then

$$\begin{aligned} x \wedge y &\leq y \\ &\leq a \vee b. \end{aligned}$$

Therefore  $x \wedge y \in I$ . Since  $x \in L$  and  $y \in I$  are arbitrary, we have that for each  $x \in L$  and  $y \in I$ ,  $x \wedge y \in I$ .

Therefore  $I \in \mathcal{I}(L)$ .

2. Similar to (1). **FINISH!!!**

□

**Definition 11.4.2.3.** Let  $L$  be a lattice. Define  $\leq_{\mathcal{I}(L)} := \subset|_{\mathcal{I}(L)}$  (maybe standardize notation).

**Exercise 11.4.2.4.** Let  $L$  be a lattice.

1. Let  $I_1, I_2 \in \mathcal{I}(L)$ .

- (a)  $\sup(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) = \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$ .
- (b)  $\inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) = I_1 \cap I_2$

2. Set  $\leq_{\mathcal{F}(L)} := \subset|_{\mathcal{F}(L)}$  (maybe standardize notation). Let  $F_1, F_2 \in \mathcal{F}(L)$ . **FIX!!!**

- (a)  $\sup(\{F_1, F_2\}, \leq_{\mathcal{F}(L)}) = \{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } x \leq a \vee b\}$ .
- (b)  $\inf(\{F_1, F_2\}, \leq_{\mathcal{F}(L)}) = F_1 \cap F_2$

*Proof.*

1. (a) Set  $I := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$ . (ref previous ex) implies that  $I \in \mathcal{I}(L)$ .

- i. • Let  $a \in I_1$ . Since  $I_2 \in \mathcal{I}(L)$ ,  $I_2 \neq \emptyset$ . Hence there exists  $b \in I_2$ . Then  $a \leq a \vee b$ . Since  $a \in I_1$ ,  $b \in I_2$  and  $a \leq a \vee b$ , we have that  $a \in I$ . Since  $a \in I_1$  is arbitrary, we have that for each  $a \in I_1$ ,  $a \in I$ . Thus  $I_1 \subset I$ .
- Similarly,  $I_2 \subset I$ .

Since  $I_1, I_2 \subset I$ , we have that  $I \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ .

- ii. Let  $K \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$  and  $x \in I$ . Then  $I_1, I_2 \subset K$  and there exist  $a \in I_1$  and  $b \in I_2$  such that  $x \leq a \vee b$ . Since  $a \in I_1$ ,  $b \in I_2$  and  $I_1, I_2 \subset K$ , we have that  $a, b \in K$ . Since  $K \in \mathcal{I}(L)$ ,  $a \vee b \in K$ . Since  $K \in \mathcal{I}(L)$  (ref a prev ex) implies that  $K$  is a lower set. Since  $K$  is a lower set,  $a \vee b \in K$  and  $x \leq a \vee b$ , we have that  $x \in K$ . Since  $x \in I$  is arbitrary, we have that for each  $x \in I$ ,  $x \in K$ . Hence  $I \subset K$ . Since  $K \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$  is arbitrary, we have that for each  $K \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ ,  $I \leq_{\mathcal{I}(L)} K$ .

Since

- i.  $I \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ ,
- ii. for each  $K \in \text{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ ,  $I \leq_{\mathcal{I}(L)} K$ ,

we have that  $I = \sup(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ .

- (b) (ref previous ex) implies that  $I_1 \cap I_2 \in \mathcal{I}(L)$ . Since  $I_1 \cap I_2 \subset I_1$  and  $I_1 \cap I_2 \subset I_2$ , we have that  $I_1 \cap I_2 \in \text{lb}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ . Since  $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$  is a subposet of  $(\mathcal{P}(L), \subset)$ , (define subproset/subposet and make exercise for the following fact) implies that

$$\begin{aligned} \inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) &\leq \inf(\{I_1, I_2\}, \subset) \\ &= I_1 \cap I_2 \\ &\leq \inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}). \end{aligned}$$

Hence  $I_1 \cap I_2 = \inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ .

2. Similar to (1). **FINISH!!!**

□

**Definition 11.4.2.5.** Let  $L$  be a lattice. Define  $\vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)} : \mathcal{I}(L) \times \mathcal{I}(L) \rightarrow \mathcal{I}(L)$  and  $\vee_{\mathcal{F}(L)}, \wedge_{\mathcal{F}(L)} : \mathcal{F}(L) \times \mathcal{F}(L) \rightarrow \mathcal{F}(L)$  by

1.
  - $I_1 \vee_{\mathcal{I}(L)} I_2 := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$
  - $I_1 \wedge_{\mathcal{I}(L)} I_2 := I_1 \cap I_2$ .
2. **FIX!!!**
  - $F_1 \vee_{\mathcal{F}(L)} F_2 := \{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } x \leq a \vee b\}$
  - $F_1 \wedge_{\mathcal{F}(L)} F_2 := F_1 \cap F_2$ .

**Exercise 11.4.2.6.** Let  $L$  be a lattice. Then

1.  $(\mathcal{I}(L), \vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)})$  is a lattice
2.  $(\mathcal{F}(L), \vee_{\mathcal{F}(L)}, \wedge_{\mathcal{F}(L)})$  is a lattice

*Proof.*

1. the previous exercise implies that  $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$  is an ordered lattice. Exercise 11.2.2.6 then implies that  $(\mathcal{I}(L), \vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)})$  is a lattice.
- 2.

□

### 11.4.3 Properties of Ideals and Filters

**Definition 11.4.3.1.** Let  $L$  be a lattice.

- Let  $I \in \mathcal{I}(L)$ . Then  $I$  is said to be
  - **proper** if  $I \neq L$ .
  - **maximal** if  $I$  is maximal in  $(\mathcal{I}(L) \setminus \{L\}, \subset)$ .
  - **prime** if

- 1.  $I$  is proper
- 2. for each  $a, b \in L$ ,  $a \wedge b \in I$  implies that  $a \in I$  or  $b \in I$ .
- **principal** if there exists  $a \in L$  such that  $I = \downarrow a$ .
- Let  $F \in \mathcal{F}(L)$ . Then  $F$  is said to be
  - **proper** if  $F \neq L$ .
  - **maximal** if  $F$  is maximal in  $(\mathcal{F}(L) \setminus \{L\}, \subset)$ .
  - **prime** if
    - 1.  $F$  is proper
    - 2. for each  $a, b \in L$ ,  $a \vee b \in F$  implies that  $a \in F$  or  $b \in F$ .
  - **principal** if there exists  $a \in L$  such that  $F = \uparrow a$ .

**Exercise 11.4.3.2.** Let  $L$  be a lattice.

1. For each  $I \in \mathcal{I}(L)$ .
  - (a) if  $I$  is proper, then for each  $x \in I$ ,  $x$  is not a one of  $L$ .
  - (b) if  $I$  is principal, then  $I$  is proper iff for each  $x \in I$ ,  $x$  is not a one of  $L$ .
2. For each  $F \in \mathcal{F}(L)$ .
  - (a) If  $F$  is proper, then for each  $x \in F$ ,  $x$  is not a zero of  $L$ .
  - (b) If  $F$  is principal, then  $F$  is proper iff for each  $x \in F$ ,  $x$  is not a zero of  $L$ .

*Proof.*

1. Let  $I \in \mathcal{I}(L)$ .
  - (a) Suppose that  $I$  is proper. Let  $x \in I$ . For the sake of contradiction, suppose that  $x$  is a one of  $L$ . (previous ex) then implies that  $(L, \leq) = \{x\}$ . Let  $y \in L$ . Then  $y \leq x$ . Since  $I$  is a lower set,  $x \in I$  and  $y \leq x$ , we have that  $y \in I$ . Since  $y \in L$  is arbitrary, we have that for each  $y \in L$ ,  $y \in I$ . Thus

$$\begin{aligned} L &\subset I \\ &\subset L. \end{aligned}$$

Thus  $I = L$  and  $I$  is not proper. Therefore  $x$  is not a one of  $L$ . Since  $x \in I$  is arbitrary, we have that for each  $x \in I$ ,  $x$  is not a one of  $L$ .

- (b) Suppose that  $I$  is principal.
  - ( $\implies$ ):  
If  $I$  is proper, then 1(a) implies that for each  $x \in I$ ,  $x$  is not a one of  $L$ .
  - ( $\impliedby$ ):  
Suppose that  $I$  is not proper. Then  $I = L$ . Since  $I$  is principal, there exists  $a \in I$  such that  $I = \downarrow a$ . Let  $x \in L$ . Since

$$\begin{aligned} x &\in L \\ &= I \\ &= \downarrow a, \end{aligned}$$

we have that  $x \leq a$ . Hence

$$\begin{aligned} x \wedge a &= \inf\{x, a\} \\ &= x. \end{aligned}$$

Since  $x \in L$  is arbitrary, we have that for each  $x \in L$ ,  $x \wedge a = x$ . Hence  $a$  is a one of  $L$ . Therefore  $I$  is not proper implies that there exists  $a \in I$  such that  $a$  is a one of  $L$ . By contrapositive, we have that if for each  $x \in I$ ,  $x$  is not a one of  $L$ , then  $I$  is proper.

2. **FINISH!!**

□

**Exercise 11.4.3.3.** Let  $L$  be a lattice and  $x \in L$ .

1. Then  $\downarrow x \in \mathcal{I}_p(L)$  iff  $x \in \text{prime}_\wedge(L)$ .
2. Then  $\uparrow x \in \mathcal{F}_p(L)$  iff  $x \in \text{prime}_\vee(L)$ .

*Proof.*

1. • ( $\implies$ ):  
 Suppose that  $\downarrow x \in \mathcal{I}_p(L)$ . Then  $\downarrow x$  is proper and for each  $a, b \in L$ ,  $a \wedge b \in \downarrow x$  implies that  $a \in \downarrow x$  or  $b \in \downarrow x$ .
  - Since  $\downarrow x$  is proper, **previous exercise** implies that  $x$  is not a one of  $L$ .
  - Let  $a, b \in L$ . Suppose that  $a \wedge b \leq x$ . Since  $\downarrow x$  is a lower set,  $x \in \downarrow x$  and  $a \wedge b \leq x$ , we have that  $a \wedge b \in \downarrow x$ . Therefore  $a \in \downarrow x$  or  $b \in \downarrow x$ . Then  $a \leq x$  or  $b \leq x$ . Since  $a, b \in L$  with  $a \wedge b \leq x$  are arbitrary, we have that for each  $a, b \in L$ ,  $a \wedge b \leq x$  implies that  $a \leq x$  or  $b \leq x$ .
 Thus  $x \in \text{prime}_\wedge(L)$ .
  - ( $\impliedby$ ):  
 Suppose that  $x \in \text{prime}_\wedge(L)$ .
    - For the sake of contradiction, suppose that  $\downarrow x$  is not proper. Then  $\downarrow x = L$ . Thus  $\text{ub } L = \{x\}$ . **A previous ex** implies that  $x$  is a one of  $L$ . This is a contradiction since  $x \in \text{prime}_\wedge(L)$ . Therefore  $\downarrow x$  is proper.
    - Let  $a, b \in L$ . Suppose that  $a \wedge b \in \downarrow x$ . Then  $a \wedge b \leq x$ . Since  $x \in \text{prime}_\wedge(L)$ ,  $a \leq x$  or  $b \leq x$ . Hence  $a \in \downarrow x$  or  $b \in \downarrow x$ . Since  $a, b \in L$  with  $a \wedge b \in \downarrow x$  are arbitrary, we have that for each  $a, b \in L$ ,  $a \wedge b \in \downarrow x$  implies that  $a \in \downarrow x$  or  $b \in \downarrow x$ .
 Therefore  $\downarrow x \in \mathcal{I}_p(L)$ .
2. Similar to (1) **FIX/CHECK/FINISH!!!**

□

**Exercise 11.4.3.4.** Let  $L$  be a lattice.

1. For each  $I \in \mathcal{I}(L)$ ,  $I \in \mathcal{I}_p(L)$  iff  $I^c \in \mathcal{F}(L)$ .
2. For each  $F \in \mathcal{F}(L)$ ,  $F \in \mathcal{F}_p(L)$  iff  $F^c \in \mathcal{I}(L)$ .

*Proof.*

1. Let  $I \in \mathcal{I}(L)$ .
  - ( $\implies$ ):  
 Suppose that  $I \in \mathcal{I}_p(L)$ . Then  $I$  is proper and for each  $a, b \in L$ ,  $a \wedge b \in I$  implies that  $a \in I$  or  $b \in I$ .
    - (a) Since  $I$  is proper,  $I \neq L$ . Therefore  $I^c \neq \emptyset$
    - (b) Let  $a, b \in I^c$ . For the sake of contradiction, suppose that  $a \wedge b \in I$ . Since  $I \in \mathcal{I}_p(L)$ ,  $a \in I$  or  $b \in I$ . This is a contradiction since  $a \in I^c$  and  $b \in I^c$ . Therefore  $a \wedge b \notin I$ . Hence  $a \wedge b \in I^c$ . Since  $a, b \in I^c$  are arbitrary, we have that for each  $a, b \in I^c$ ,  $a \wedge b \in I^c$ .
    - (c) Since  $I \in \mathcal{I}(L)$ , **a previous exercise** implies that  $I$  is a lower set. **an exercise in proset section** implies that  $I^c$  is an upper set. Let  $x \in L$  and  $a \in I^c$ . Since

$$\begin{aligned} x \vee a &\geq a \\ &\in I^c \end{aligned}$$

and  $I^c$  is an upper set, we have that  $x \vee a \in I^c$ . Since  $x \in L$  and  $a \in I^c$  are arbitrary, we have that for each  $x \in L$  and  $a \in I^c$ ,  $x \vee a \in I^c$ .<sup>1</sup>

Thus  $I^c \in \mathcal{F}(L)$ .

- ( $\Leftarrow$ ):

Suppose that  $I^c \in \mathcal{F}(L)$ .

- Since  $I^c \in \mathcal{F}(L)$ ,  $I^c \neq \emptyset$ . Then  $I \neq L$  and therefore  $I$  is proper.
- Let  $a, b \in L$ . Suppose that  $a \wedge b \in I$ . For the sake of contradiction, suppose that  $a \notin I$  and  $b \notin I$ . Then  $a \in I^c$  and  $b \in I^c$ . Since  $I^c \in \mathcal{F}(L)$ , we have that  $a \wedge b \in I^c$ . This is a contradiction since  $a \wedge b \in I$ . Hence  $a \in I$  or  $b \in I$ . Since  $a, b \in L$  with  $a \wedge b \in I$  are arbitrary, we have that for each  $a, b \in L$ ,  $a \wedge b \in I$  implies that  $a \in I$  or  $b \in I$ .

Thus  $I \in \mathcal{I}_p(L)$ .

2. use duality.

□

## 11.5 Contructions with Lattices

### 11.5.1 Sublattices

**Definition 11.5.1.1.** Let  $(L, \vee_L, \wedge_L)$  and  $(M, \vee_M, \wedge_M)$  be lattices. Then  $(M, \vee_M, \wedge_M)$  is said to be a **sublattice of**  $(L, \vee_L, \wedge_L)$  if

1.  $M \subset L$
2.  $\iota_M$  is a  $((\vee_L, \wedge_L), (\vee_M, \wedge_M))$ -homomorphism.

**Exercise 11.5.1.2.** Let  $(L, \vee_L, \wedge_L)$  and  $(M, \vee_M, \wedge_M)$  be lattices. Then  $(M, \vee_M, \wedge_M)$  is a sublattice of  $L$  iff

1.  $M \subset L$
2.  $\vee_L|_{M^2} = \vee_M$  and  $\wedge_L|_{M^2} = \wedge_M$ .

*Proof.*

- ( $\Rightarrow$ ):

Suppose that  $(M, \vee_M, \wedge_M)$  is a sublattice of  $L$ .

1. By definition,  $M \subset L$ .
2. By definition,  $\iota_M$  is a  $((\vee_L, \wedge_L), (\vee_M, \wedge_M))$ -homomorphism. Then for each  $a, b \in M$ ,

$$\begin{aligned} \vee_M(a, b) &= a \vee_M b \\ &= \iota_M(a \vee_M b) \\ &= \iota_M(a) \vee_L \iota_M(b) \\ &= a \vee_L b \\ &= \vee_L|_{M^2}(a, b). \end{aligned}$$

Hence  $\vee_L|_{M^2} = \vee_M$ . Similarly,  $\wedge_L|_{M^2} = \wedge_M$ .

- ( $\Leftarrow$ ):

Suppose that

1.  $M \subset L$
2.  $\vee_L|_{M^2} = \vee_M$  and  $\wedge_L|_{M^2} = \wedge_M$ .

1. By assumption  $M \subset L$ .



2. Since  $\vee_L|_{M^2} = \vee_M$ , we have that for each  $a, b \in M$ ,

$$\begin{aligned}\iota_M(a \vee_M b) &= a \vee_M b \\ &= \vee_M(a, b) \\ &= \vee_L|_{M^2}(a, b) \\ &= \vee_L(a, b) \\ &= a \vee_L b.\end{aligned}$$

Hence  $\iota_M$  is a  $(\vee_M, \vee_L)$ -homomorphism. Similarly,  $\iota_M$  is a  $(\wedge_M, \wedge_L)$ -homomorphism. Therefore  $\iota_M$  is a  $((\vee_L, \wedge_L), (\vee_M, \wedge_M))$ -homomorphism.

Thus  $(M, \vee_M, \wedge_M)$  is a sublattice of  $L$ .

**FINISH!!!!** □

need to make preliminary ex showing that if  $M \subset L$ ,  $M \neq \emptyset$  and  $\vee(M^2), \wedge(M^2) \subset M$ , then  $(M, \vee|_{M^2}, \wedge|_{M^2})$  is a lattice

**Exercise 11.5.1.3.** Let  $(L, \vee, \wedge)$  be a lattice and  $M \subset L$ . Then  $(M, \vee|_{M^2}, \wedge|_{M^2})$  is a sublattice of  $(L, \vee, \wedge)$  iff

1.  $M \neq \emptyset$
2.  $\vee(M^2), \wedge(M^2) \subset M$ .

*Proof.* Set  $\vee_M := \vee|_{M^2}$  and  $\wedge_M := \wedge|_{M^2}$

- ( $\implies$ ):  
Suppose that  $(M, \vee_M, \wedge_M)$  is a sublattice of  $(L, \vee, \wedge)$ . Since  $(M, \vee_M, \wedge_M)$  is a sublattice of  $(L, \vee, \wedge)$ , we have that  $(M, \vee_M, \wedge_M)$  is a lattice.
  1. Since  $(M, \vee_M, \wedge_M)$  is a lattice,  $M \neq \emptyset$ .
  2. Since  $(M, \vee_M, \wedge_M)$  is a lattice,  $\vee_M(M^2), \wedge_M(M^2) \subset M$ .
- ( $\impliedby$ ):  
Suppose that
  1.  $M \neq \emptyset$
  2.  $\vee(M^2), \wedge(M^2) \subset M$ .
  1. By assumption  $M \neq \emptyset$ ,
  2. Since  $\vee, \wedge$  are semilattice operators,  $\vee, \wedge$  satisfy the lattice absorption identities and  $\vee(M^2), \wedge(M^2) \subset M$ , we have that  $\vee_M, \wedge_M$  are semilattice operators and  $\vee|_M, \wedge|_M$  satisfy the lattice absorption identities.

Thus  $(M, \vee_M, \wedge_M)$  is a lattice. By assumption,  $M \subset L$  and  $\vee_M = \vee|_{M^2}$ ,  $\wedge_M = \wedge|_{M^2}$ , we have that  $(M, \vee_M, \wedge_M)$  is a sublattice of  $(L, \vee, \wedge)$ . □

## 11.5.2 Product Lattices

## 11.5.3 Coproduct Lattices

## 11.5.4 Quotient Lattices



# Chapter 12

## Complete Lattices

maybe move complete lattices to order chapter? two choices: a) cover complete lattices after order and before other structures

(including lattices?) so that we can define the 'subgroup/ring/lattice generated by a subset' or a 'subquiver generated by a sub set of vertices/edges' using the ideas of galois connection between two posets, one of which has extra structure b) cover complete lattices

after algebraic structures with lattices and note that we can capture all the subobjects with the notion of galois connection between complete lattices and give groups, rings etc as fundamental examples of complete lattices.

idk which is best option, it might make more sense to keep complete lattices after lattices, since we can talk about completely

prime ideals and other lattice-type objects, and we can give fundamental examples of complete lattices that were discussed prior like groups/rings/lattice ideals/etc

### 12.1 Introduction

**Exercise 12.1.0.1.** Let  $(X, \leq)$  be a poset. Then for each  $A \subset X$ ,  $\sup A$  exists iff for each  $A \subset X$ ,  $\inf A$  exists.

*Proof.*

- $(\implies)$ :  
Suppose that for each  $A \subset X$ ,  $\sup A$  exists. Let  $A \subset X$ . We note that

$$\sup(\text{lb } A) \text{ exists} \iff \inf A \text{ exists} \quad (\text{by Exercise 9.3.0.2}).$$

Since  $\text{lb } A \subset X$ , then by assumption,  $\sup(\text{lb } A)$  exists. Therefore  $\inf A$  exists. Since  $A \subset X$  is arbitrary, we have that for each  $A \subset X$ ,  $\inf A$  exists.

- $(\impliedby)$ :  
Similar to  $(\implies)$ .

□

**Definition 12.1.0.2.** Let  $(X, \leq)$  be a poset.

- Then  $(X, \leq)$  is said to be a **complete lattice** if for each  $A \subset X$ ,  $\sup A$  exists.
- If  $(X, \leq)$  is a complete lattice, we define the **arbitrary join and meet operators on  $X$** , denoted  $\bigvee_{\leq} : \mathcal{P}(X) \rightarrow X$  and  $\bigwedge_{\leq} : \mathcal{P}(X) \rightarrow X$  respectively, by  $\bigvee_{\leq} A := \sup(A, \leq)$  and  $\bigwedge_{\leq} A := \inf(A, \leq)$ .

**Note 12.1.0.3.** When the context is clear, we may write  $\bigvee$  and  $\bigwedge$  in place of  $\bigvee_{\leq}$  and  $\bigwedge_{\leq}$  respectively.

**Exercise 12.1.0.4.** Let  $(X, \leq)$  be a complete lattice. Then  $(X, \vee_{\leq}, \wedge_{\leq}, \inf(L, \leq), \sup(L, \leq), )$  is a bounded lattice.

*Proof.* **FIX!!!**

1. **make previous exercise** implies that  $\text{lb}(\emptyset, \leq) = X$ . Since  $(X, \leq)$  is a complete lattice and  $\emptyset \subset X$ , there exists  $x_1 \in X$  such that  $x_1 := \inf \emptyset$ . In particular,  $X \neq \emptyset$ .

2.
  - **make previous ex saying  $\inf \emptyset = \sup X$  and  $\sup \emptyset = \inf X$  with the following logic**

$$\begin{aligned} \inf \emptyset &\in \text{lb } \emptyset \cap \text{ub}(\text{lb } \emptyset) \\ &= X \cap \text{lb}(X) \\ &= \text{lb } X. \end{aligned}$$

**ref an ex** implying that  $\text{lb } X = \inf X$ . Hence  $X$  is a one of  $X$ .

- similar  $x_0$

Clearly  $\bigvee X$  is a one of  $X$  and  $\bigwedge X$  is a zero of  $X$ . □

## 12.2 Complete Lattice Homomorphisms

**Definition 12.2.0.1.** Let  $(X, \leq_X), (Y, \leq_Y)$  be complete lattices and  $f : X \rightarrow Y$ . Then  $f$  is said to

1.  **$(\leq_X, \leq_Y)$ -arbitrary join preserving** if for each  $A \subset X$ ,  $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$
2.  **$(\leq_X, \leq_Y)$ -arbitrary meet preserving** if for each  $A \subset X$ ,  $f(\inf(A, \leq_X)) = \inf(f(A), \leq_Y)$

**Exercise 12.2.0.2.** Let  $(X, \leq_X), (Y, \leq_Y)$  be complete lattices and  $f : X \rightarrow Y$ . Then

1.  $f$  is  $(\leq_X, \leq_Y)$ -arbitrary join preserving iff  $f$  is  $(\leq_X^*, \leq_Y^*)$ -arbitrary meet preserving.
2.  $f$  is  $(\leq_X, \leq_Y)$ -arbitrary meet preserving iff  $f$  is  $(\leq_X^*, \leq_Y^*)$ -arbitrary join preserving.

*Proof.*

1.
  - **$(\implies)$ :**  
Suppose that  $f$  is  $(\leq_X, \leq_Y)$ -arbitrary join preserving. Then **a prev ex in poset section on suprema** implies that for each  $A \subset X$ ,

$$\begin{aligned} f(\inf(A, \leq_X^*)) &= f(\sup(A, \leq_X)) \\ &= \sup(f(A), \leq_Y) \\ &= \inf(f(A), \leq_Y^*). \end{aligned}$$

Hence  $f$  is  $(\leq_X^*, \leq_Y^*)$ -arbitrary meet preserving.

- **$(\impliedby)$ :**  
Similar to  $(\implies)$ . **ref.**

2. Similar to (1). **ref.** □

**Definition 12.2.0.3. RENAME FINISH!!!! , maybe call right/left subinverse?** Let  $(X, \leq_X), (Y, \leq_Y)$  be complete lattices,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . We define the

- **right  $(\leq_X, \leq_Y)$ -quasiadjoint of  $f$** , denoted  $\text{rad}_{\leq_X}^{\leq_Y} f : Y \rightarrow X$ , by  $[\text{rad}_{\leq_X}^{\leq_Y} f](y) := \sup(f^{-1}(\downarrow(y, \leq_Y))), \leq_X$ .
- **left  $(\leq_X, \leq_Y)$ -quasiadjoint of  $g$** , denoted  $\text{lad}_{\leq_X}^{\leq_Y} g : X \rightarrow Y$ , by  $[\text{lad}_{\leq_X}^{\leq_Y} g](x) := \inf(f^{-1}(\uparrow(x, \leq_X))), \leq_Y$ .

**Exercise 12.2.0.4.** Let  $(X, \leq_X), (Y, \leq_Y)$  be complete lattices,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Define  $r_f : Y \rightarrow X$  and  $l_g : X \rightarrow Y$ , by  $r_f(y) := \sup f^{-1}(\downarrow y, \leq_X)$

1. Then

- (a)  $r_f$  is monotonic
- (b)  $l_g$  is monotonic

- 2. (a)  $f$  is arbitrary join preserving iff  $f$  has a right adjoint and  $\text{rad } f = r_f$ .
- (b)  $g$  is arbitrary meet preserving iff  $g$  has a left adjoint and  $\text{lad } g = l_g$ .

*Proof.*

- 1. (a) For each  $y_1, y_2 \in Y$ ,

$$\begin{aligned}
 y_1 \leq_Y y_2 &\implies \downarrow y_1 \subset \downarrow y_2 \\
 &\implies f^{-1}(\downarrow y_1) \subset f^{-1}(\downarrow y_2) \\
 &\implies \sup f^{-1}(\downarrow y_1) \leq_X \sup f^{-1}(\downarrow y_2) \quad (\text{by prev ex}) \\
 &\implies r_f(y_1) \leq_X r_f(y_2).
 \end{aligned}$$

Hence  $r_f$  is  $(\leq_Y, \leq_X)$ -monotone.

- (b) Similar to prev part.

- 2. (a) • ( $\implies$ ):

Suppose that  $f$  is arbitrary join preserving. an ex in subsection on semilattice homomorphisms Exercise ?? implies that  $f$  is monotone. The prev ex implies that  $r_f$  is monotone.

- For each  $x \in X$ ,

$$\begin{aligned}
 f(x) \leq_Y f(x) &\implies f(x) \in \downarrow f(x) \\
 &\implies x \in f^{-1}(\downarrow f(x)) \\
 &\implies x \leq_X \sup f^{-1}(\downarrow f(x)) \\
 &\implies x \leq_X [r_f \circ f](x) \\
 &\implies \text{id}_X(x) \leq_X [r_f \circ f](x).
 \end{aligned}$$

Hence  $\text{id}_X \leq_X r_f \circ f$ .

- For each  $y \in Y$ ,

$$\begin{aligned}
 [f \circ r_f](y) &= f(\sup f^{-1}(\downarrow y)) \\
 &= \sup f(f^{-1}(\downarrow y)) \quad (\text{since } f \text{ is arbitrary join preserving}) \\
 &\leq_Y \sup \downarrow y \quad (\text{by ex on image of preimage and ex in section on suprema and subsets}) \\
 &= y \quad (\text{by ex in section on suprema}) \\
 &= \text{id}_Y(y).
 \end{aligned}$$

Hence  $f \circ r_f \leq_Y \text{id}_Y$ .

Exercise 7.2.0.6 then implies that  $(f, r_f)$  is a monotone Galois connection. Thus  $f$  has a right adjoint and  $\text{rad } f = r_f$ .

- ( $\impliedby$ ):

Suppose that  $f$  has a right adjoint and  $\text{rad } f = r_f$ . Then  $(f, \text{rad } f)$  is a monotone Galois connection. Exercise 9.3.0.9 implies that for each  $A \subset X$ ,  $f(\sup A) = \sup f(A)$ . Hence  $f$  is arbitrary join preserving.

- (b) Similar to the prev part FINISH!!!, rewrite

□

Comment on how the prev ex connects to probability theory in that the cdf of a random variable is a right continuous increasing function and therefore preserves infima. Its associated quantile function is actually its left adjoint whose most general definition is given above.

**Exercise 12.2.0.5.** Let  $(X, \leq_X), (Y, \leq_Y)$  be complete lattices,  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$ . Suppose that  $(f, g)$  is a monotone Galois connection.

1. (a)  
(b)
2. (a)  $g(1_Y) = 1_X$  iff  
(b)

**Exercise 12.2.0.6.** show that if  $g$  is a right adjoint, then  $g(\inf \uparrow x) = \inf g(\uparrow x) = \inf \uparrow(g(x)) = g(x)$

**Exercise 12.2.0.7.**

try to capture idea of a forgetful functor in terms of posets. If we go from groups to sets, we 'lose information' a group with a relations (i.e. a quotient group) is somehow 'more informative' than the free group. So if I send a set to a group in the least informative way (i.e. in a way which preserves as much entropy as possible) then we should get the free group, anything else would be more informative, with a loss of entropy. So quotient objects of a free object can maybe be characterized by some loss of entropy.

For example in **Set**,  $A \subset B$  (i.e.  $A$  is a subobject of  $B$ ) is equivalent to saying that  $A$  is a quotient of  $B$  in the opposite category, which is equivalent to the complete atomic boolean algebras, or the power set, so that if  $\iota : A \rightarrow B$  is a monomorphism in **Set**, then  $\iota^{-1} : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$  is an epimorphism in complete atomic boolean lattices and we think of  $\mathcal{P}(A)$  as a quotient of  $\mathcal{P}(B)$ . So in this sense  $\mathcal{P}(A)$  has less information or is less informative than  $\mathcal{P}(B)$  (i.e.  $A$  is less informative than  $B$  in **Set**<sup>op</sup>.)

If  $(X, x)$  is a set  $X$  with some structure  $x$ , and  $(A, a)$  is in the same category  $\mathcal{C}$  (which we order by information level) and we can think of  $(A, a)$  as a quotient of  $(X, x)$ , so it is less informative as a structured object  $(A, a) \leq_{\text{info}\mathcal{C}} (X, x)$ , then if we change the context (apply a covariant functor) to land in  $\text{Set}^{\text{op}}$  and think of  $A$  quotient of  $X$ , (i.e.  $A$  is subset of  $X$  in **Set**), so that  $A \leq_{\text{info}\text{Set}^{\text{op}}} B$ . Then we would like an adjoint of  $F$  in the sense of a poset so that we obtain the other half a monotone galois connection  $G : \text{Set}^{\text{op}} \rightarrow \mathcal{C}$ . The idea here is that the order  $\leq_{\text{info}\text{Set}^{\text{op}}}$  on  $\text{Set}^{\text{op}}$  is a part of  $\leq_{\text{info}\mathcal{C}}$ , that is  $\leq_{\text{info}\mathcal{C}}$  is  $\leq_{\text{info}\text{Set}^{\text{op}}}$  plus some extra order information

i.e. a restriction in some sense so  $(A, a) \leq_{\mathcal{C}} (B, b)$  (i.e. there is a quotient from  $B$  to  $A$ , like free group to abelianization) implies  $A \leq_{\text{Set}^{\text{op}}} B$ . So then that  $\leq_{\mathcal{C}} \subset \leq_{\text{Set}^{\text{op}}}$  (in some sense), so that  $\text{ub}((A, a), \leq_{\mathcal{C}}) \subset \text{ub}(A, \leq_{\text{Set}^{\text{op}}})$  in some sense as sets, so that  $\inf(\text{ub}(A, \leq_{\text{Set}^{\text{op}}}) \leq_{\text{Set}^{\text{op}}} \inf(\text{ub}((A, a), \leq_{\mathcal{C}}))) = A$ . Then if we have a right adjoint  $G : (\text{Set}^{\text{op}}, \leq_{\text{Set}^{\text{op}}}) \rightarrow (\mathcal{C}, \leq_{\mathcal{C}})$ , which preserves meets and is monotone, we would have that (considering  $\leq_{\mathcal{C}}$  as a subset of  $\leq_{\text{Set}^{\text{op}}}$ )

$$\begin{aligned} G(A) &= G(\inf \text{ub}(A, \leq_{\mathcal{C}}), \leq_{\mathcal{C}}) \\ &= \inf(G(\text{ub}(A, \leq_{\mathcal{C}}), \leq_{\mathcal{C}})) \\ &\geq \inf(G(\text{ub}(A, \leq_{\text{Set}^{\text{op}}}), \leq_{\mathcal{C}})) \end{aligned}$$

idk, I think we can show this aligns with the (least informative object  $(B, b)$  in  $\mathcal{C}$  which is "above"  $A$ , i.e. the structured object  $(B, b)$  generated by  $A$ ), i.e. the structured object "above"  $A$  with the least entropy gained from the minimum or zero structured element in the poset.

Try to show that a disintegration of measure, i.e. a conditional probability is the kind of the opposite or dual of this, i.e. it is the highest structured object (markov kernel + data) sitting below an unstructured object (markov kernel of a measure), so that going from (kernel + data) to disintegration results in the least entropy lost, maybe this is related to kl divergence as a measure of entropy loss, see baez/leinster/fritz paper.

In the case of statistics and say maximum likelihood estimation, and kl divergence as loss of estimate  $\hat{\theta}(x_j : j \in \mathbb{N}) : (X, P_{\theta}) \rightarrow (X, P_{\hat{\theta}})$ , there should be a categorical way of rephrasing so that the observed data  $(x_j)_{j \in \mathbb{N}} \subset X$  could perhaps be viewed as maps  $x_j : (\Lambda, \mu) \rightarrow (X, P_{\theta})$ , so that  $\hat{\theta} : \left[ \prod_{j \in \mathbb{N}} \text{Hom}((\Lambda, \mu), (X, P_{\theta})) \right] \times (X, P_{\theta}) \rightarrow (X, P_{\hat{\theta}})$

We should be able to do the same in a different context, say the true group  $(G, \cdot)$  is unknown, but we observe the elements of  $G$  and the fact that for each  $a, b \in G$ ,  $a \cdot b = b \cdot a$  (i.e. we are given the proposition that  $G$  is commutative), then what would be the best estimate for  $G$  under this information in the sense of minimizing info loss (i.e. maybe occams razor?)? Well it would probably be the free abelian group. Any further processed abelian group, (i.e. a quotient of the free abelian group) would lose more info, so even though the trivial group also satisfies the observation, it is suboptimal because we lose basically all info about  $G$ . Need a way to quantify info loss in this way. What would the analogouse 'data' be? here it seems we need a way to view relations about elements as a morphism into  $G$  from some other object, like in the case of statistics.

Vague idea: in two different contexts, say two different posets, we want to relate two objects in the ordering induced by  $a \leq b$  iff  $b$  is a quotient of  $a$ , i.e.  $b$  is more informative, i.e. there is an arrow from  $a$  to  $b$  which exhibits entropy loss.

talk about the idea of the element in a complete lattice with more structure generated another element in a complete lattice with less structure. i.e. subgroup generated subset where subgroups ordered by inclusion, power set ordered by inclusion. Or topology generated by collection of subsets (i.e. subset of powerset) where topologies ordered by inclusion and double power set ordered by inclusion. In these cases, both posets are complete lattices since arbitrary meets exist. If we consider the map  $g$  from the more structured space  $P_2$  to the less structured space  $P_1$ , by sending an object to itself minus the structure, so a group to its underlying set, or a topology to itself considered as a subset of the power set, then meets are preserved. Thus  $g$  is a right adjoint to some left adjoint  $f$  from the less structured space  $P_1$  to the more structured space  $P_2$ , so  $f$  preserves arbitrary joins. For an object  $1$  in the less structured poset, we define the object in the more structured poset generated by  $\text{obj}1$  to be the left adjoint of  $\text{obj}1$ , i.e.  $f(\text{obj}1)$ . we can find this in many cases by noting that in  $P_1$ ,  $\text{obj}1$  is the sup of its upper bounds. Since sups are preserved by  $f$ , the generated structured object in  $P_2$  is  $f(\text{obj}1)$  which (from a prev ex) is the sup of  $f$ (of the lower bounds of  $\text{obj}1$ ) i.e. the sup of  $f(\downarrow o1)$ , which should be the sup of all upper bounds of  $f(\text{obj}1)$ , which is the inf

ok in  $P_1$  is less structured,  $P_2$  is more structured,  $f : P_1 \rightarrow P_2$  preserves joins, adjoint  $g$  preserves meets, then the object in  $P_2$  generated by  $a \in P_1$  is  $f(a) = f(\sup \downarrow a) = \sup f(\downarrow a)$ , or something like this

## 12.3 Properties of Ideals and Filters

### 12.3.1 Prime Ideals and Filters in Complete Lattices

**Definition 12.3.1.1.** Let  $L$  be a complete lattice.

- Let  $I \in \mathcal{I}(L)$ . Then  $I$  is said to be
  - **completely prime** if
    1.  $I$  is proper
    2. for each  $A \subset L$ ,  $\bigwedge A \in I$  implies that there exists  $a \in A$  such that  $a \in I$ .
- Let  $F \in \mathcal{F}(L)$ . Then  $F$  is said to be
  - **completely prime** if
    1.  $F$  is proper
    2. for each  $A \subset L$ ,  $\bigvee A \in F$  implies that there exists  $a \in A$  such that  $a \in F$ .

**Exercise 12.3.1.2.** Let  $L$  be a complete lattice.

1. For each  $I \in \mathcal{I}(L)$ , if  $I$  is completely prime, then  $I \in \mathcal{I}_p(L)$ .
2. For each  $F \in \mathcal{F}(L)$ , if  $F$  is completely prime, then  $F \in \mathcal{F}_p(L)$ .

*Proof.* Clear. fill out

□

**Exercise 12.3.1.3.** Let  $L$  be a complete lattice.

1. For each  $I \in \mathcal{I}_{cp}(L)$ ,
  - (a)  $\bigwedge I^c \in I^c$ ,
  - (b)  $\bigwedge I^c \in \text{prime}_\vee(L)$ .
2. For each  $F \in \mathcal{F}_{cp}(L)$ ,
  - (a)  $\bigvee F^c \in F^c$ ,
  - (b)  $\bigvee F^c \in \text{prime}_\wedge(L)$ .

*Proof.*

1. Let  $I \in \mathcal{I}_{cp}(L)$ .

- (a) For the sake of contradiction, suppose that  $\bigwedge I^c \in I$ . Since  $I$  is completely prime, there exists  $a \in I^c$  such that  $a \in I$ . This is a contradiction. Hence  $\bigwedge I^c \notin I$ . Thus  $\bigwedge I^c \in I^c$ .
- (b) i. Since  $I \in \mathcal{I}(L)$ , **a previous ex** implies that  $0 \in I$ . Since  $\bigwedge I^c \in I^c$ , we have that  $\bigwedge I^c \neq 0$ .
- ii. Define  $x \in L$  by  $x := \bigwedge I^c$ . Since  $I \in \mathcal{I}(L)$ , **a prev ex** implies that  $I$  is a lower set. Let  $a, b \in L$ . Suppose that  $x \leq a \vee b$ . For the sake of contradiction, suppose that  $a \in I$  and  $b \in I$ . Since  $I \in \mathcal{I}(L)$ , we have that  $a \vee b \in I$ . Since  $I$  is a lower set and

$$\begin{aligned} x &\leq a \vee b \\ &\in I, \end{aligned}$$

we have that  $x \in I$ . This is a contradiction since  $x \in I^c$ . Hence  $a \notin I$  or  $b \notin I$ . Thus  $a \in I^c$  or  $b \in I^c$ . Since  $x = \bigwedge I^c$ , we have that  $x \leq a$  or  $x \leq b$ . Since  $a, b \in L$  with  $x \leq a \vee b$  are arbitrary, we have that for each  $a, b \in L$ ,  $x \leq a \vee b$  implies that  $x \leq a$  or  $x \leq b$ .

Hence  $x \in \text{prime}_\vee(L)$ .

2. **use duality**

□

**Exercise 12.3.1.4.** Let  $L$  be a complete lattice. Then

1. for each  $x \in \text{prime}_\vee(L)$ ,  $(\uparrow x)^c \in \mathcal{I}_{cp}(L)$ .
2. for each  $x \in \text{prime}_\wedge(L)$ ,  $(\downarrow x)^c \in \mathcal{F}_{cp}(L)$ .

*Proof.*

1. Let  $x \in \text{prime}_\vee(L)$ . **a prev ex** implies that  $\uparrow x \in \mathcal{F}(L)$ .

- (a) Since  $x \in \text{prime}_\vee(L)$ ,  $x \neq 0$ . Hence  $0 \in (\uparrow x)^c$ . Thus  $(\uparrow x)^c \neq \emptyset$ .
- (b) Let  $a, b \in (\uparrow x)^c$ . For the sake of contradiction, suppose that  $a \vee b \in \uparrow x$ . Then  $x \leq a \vee b$ . Since  $x \in \text{prime}_\vee(L)$ ,  $x \leq a$  or  $x \leq b$ . Hence  $a \in \uparrow x$  or  $b \in \uparrow x$ . This is a contraction since  $a \in (\uparrow x)^c$  and  $b \in (\uparrow x)^c$ . Therefore  $a \vee b \in (\uparrow x)^c$ . Since  $a, b \in (\uparrow x)^c$  are arbitrary, we have that for each  $a, b \in (\uparrow x)^c$ ,  $a \vee b \in (\uparrow x)^c$ .
- (c) Let  $y \in L$  and  $a \in (\uparrow x)^c$ . For the sake of contradiction, suppose that  $y \wedge a \in \uparrow x$ . Then

$$\begin{aligned} x &\leq y \wedge a \\ &\leq a. \end{aligned}$$

Thus  $a \in \uparrow x$ , which is a contradiction. Hence  $y \wedge a \in (\uparrow x)^c$ . Since  $y \in L$  and  $a \in (\uparrow x)^c$  are arbitrary, we have that for each  $y \in L$  and  $a \in (\uparrow x)^c$ ,  $y \wedge a \in (\uparrow x)^c$ .

Thus  $(\uparrow x)^c \in \mathcal{I}(L)$ .

- (a) Since  $\uparrow x \neq \emptyset$ , we have that  $(\uparrow x)^c \neq L$ . Thus  $(\uparrow x)^c$  is proper.
- (b) Let  $A \subset L$ . Suppose that  $\bigwedge A \in (\uparrow x)^c$ . For the sake of contradiction, suppose that for each  $a \in A$ ,  $a \notin (\uparrow x)^c$ . Then for each  $a \in A$ ,  $a \in \uparrow x$ . Hence for each  $a \in A$ ,  $x \leq a$ . Therefore  $x \in \text{lb } A$ . Thus

$$\begin{aligned} x &\leq \inf A \\ &= \bigwedge A. \end{aligned}$$

Then  $\bigwedge A \in \uparrow x$ , which is a contradiction. Hence there exists  $a \in A$  such that  $a \in (\uparrow x)^c$ . Since  $A \subset L$  with  $\bigwedge A \in (\uparrow x)^c$  is arbitrary, we have that for each  $A \subset L$ ,  $\bigwedge A \in (\uparrow x)^c$  implies that there exists  $a \in A$  such that  $a \in (\uparrow x)^c$ .

Thus  $(\uparrow x)^c \in \mathcal{I}_{cp}(L)$ .

2. **similar or duality**

□

**maybe make ex in earlier section on ideals that  $\downarrow x = \downarrow y$  iff  $x = y$  and similar for  $\uparrow x$ . Then in this section, show that for complete lattice  $L$ ,  $\mathcal{F}_{cp}(L) \cong \text{prime}_\wedge(L)$  bijective.**



## 12.4 Constructions with Complete Lattices

### 12.4.1 Complete Sublattices

**Definition 12.4.1.1.** **define** complete sublattice Let  $L, M$  be lattices

### 12.4.2 Products of Complete Lattices

### 12.4.3 Coproducts of Complete Lattices

### 12.4.4 Quotients of Complete Lattices

## 12.5 Fundamental Examples

### 12.5.1 Introduction

**Exercise 12.5.1.1.** Let  $L$  be a complete lattice.

### 12.5.2 Sets

### 12.5.3 Equivalence Relations

*Proof.* Let  $X$  be a set. Suppose that  $X \neq \emptyset$ . Then  $(\text{Equiv}, \subset)$  is a complete lattice. □

### 12.5.4 Ideal and Filter Lattices

**Exercise 12.5.4.1.** Let  $L$  be a lattice. If  $L$  has a zero, then

1.  $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$  is a complete lattice.
2.  $(\mathcal{F}(L), \leq_{\mathcal{F}(L)})$  is a complete lattice.

### 12.5.5 Topological Spaces

## 12.6 Least Upper Bound Property

**Definition 12.6.0.1.** Let  $(X, \leq)$  be a poset. Then  $(X, \leq)$  is said to satisfy the

- **least upper bound (LUB) property** if for each  $A \subset X$ , if  $A \neq \emptyset$  and  $A$  is bounded above, then there exists  $x \in X$  such that  $x = \sup A$
- **greatest lower bound (GLB) property** if for each  $A \subset X$ , if  $A \neq \emptyset$  and  $A$  is bounded below, then there exists  $x \in X$  such that  $x = \inf A$ .

**Exercise 12.6.0.2.** LUB iff GLB

*Proof.* **FINISH!!!!** □

**Definition 12.6.0.3. Suplattice:**

Let  $(L, \leq)$  be a poset. Then  $(L, \leq)$  is said to be a **suplattice** if for each  $A \subset L$ , there exists  $x \in L$  such that  $x = \sup A$ .

## 12.7 Complete Lattice Homomorphisms

**Part IV**

**Continuous Lattices**



pulling from [21]



# Chapter 13

# Scott

Definition 13.0.0.1.





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